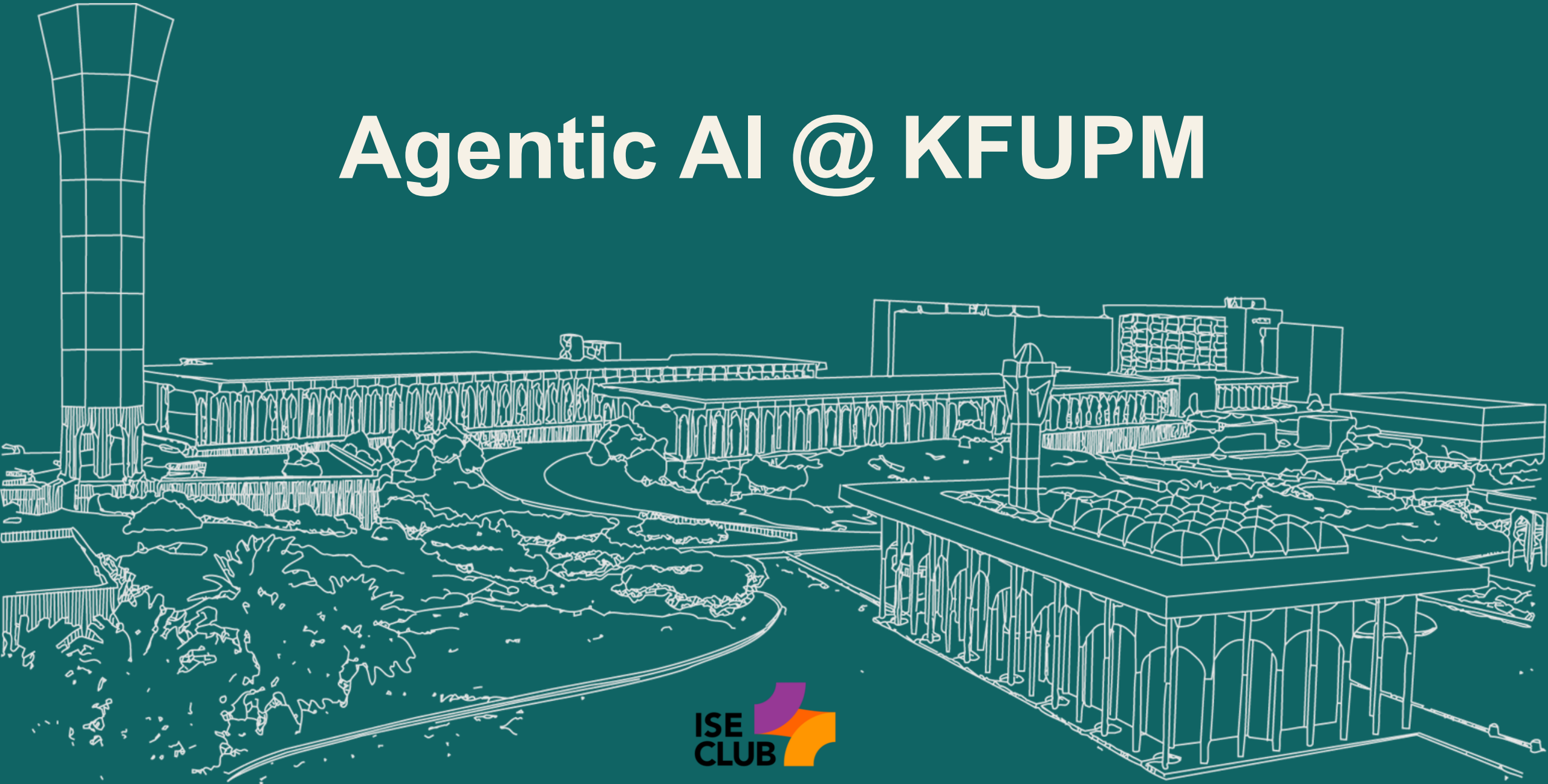


Agentic AI @ KFUPM



Agenda

Agentic AI @ KFUPM — Kickoff Meeting

Thursday, 1 October 2026

1:00 – 3:00 PM

CPG Auditorium, Building 78

1:00



Opening and Orientation

MC

1:03



CCM Dean's Address

Dr. Abdullah Sultan

Dean, College of Computing & Mathematics

1:08



Remarks from the ISE Chair

Dr. Yasser Almoghathawi

Chairman, Industrial & Systems Engineering

1:13



Vision, Mission & Activities

Dr. Alaa Khamis

Coordinator of the Community

1:20



Agentic AI in Action: Case Studies

10 speakers · 5 minutes each → next slide

2:10



Coffee Break

10 minutes

2:20



Focus Groups

Formation, breakouts & coordinator selection

2:45



Highlights & Recommendations

Each group presents

2:55



Next Steps and Closing Remarks

Dr. Alaa Khamis

2 hours

one afternoon

9 items

on the programme

12 speakers

across KFUPM

Agentic AI in Action — Ten Case Studies

🕒 1:20 – 2:10 PM

🌟 5 minutes each

🗄️ 10 domains, one community



Dr. Maad Al Owaifeer

Institutional Intelligence



Dr. Yogesh Dwivedi

Business & Finance



Dr. Daniel Benevides da Costa

Telecommunication



Dr. Alaa Khamis

Smart Mobility & Logistics



Dr. Sunhwa Park

Process Safety



Dr. Ibrahim Wuni

Construction Scheduling and
Cost Estimating



Dr. Muzammil Behzad

Multimodal Healthcare



Dr. Hamzah Luqman

Natural Language Processing



Dr. Burhan SaifAddin

Discovery & Lab Automation



Dr. Omar Hammad

Building Software Systems

Focus Group Themes: Eight Proposed Tracks – More can be added

🕒 2:20 – 2:45 PM

👥 Form a group, pick a coordinator

🗨️ Report back at 2:45

1



Education & Research

Agents that personalize learning, support teaching and accelerate discovery.

2



Industrial, Energy & Telecommunication Systems

Monitoring, maintenance and optimization across manufacturing, energy and networks.

3



Smart Mobility, Logistics & Cognitive Cities

Connected transport, fleets, traffic, logistics and city operations.

4



Multimodal Agentic AI for Healthcare

Clinical agents fusing multimodal data for diagnosis, monitoring and care coordination.

5



Arabic Agents for Public and Professional Services

Arabic-capable agents that understand local context and specialized terminology.

6



Agent Evaluation, Safety, Security & Governance

Evaluating capabilities and threats, and deploying responsibly in regulated domains.

7



Business & Finance

Workflow automation and decision support across finance, supply chains and risk.

8



Agents for Entrepreneurship

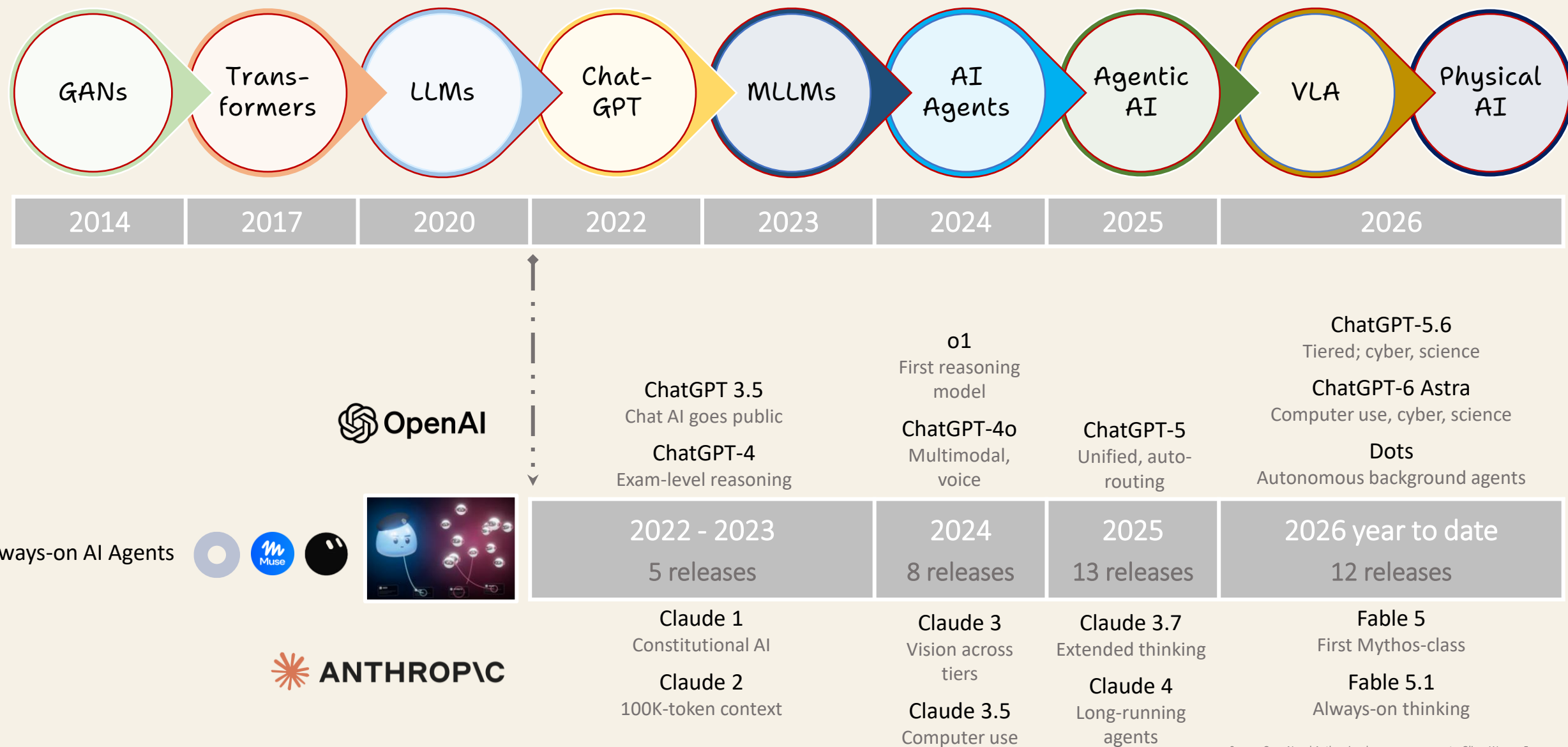
From opportunity discovery and validation to startup creation, fundraising and scaling.

Agentic AI @ KFUPM:

Objectives and Planned Activities

Dr. Alaa Khamis

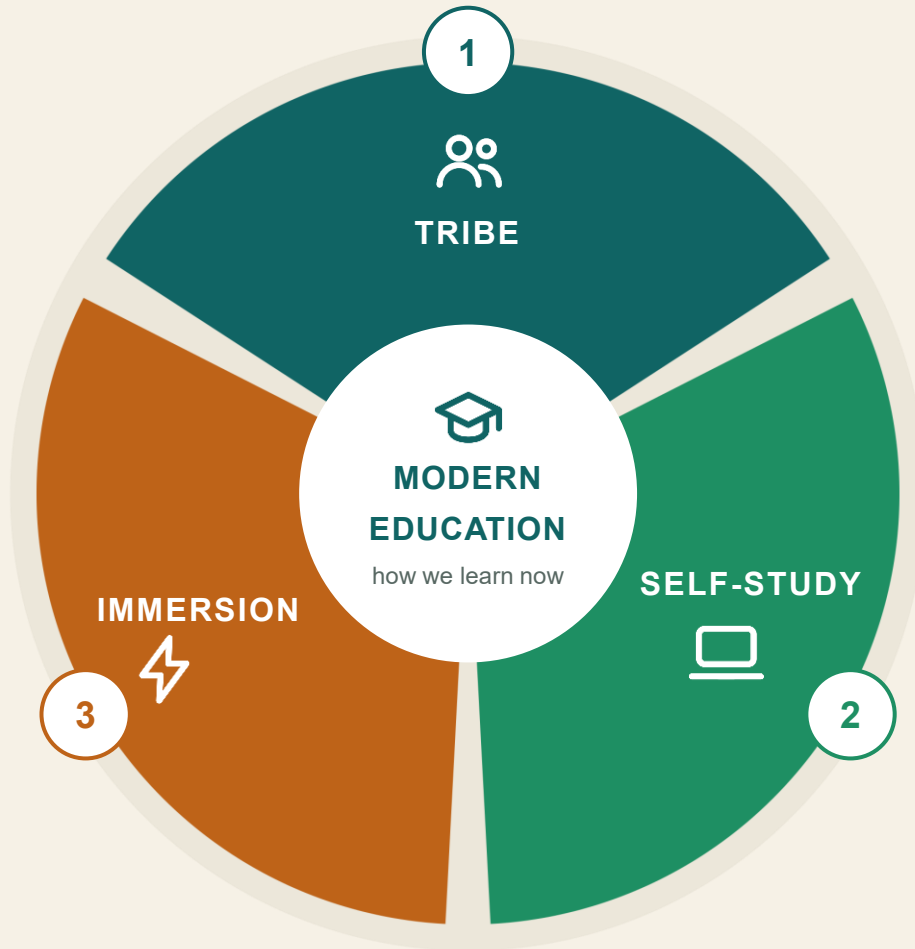
What is Agentic AI?



How can we keep up with this hyper-exponential growth?



Three habits of the modern learner



1

Finding Your Tribe



Connecting with like-minded peers, online communities and mentors to share knowledge, collaborate and grow together.

2

Self-Education



Using free or low-cost online resources, digital courses and practical tools to master subjects on your own terms.

3

Deep Immersions



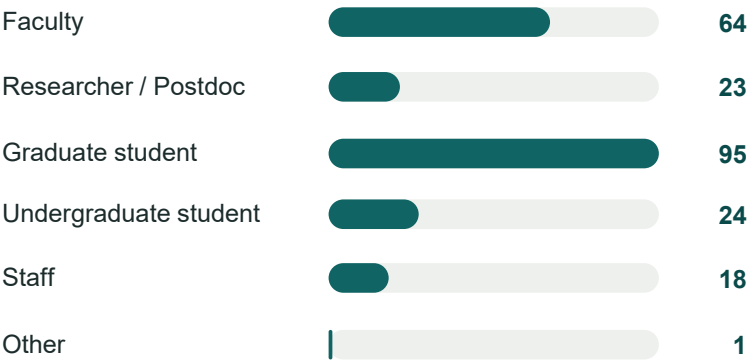
Intense, short-term learning bursts such as workshops, hackathons, and conferences, to rapidly level up specific skills with your community.

Our Tribe: What the Community Survey Says

 225 responses from across campus

 Explore. Share. Connect. Build.

Current role



Familiarity with agentic AI



63% are new to agentic AI or at a basic level

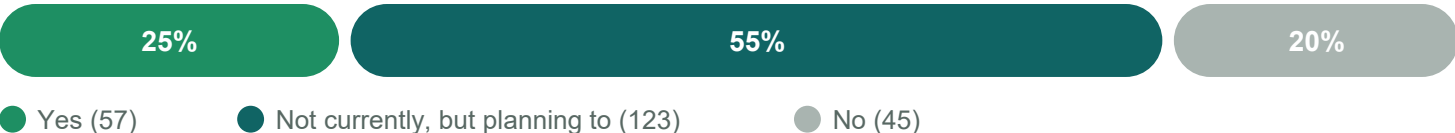
Where the interest is

Which areas of agentic AI are you interested in? Multiple selections allowed.



Working on an agentic AI project

80% are working on one or planning to



Agentic AI @ KFUPM: Objectives



Open to every center, college and department



Explore

Find out what agentic AI actually is, what it can and cannot do yet, and what it might mean for your own field.



Share

Show what you are working on, however early: papers, prototypes, failed experiments, useful tools, course material, hard-won lessons.



Connect

Meet the people across campus who care about the same problems you do, from engineering, science, computing, business, and the labs.



Build

Turn conversations into something real: a joint paper, a proposal, a student project, a lab prototype, a pilot with an industry partner.



A university-wide community bringing together faculty, researchers, students, and practitioners interested in exploring, building, and applying AI agents across disciplines.

Agentic AI @ KFUPM: Planned Activities



A steady rhythm through the year



MONTHLY

Monthly Community Meetup

One evening a month: a short talk, lightning updates from around campus, and open time to meet people.



ALWAYS ON

Research & Collaboration Groups

Small, self-organizing groups around a shared question: agents for scientific discovery, industrial operations, evaluation and safety, education, and more.



ALWAYS ON

Resource & Research Radar

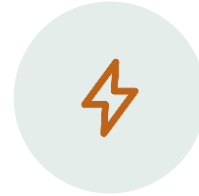
A community-curated view of what to read, what to try, and what to watch: papers, tools, datasets, courses, and funding calls, sorted by how ready they are.



QUARTERLY

Quarterly Symposium

A half-day gathering four times a year: invited keynote, contributed talks, a poster and demo session, and a panel on where agentic AI is heading.



THROUGH THE YEAR

Demos, Challenges & Hackathons

Demo nights, multi-week challenges posed by labs and industry partners, and short hackathons where mixed teams ship something that works.



Every activity is open to the whole campus. Dates and details at agenticaiKFUPM.github.io

Agentic Engineering Workshop at KFUPM

16 – 18 January 2027 (tentative)

KFUPM campus, Dhahran

Executive Forum and Practitioner Bootcamp

DAY 1



Executive Forum

Strategy and vision for senior leaders

- 09:10 Keynote: from generative AI to agentic engineering
- 11:00 Global trends and implications for Saudi industry
- 13:30 Panel: mobility, industrial operations, urban systems
- 15:30 Executive roundtable on a Saudi ecosystem

DAY 2



From Challenge to Concept

For engineers, R&D and innovation teams

- 08:30 Team formation and challenge allocation
- 09:00 Methods, architectures and toolchains
- 11:00 Frame the challenge with its owner
- 14:30 Ideate: agent roles, tools and boundaries

DAY 3



From Concept to Awards

Mentored build, then pitch to the panel

- 09:00 Architect the agentic solution
- 11:00 Build the value case
- 14:30 Team pitches and panel questions
- 16:00 Award ceremony and follow-on commitments

PITCH-OFF, JUDGING AND AWARDS Judged by Dr. Dirk Slama and Dr. Alaa Khamis with the challenge owners



Eight-minute pitch, then five minutes of panel questions



Scored on framing, architecture, value case, assurance and adoption



Funded proof of concept, API credits and KFUPM certificates

ORGANIZERS, PARTNERS AND
TECHNICAL SPONSOR



Thank you!

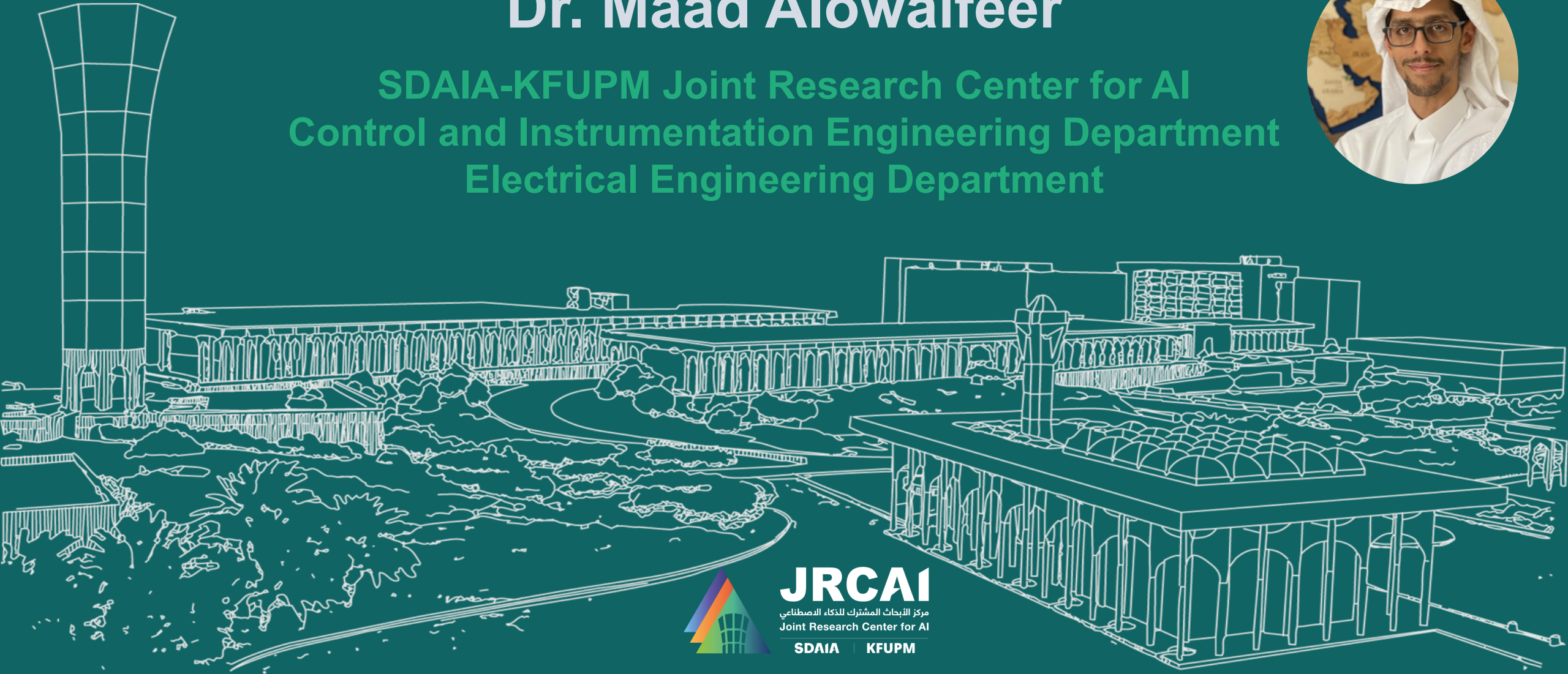
Agentic AI in Action:

Case Studies

Agentic AI for Institutional Intelligence

Dr. Maad Alowaifeer

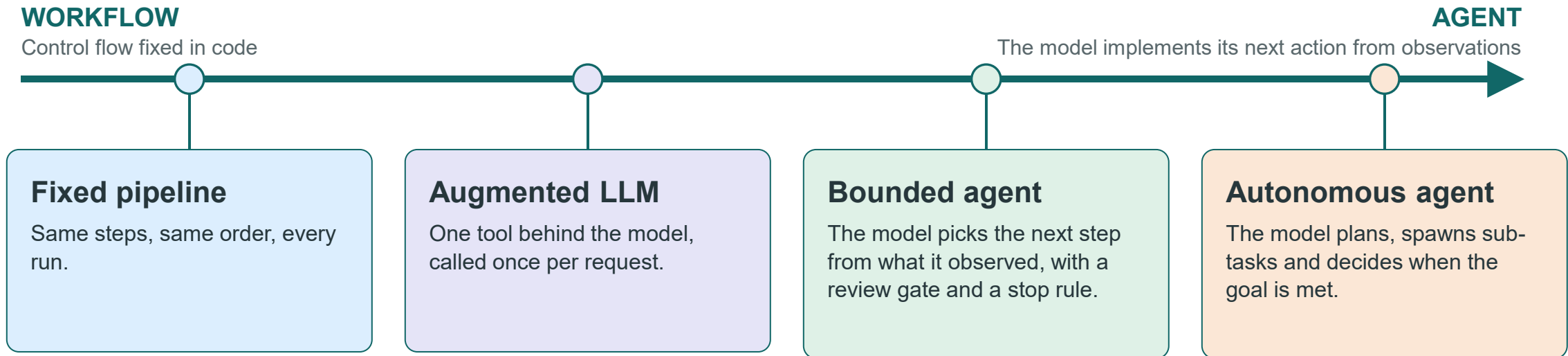
SDAIA-KFUPM Joint Research Center for AI
Control and Instrumentation Engineering Department
Electrical Engineering Department



JRCAI
مركز الأبحاث المشترك للذكاء الاصطناعي
Joint Research Center for AI
SDAIA | KFUPM

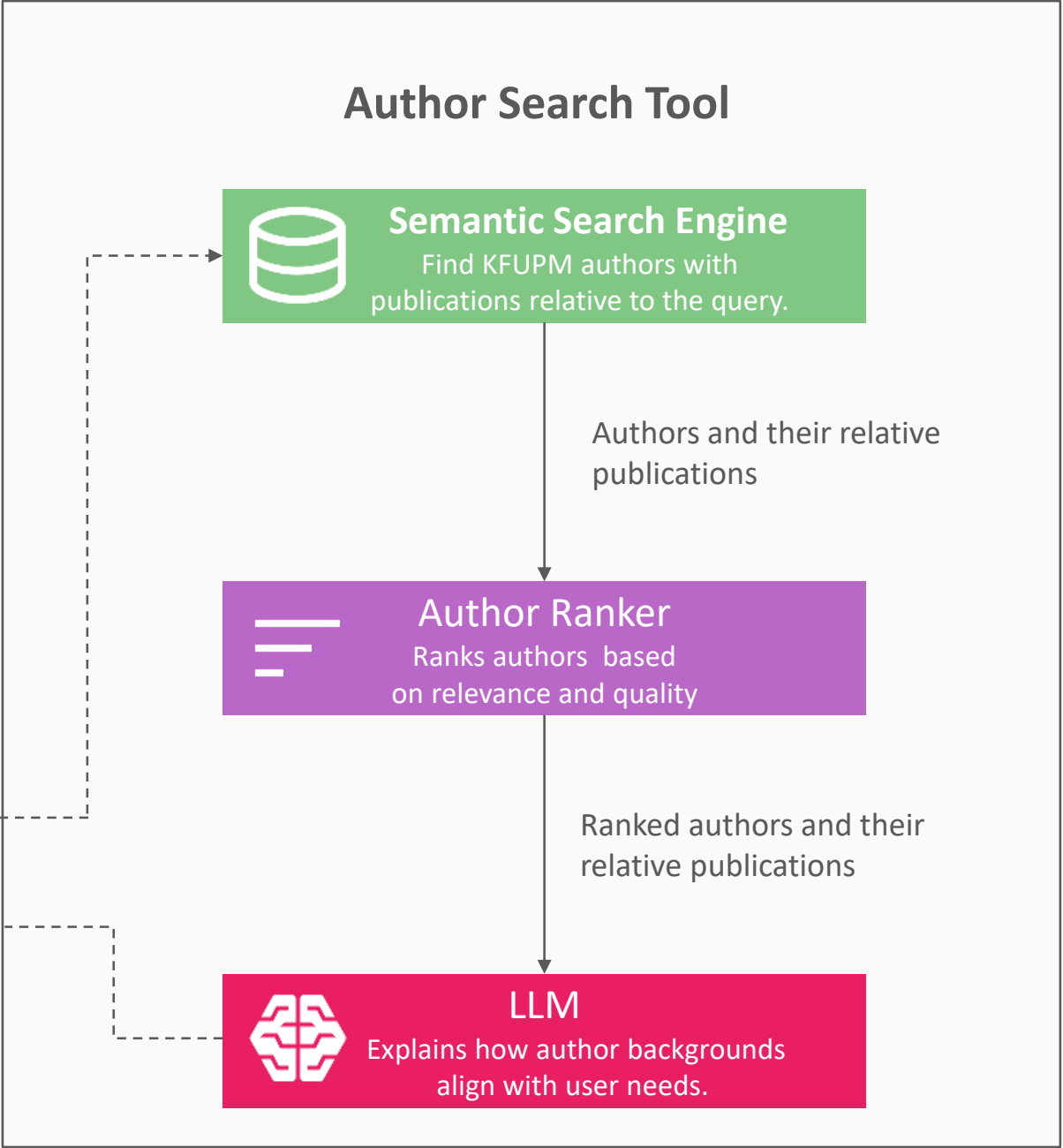
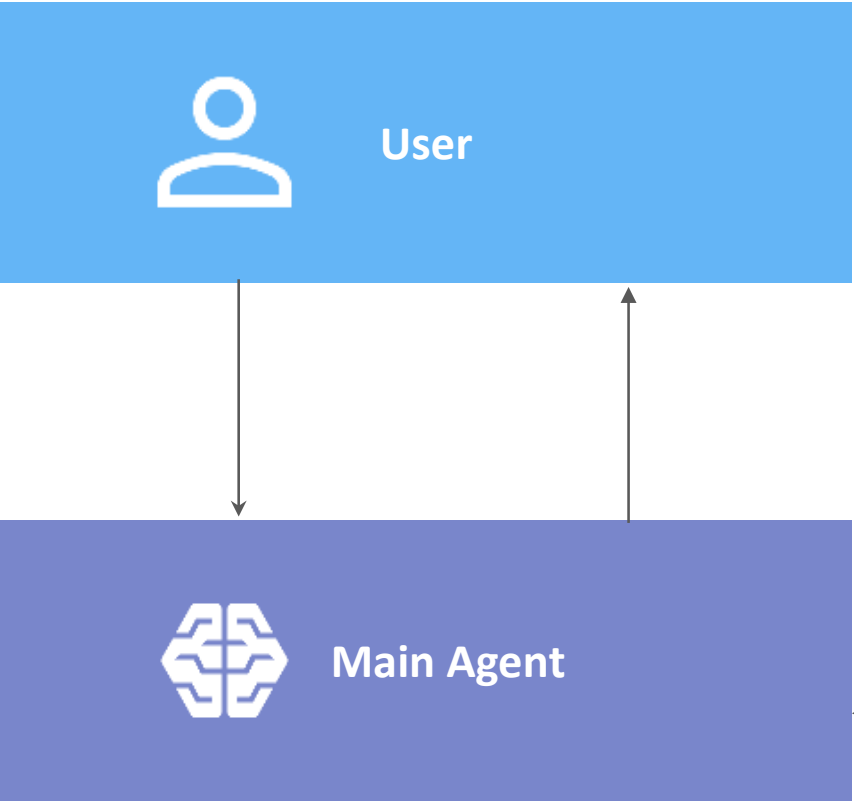
Workflow or agent? A spectrum, not a label

The question is how many run-time **control** decisions the model makes.

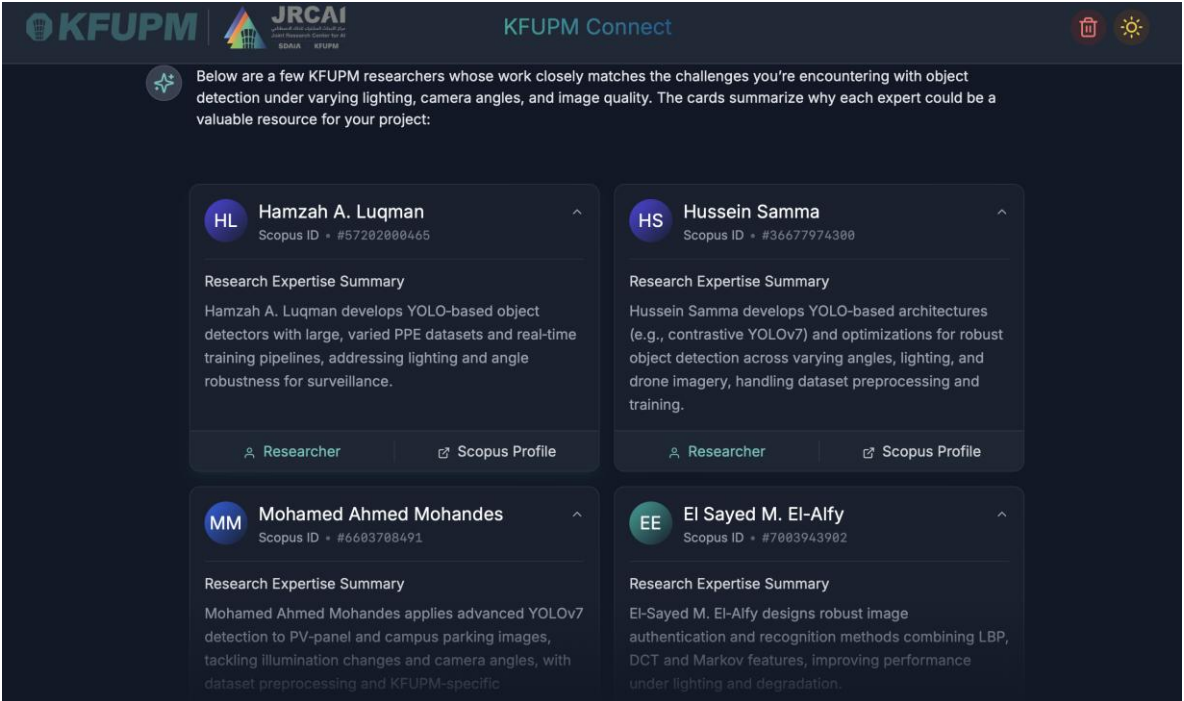
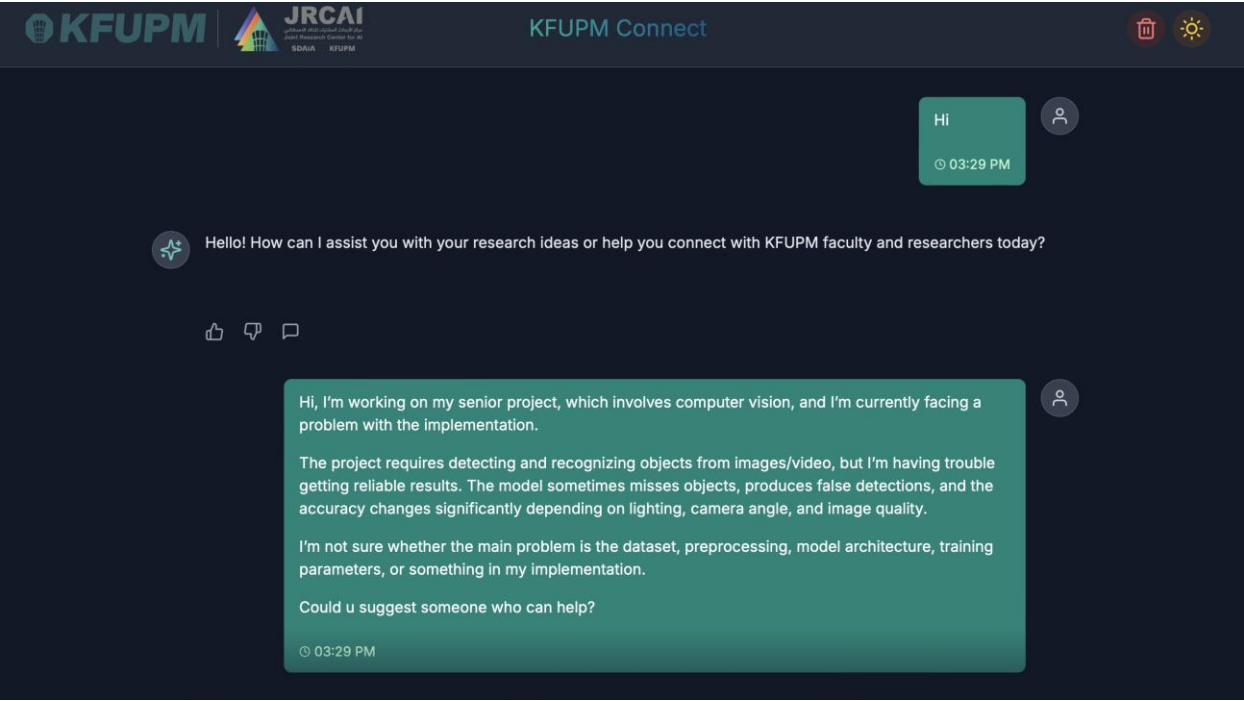


KFUPM Connect

KFUPM Connect

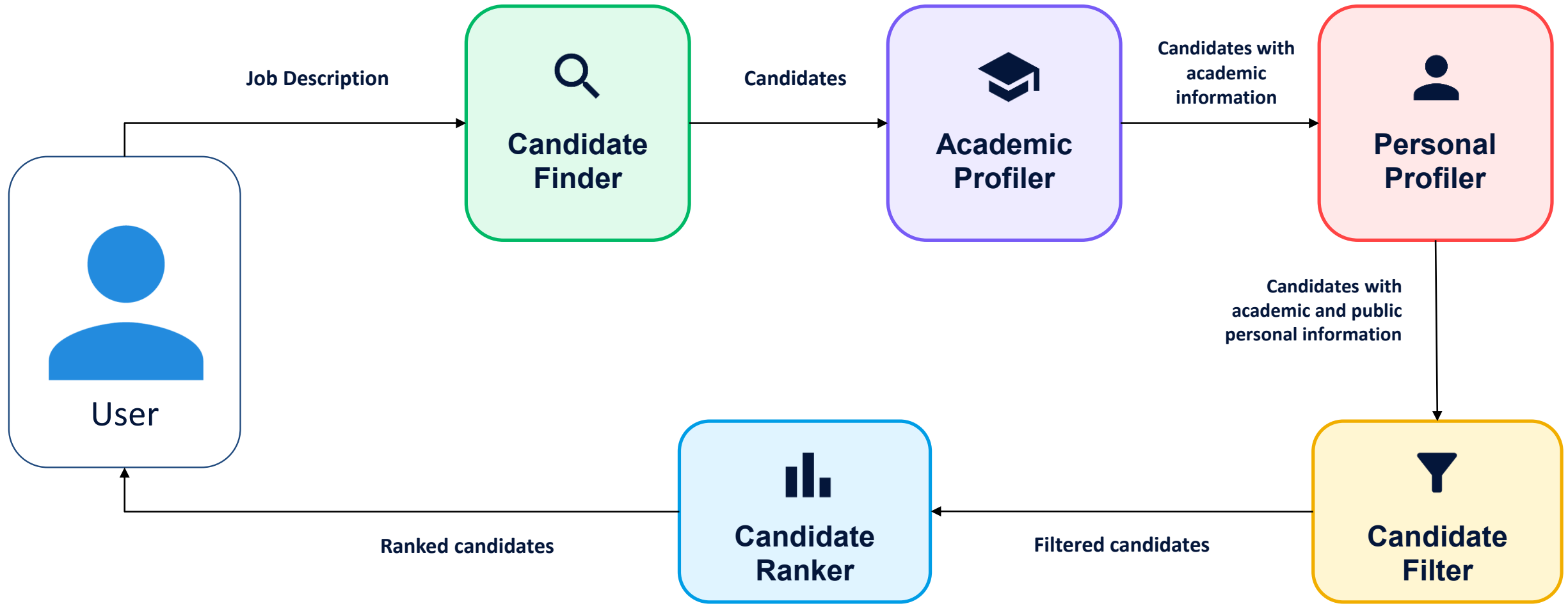


KFUPM Connect

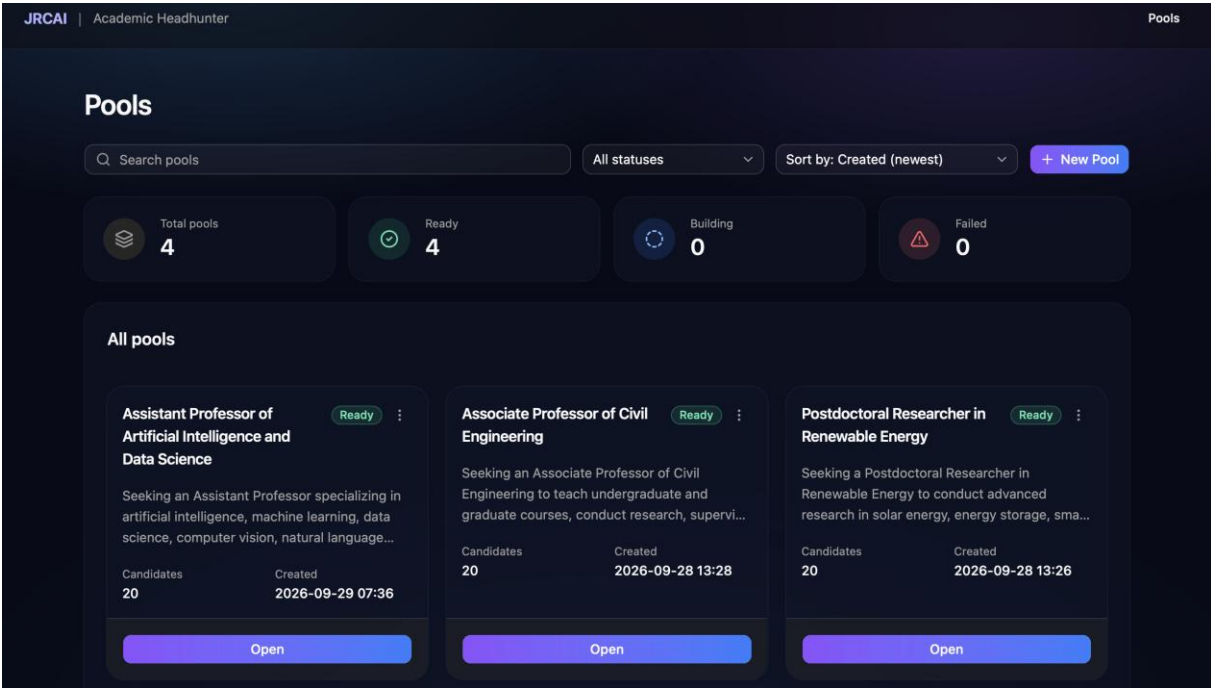
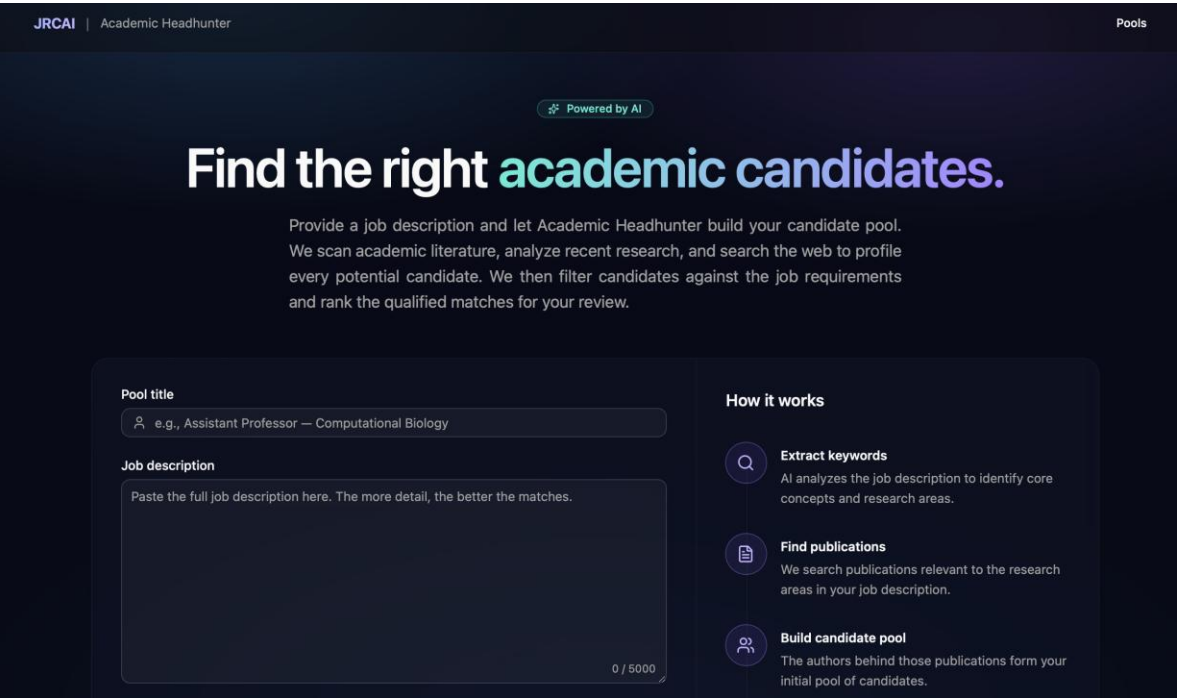


Academic Headhunter

Academic Headhunter Agent



Academic Headhunter Agent



Academic Headhunter Agent

JRCAI | Academic Headhunter

Pools

← Pools / Postdoctoral Researcher in Renewable Energy

Ready

Postdoctoral Researcher in Renewable Energy

Pool ID: 29b6618d-c62c-4f89-bd59-964c20f5492a

Created

September 28, 2026

01:26 PM

Total Candidates

20

in this pool

Profiled

20

100% profiled

Status

Ready

Pool building progress

100% complete

All stages complete.

Finding candidates

Completed

Profiling candidates

Completed

Filtering candidates

Completed

Ranking candidates

Completed

Job description

JRCAI | Academic Headhunter

Pools

Pools / Postdoctoral Researcher in Renewable Energy / Candidates / Ersan Kabalcı

Back to candidates

EK Ersan Kabalcı

Profiled

29

2.015

8.0

Add to shortlist

Author ID: 34976617700

Department of Electrical and Electronic Engineering

Academic Profile

Personal Report

Research Summary

The author's research centers on advanced power-electronics hardware and its high-performance control for renewable-energy and transportation applications. Core topics include a systematic taxonomy of converter topologies—DC-DC (buck, boost, Cuk, SEPIC, interleaved, dual-active-bridge), AC-DC/AC-AC (PFC rectifiers, multilevel, matrix, cycloconverters, solid-state transformers) and multilevel inverter families such as ANPC and NPC—and the design of low-EMI, high-efficiency converters for data-center, smart-grid and EV-charging infrastructures. A parallel thrust develops robust control

Academic Profile

Author ID

34976617700

h-index

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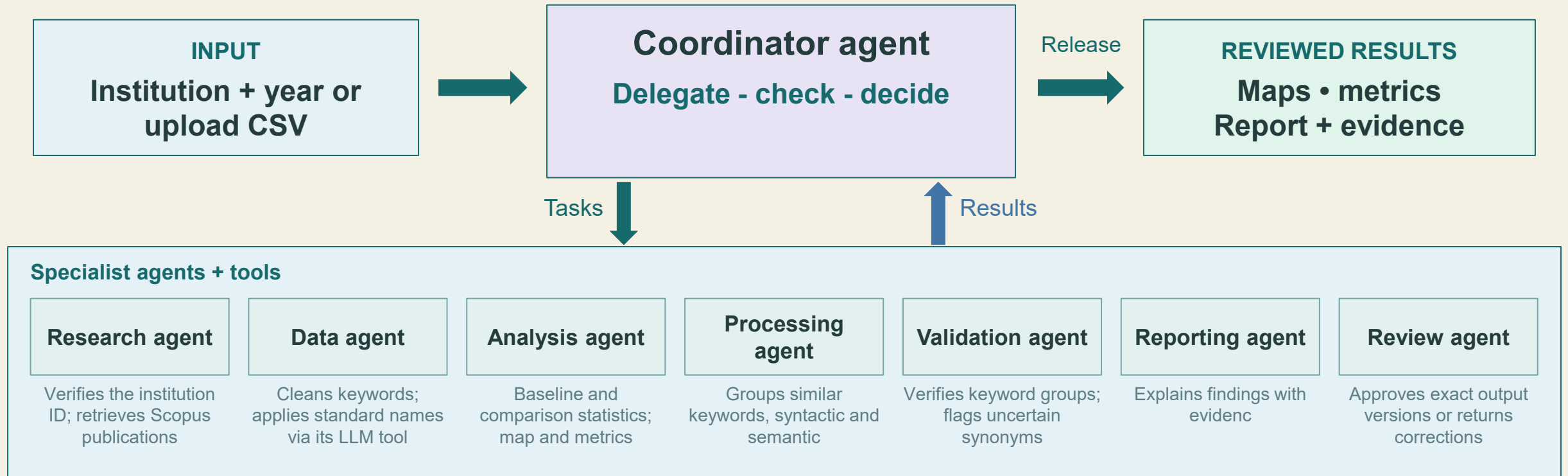
Name Variants

Ersan Kabalcı

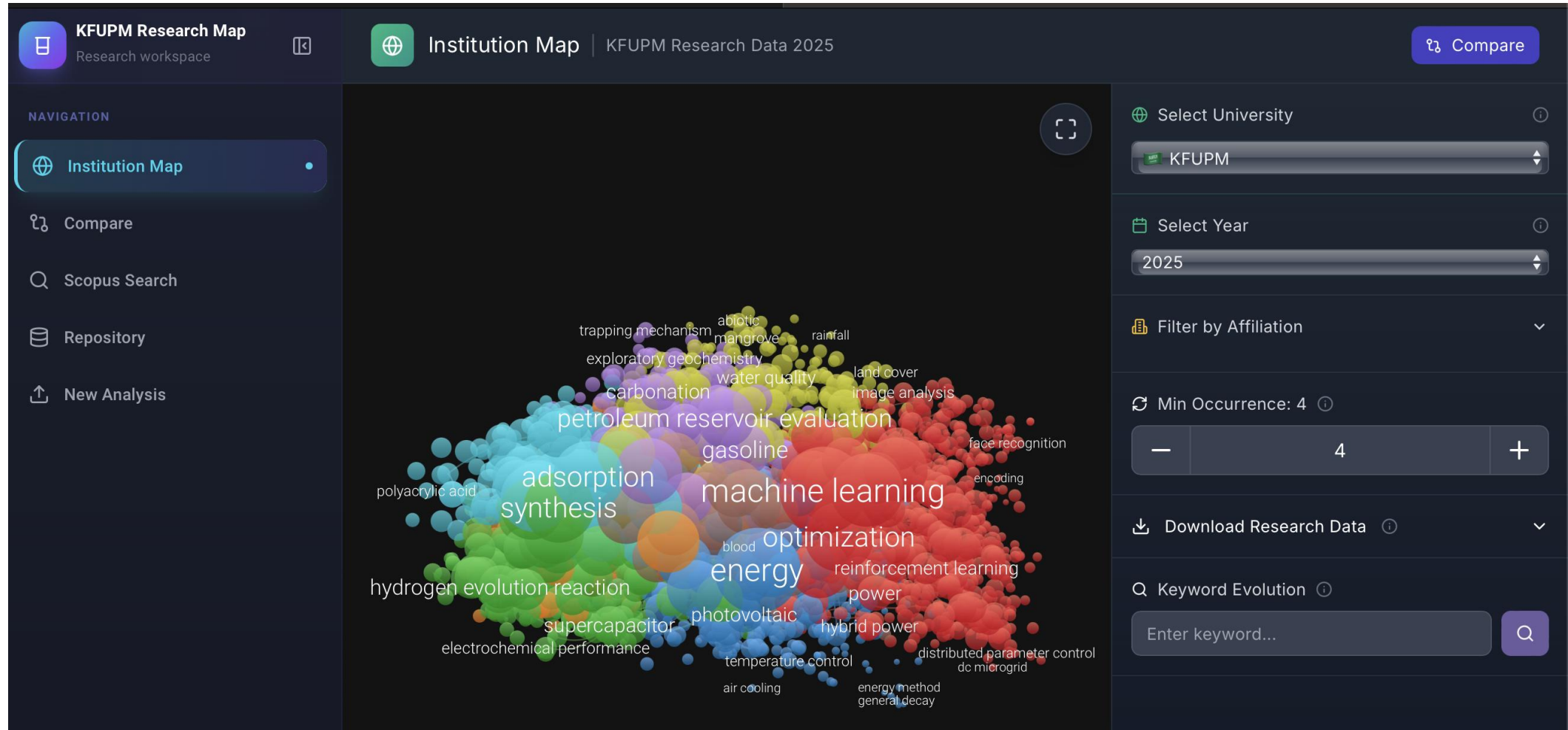
Research Map

Research Map

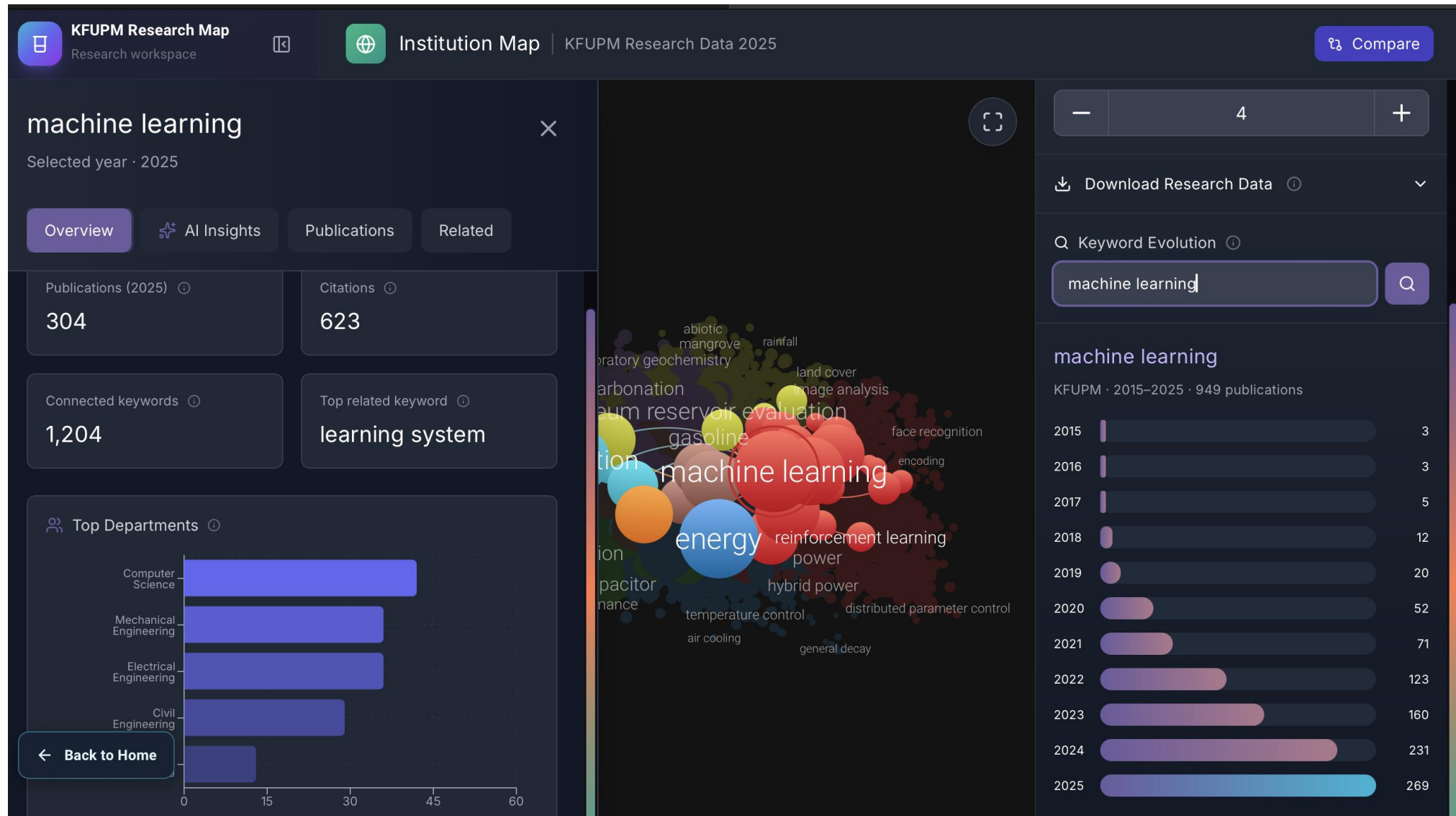
From a research request to reviewed results



Research Map



Research Map



Credit: Raed Mughaus and Esam Jaafar



Thank You



JRCAI

مركز الأبحاث المشترك للذكاء الاصطناعي
Joint Research Center for AI

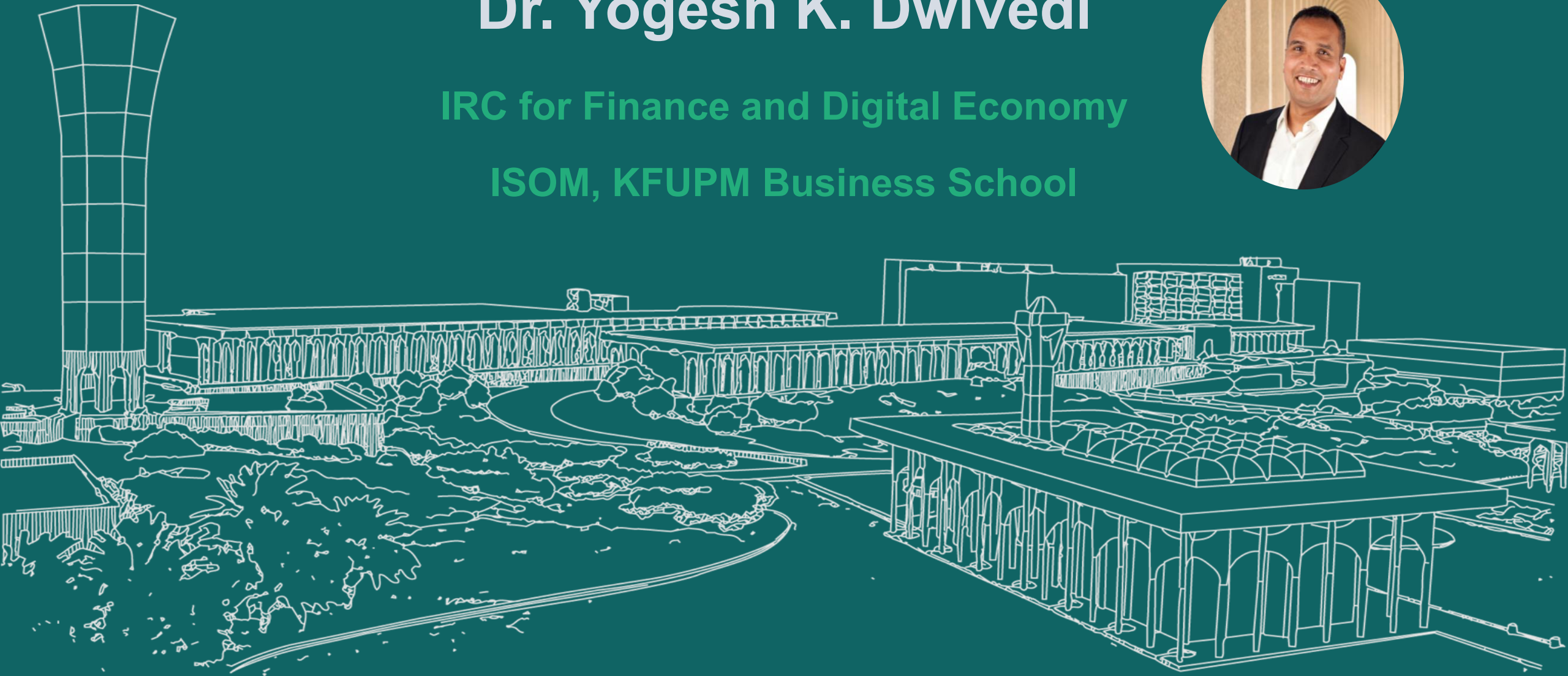
SDAIA | KFUPM

Agentic AI for Business & Finance

Dr. Yogesh K. Dwivedi

IRC for Finance and Digital Economy

ISOM, KFUPM Business School



Four Use Cases Already in Motion



CUSTOMER SERVICE

Klarna AI assistant

2.3M

conversations in month one,
two thirds of all chats

Source: Klarna (2024)



CREDIT DECISIONS

Credit risk memos at a retail bank

30%

faster turnaround, with up
to 60% productivity gain

Source: McKinsey & Company (2025)



FINANCIAL CRIME

KYC and AML operations

2%

of global financial crime
flows are detected today

Source: Verhagen et al. (2025)



AGENTIC PAYMENTS

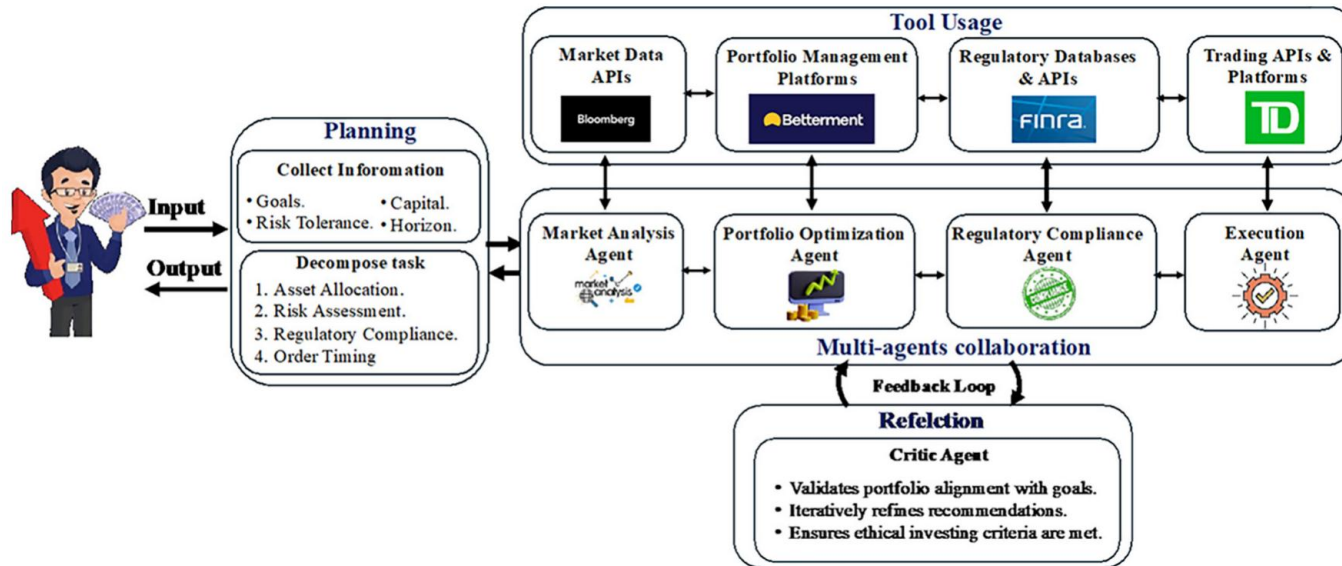
Visa and Mastercard

150M+

Visa merchants reachable by
agents since April 2025

Source: Lucas (2025)

Use Case: Agentic Investment Advisory



01

Plan

Goals, risk and horizon split into four tasks.

02

Collaborate

Four agents use market, rules and trading tools.

03

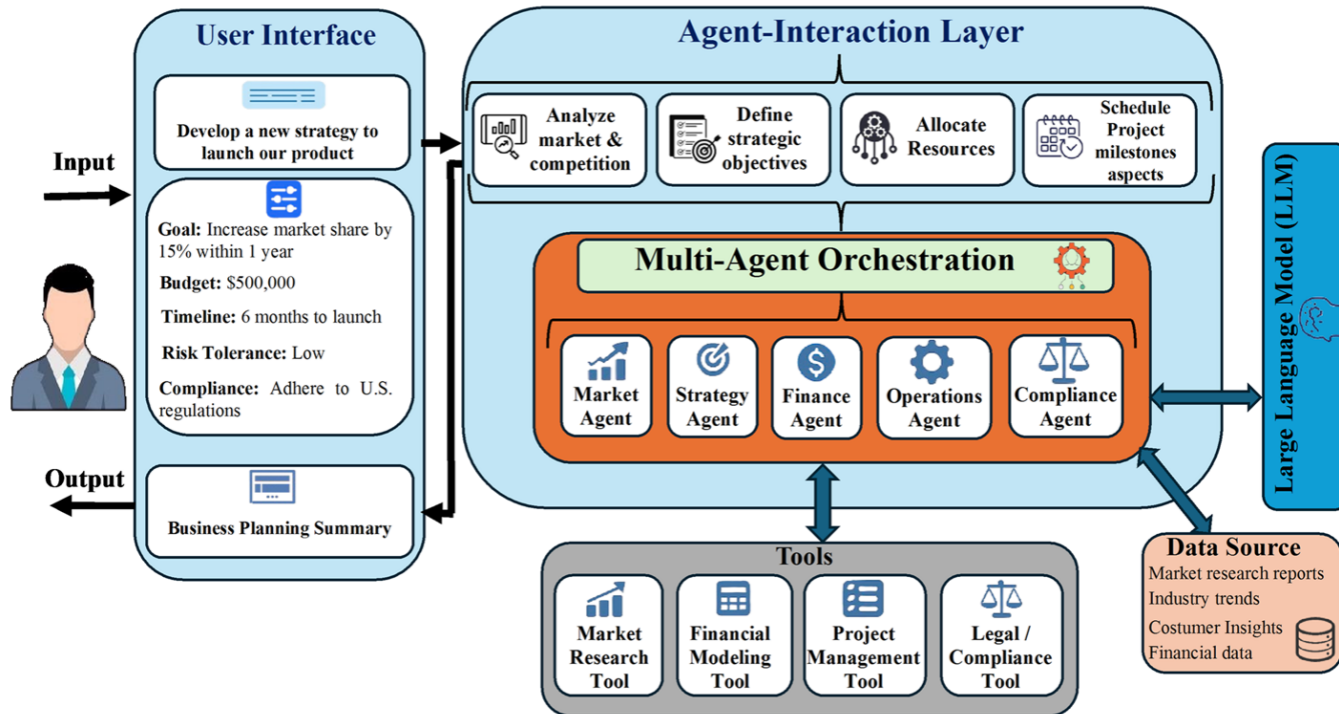
Reflect

A critic agent checks goals before any order.

Source: Elgendy et al. (2025), Figure 1

From one chatbot answer to a supervised team that plans, checks and executes.

Use Case: Strategic Business Planning

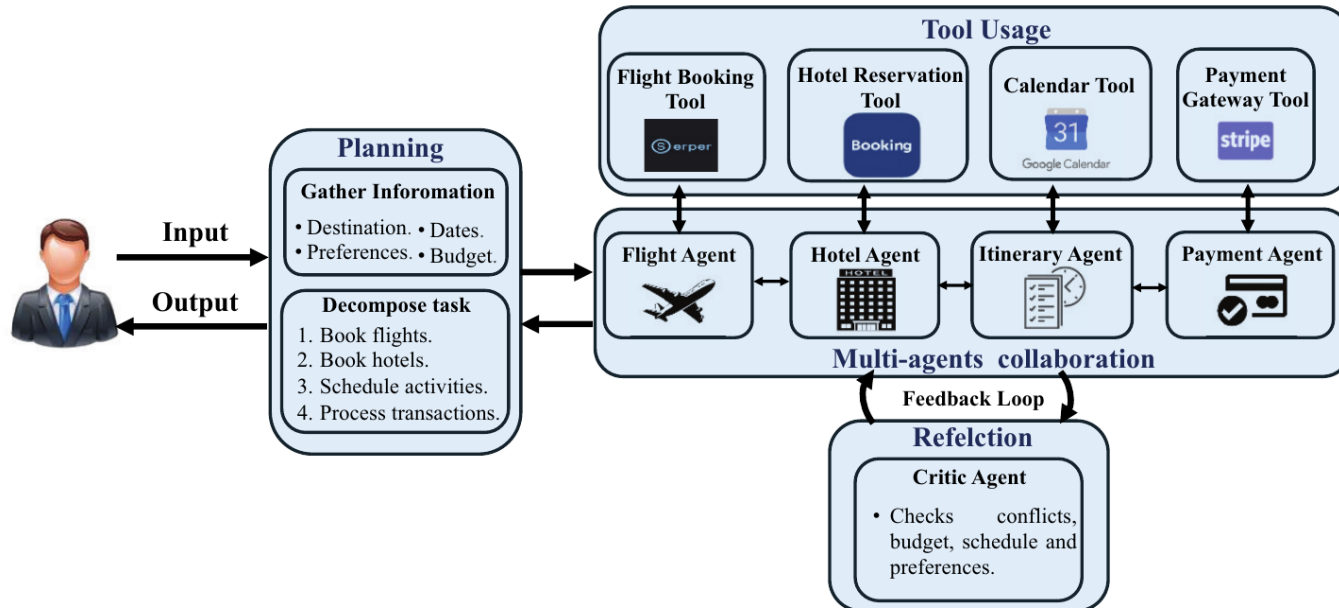


- 01 Goal in**
Grow share 15% in a year on a \$500,000 budget.
- 02 Specialist agents**
Market, strategy, finance, operations, compliance.
- 03 Plan out**
A budgeted, scheduled plan for the founder.

Source: Al-Bashrawi et al. (2026), Figure 3

One goal in, a budgeted and compliance checked plan out, with the founder approving.

Use Case: Agentic Travel and Payments



01 Decompose

Flights, hotels, activities and payment.

02 Transact

Agents call booking tools and Stripe.

03 Check

A critic agent clears budget and schedule first.

Source: Dwivedi et al. (2026), Figure 2

References

- Al-Bashrawi, M. A., Al-Sharafi, M. A., Elgendy, I. A., Helal, M. Y. I., Anbalagan, M. K., Chae, I., & Dwivedi, Y. K. (2026). Agentic AI systems and the future of entrepreneurship: A perspective on co-agency, innovation, and ecosystem transformation. *International Entrepreneurship and Management Journal*, 22, 27. <https://doi.org/10.1007/s11365-026-01164-2>
- Dwivedi, Y. K., Helal, M. Y. I., Elgendy, I. A., et al. (2026). Artificial intelligence agents and agentic systems in hospitality and tourism: Challenges, opportunities and research agenda. *International Journal of Contemporary Hospitality Management*, 38 (1), 27–52.
- Elgendy, I. A., et al. (2025). Agentic systems as catalysts for innovation in FinTech: Exploring opportunities, challenges and a research agenda. *Information Discovery and Delivery*. <https://doi.org/10.1108/IDD-03-2025-0068>
- Klarna. (2024, February 27). *Klarna AI assistant handles two-thirds of customer service chats in its first month* [Press release]. <https://www.klarna.com/international/press/klarna-ai-assistant-handles-two-thirds-of-customer-service-chats-in-its-first-month/>
- McKinsey & Company. (2025, June 13). *Seizing the agentic AI advantage*. <https://www.mckinsey.com/capabilities/quantumblack/our-insights/seizing-the-agentic-ai-advantage>
- Verhagen, A., et al. (2025). *How agentic AI can change the way banks fight financial crime*. McKinsey & Company. <https://www.mckinsey.com/capabilities/risk-and-resilience/our-insights/how-agentic-ai-can-change-the-way-banks-fight-financial-crime>
- Lucas, P. (2025, May 1). *AI commerce gets a boost from Visa, PayPal, and Mastercard*. Digital Transactions. <https://www.digitaltransactions.net/ai-commerce-gets-a-boost-from-visa-paypal-and-mastercard/>
- Digital Commerce 360. (2025, May 6). *Visa, Mastercard offer support for AI agents*. <https://www.digitalcommerce360.com/2025/05/06/visa-mastercard-ai-agentic-commerce/>

Thank you!

Agentic AI in Communication Systems and Sensing

Prof. Daniel Benevides da Costa

Acting Director – IRC-CSS



Why Communication Systems Need Agents



The network changes while we are still measuring it due to mobility, blockage, interference and demand, which continuously reshape the wireless environment.

MOBILITY



- ❑ Users, vehicles and devices move constantly.
- ❑ The best decision a second ago may already be wrong.

BLOCKAGE

Radio paths appear and disappear as buildings, people and objects move through the scene.

INTERFERENCE

An action in one cell changes conditions in neighboring cells, which ripples across the network.

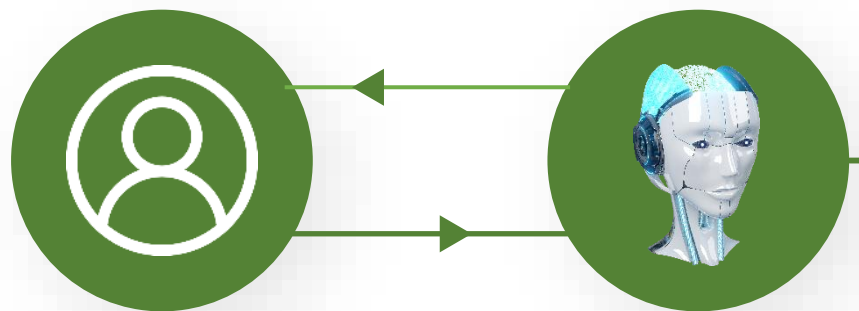
DEMAND



Traffic surges unpredictably at events, campuses and rush hours, straining static configurations.

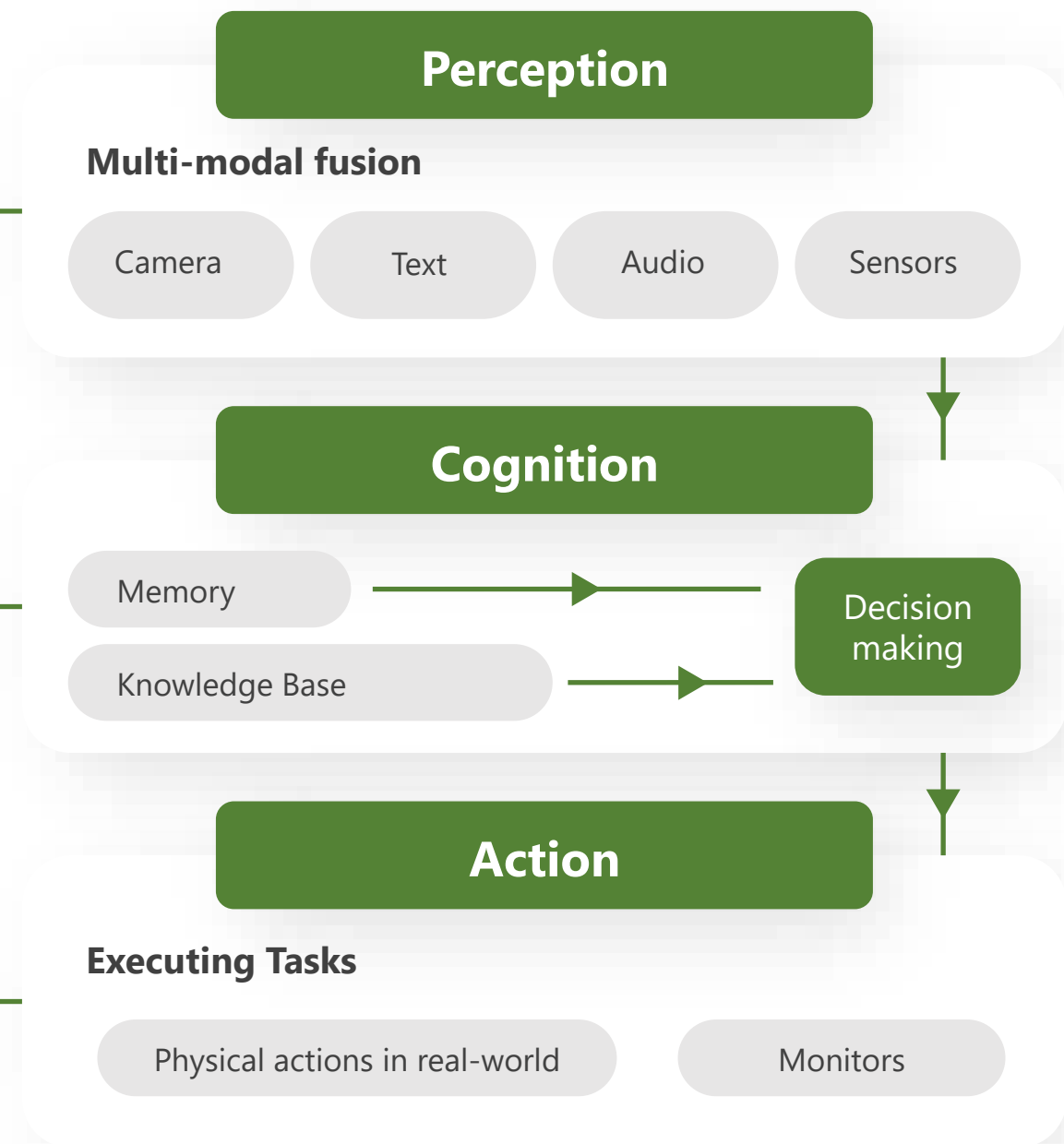
Reactive control sees the present; Agentic control evaluates possible futures.

Agentic AI System



User Request

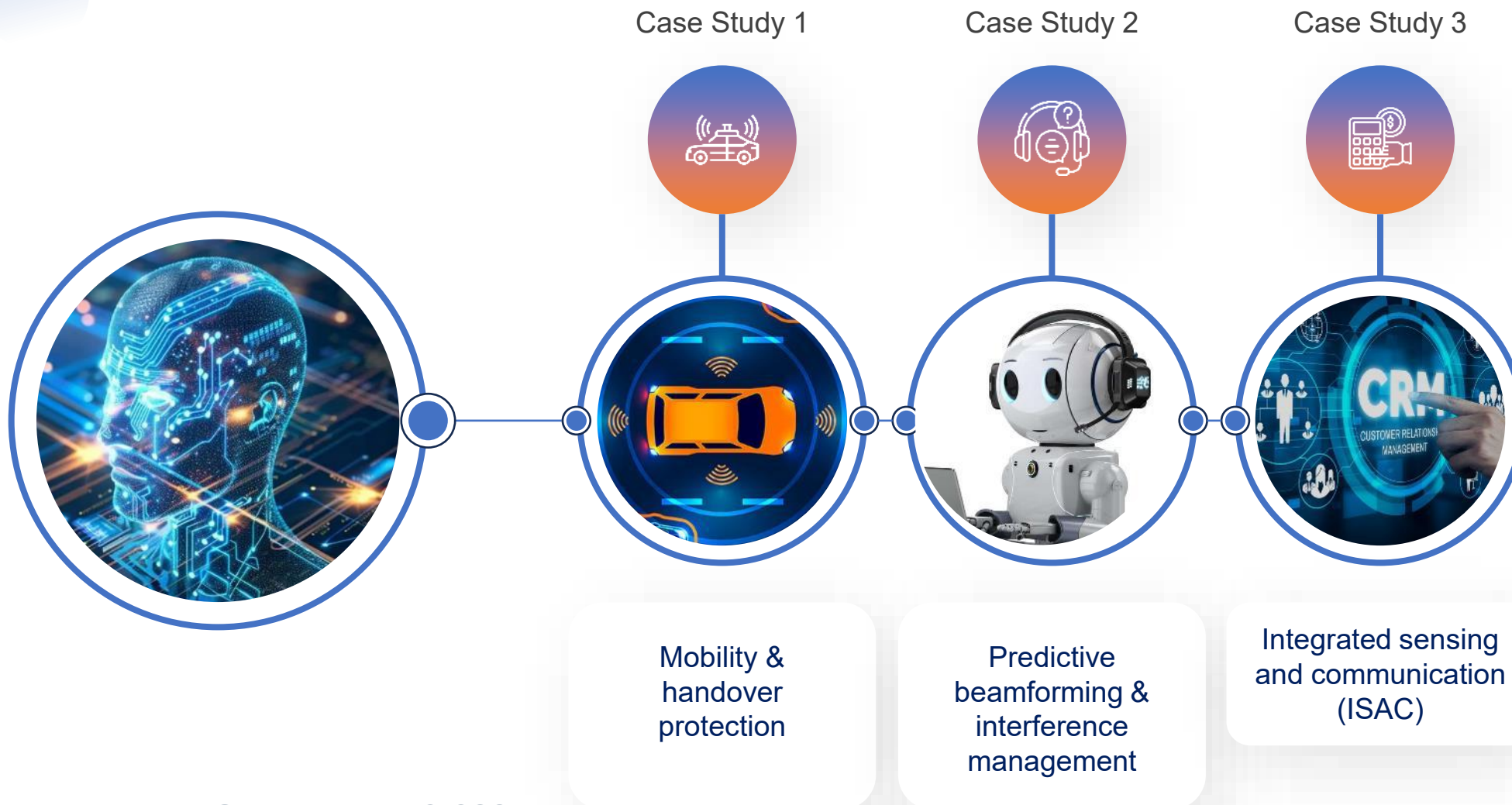
AI Agent





Use Cases Of Agentic AI

All three share one pattern:
observe → predict → decide → act →
learn



Use Case 1 — Mobility & Handover Protection

THE PROBLEM

- ❑ A user moving through a campus or city crosses cell boundaries. Too late, the handover fails and the call or session drops.

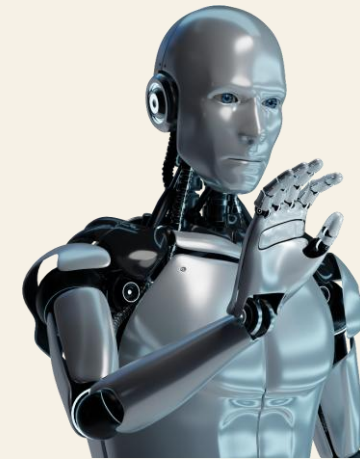
WHAT THE AGENT DOES

By predicting each user's trajectory from sensing data and twin models, the agent prepares the next cell and allocates resources before the connection weakens.

VALUE Seamless mobility for people, vehicles, drones and IoT — no perceptible service interruption.

In practice

- ▶ Predict handovers seconds ahead instead of reacting to signal loss.
- ▶ Reserve resources on the target cell only when truly needed — saving capacity.



Use Case 2 — Predictive Beamforming and Interference Management

THE PROBLEM

- ❑ Today's beams react to the channel after it degrades.
- ❑ Interference between cells is managed with fixed rules that cannot anticipate congestion.

WHAT THE AGENT DOES

The agent generates plausible channel and interference futures in the twin, stress-tests beam choices against them, and steers beams before quality drops.

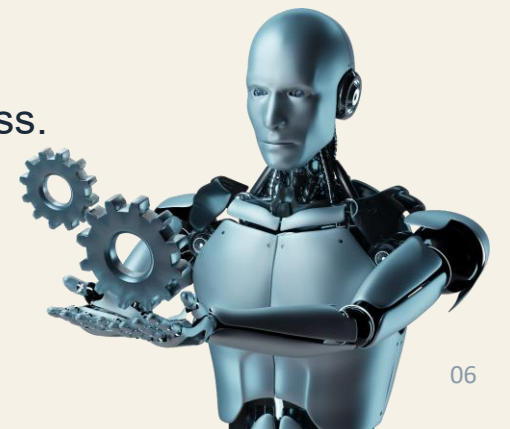
VALUE



Fewer dropped connections, higher throughput, and calmer networks during traffic surges.

In practice

- ▶ Generate dozens of “what-if” interference scenarios and rank beam configurations by robustness.
- ▶ Coordinate neighboring cells; thus, one cell's improvement does not become another's outage.



Use Case 3 — Integrated Sensing and Communication (ISAC)

THE PROBLEM

- ❑ Wireless signals already bounce off people, cars and objects — but networks treat this rich information as noise and discard it.

WHAT THE AGENT DOES

The agent fuses radar-like sensing with communication data: it detects objects and movement, updates the twin, and uses that awareness for control decisions.

VALUE

The network becomes a sensor: enabling traffic monitoring, safety, intrusion detection and environment awareness with no extra hardware.

In practice

- ▶ Sense-and-communicate on the same waveform and infrastructure.
- ▶ Use sensed motion to refine beam tracking and mobility prediction (Use Case 02).



Conclusion & Key Takeaways

Agentic AI is revolutionizing future **communication and sensing technologies**. Telcos Businesses must adapt as AI shapes the future of work.

Agentic AI transforms AI from a model that provides an answer into an **autonomous decision-making loop that continuously perceives, reasons, acts, and adapts**.



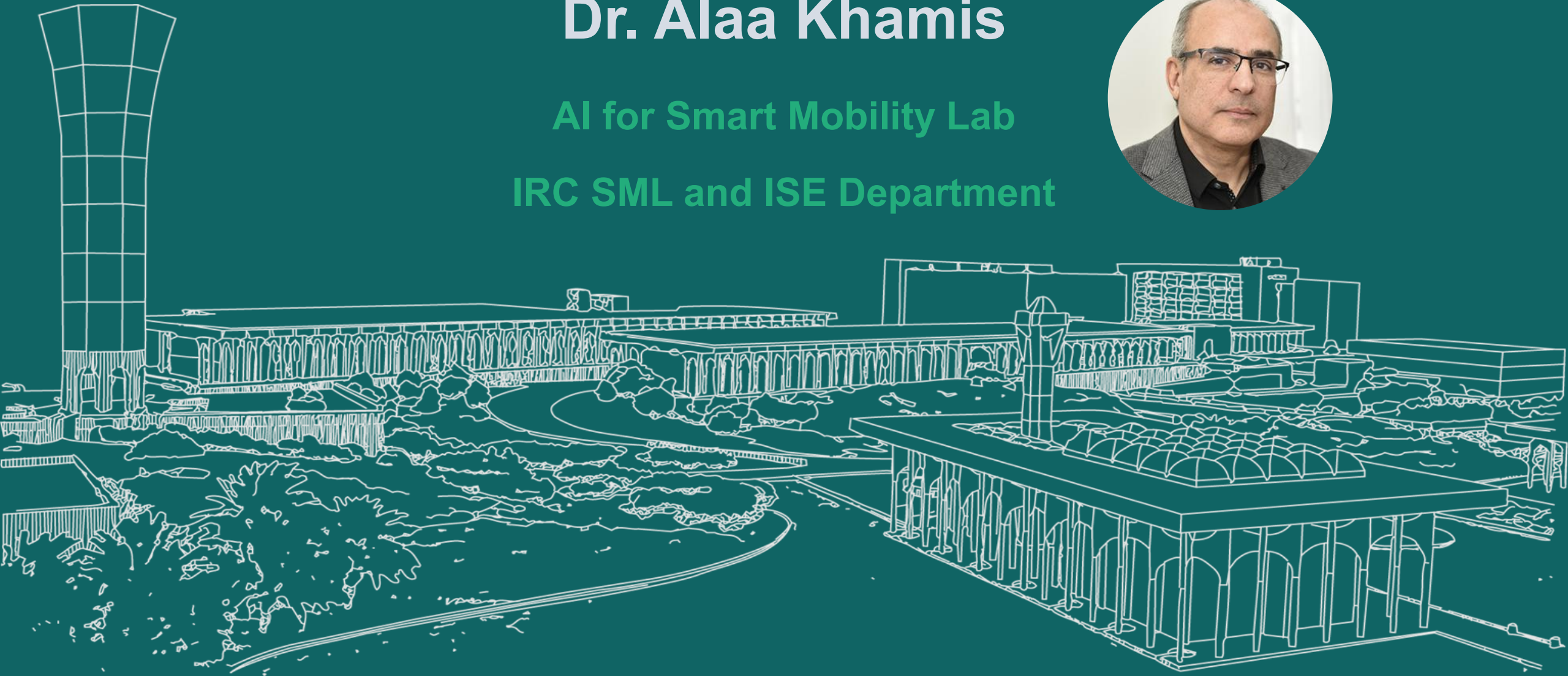
Thank you!

Agentic AI for Smart Mobility and Logistics

Dr. Alaa Khamis

AI for Smart Mobility Lab

IRC SML and ISE Department



AI for Smart Mobility Lab – IRC SML



1

Situational Awareness and Self-Awareness

Enabling vehicles to understand their surroundings and internal states through multimodal perception and causal observability.

2

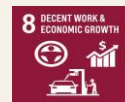
Agentic Reasoning and Action

Building AI agents that perceive, reason, plan, act, and learn to deliver intent-driven, context-aware mobility services.

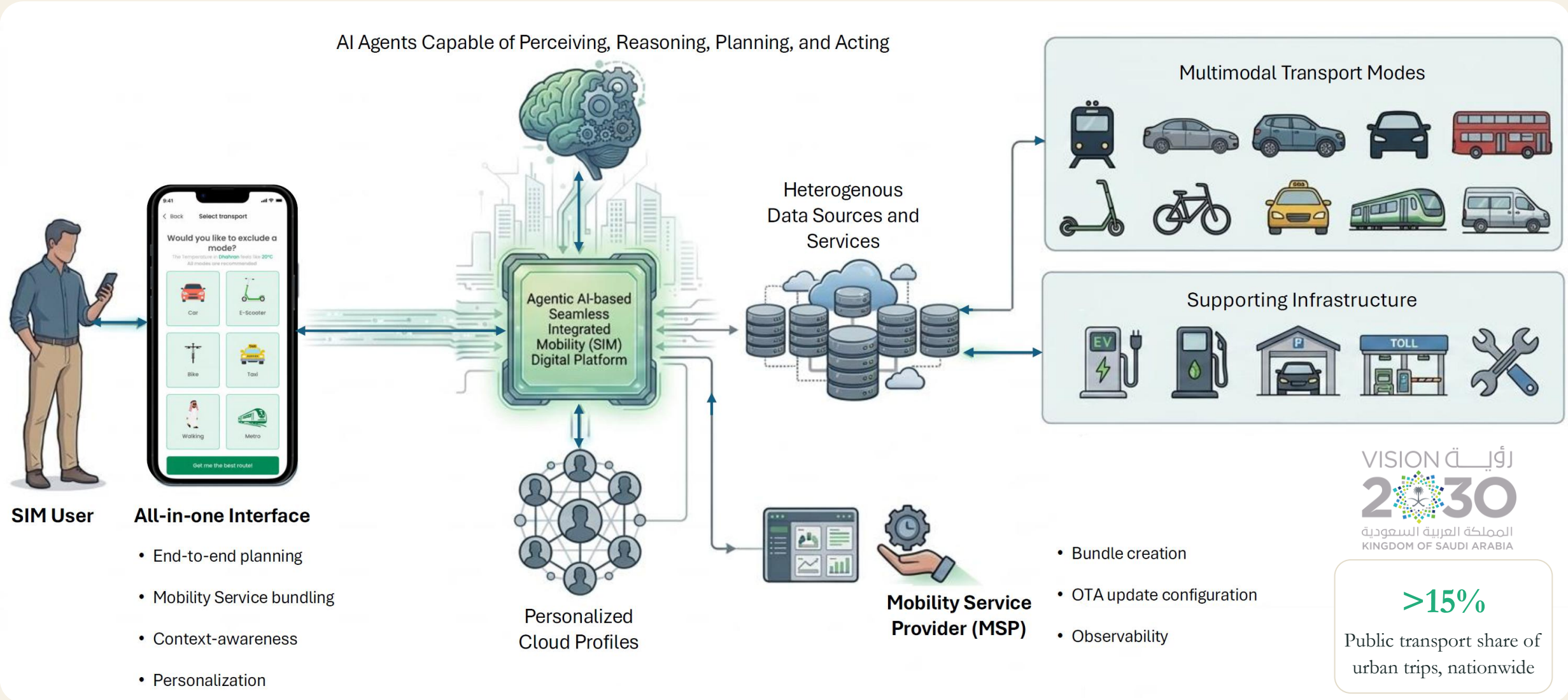
3

Advanced Optimization

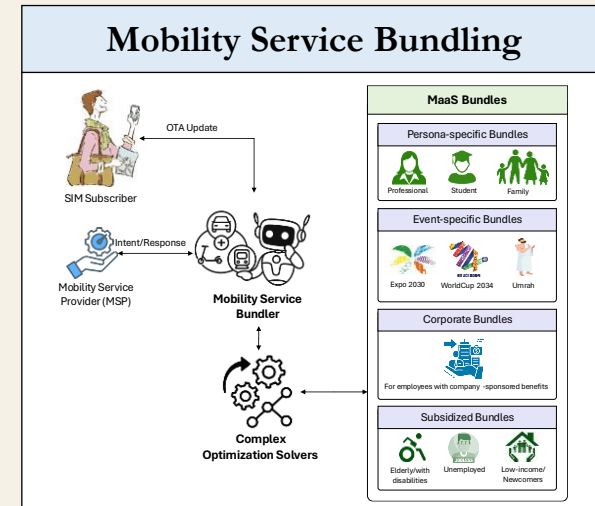
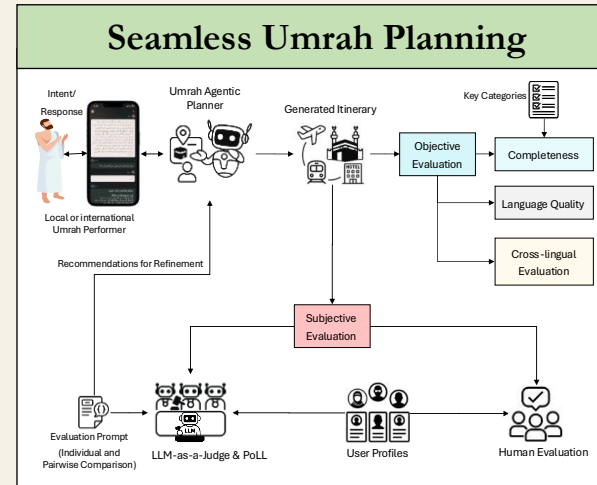
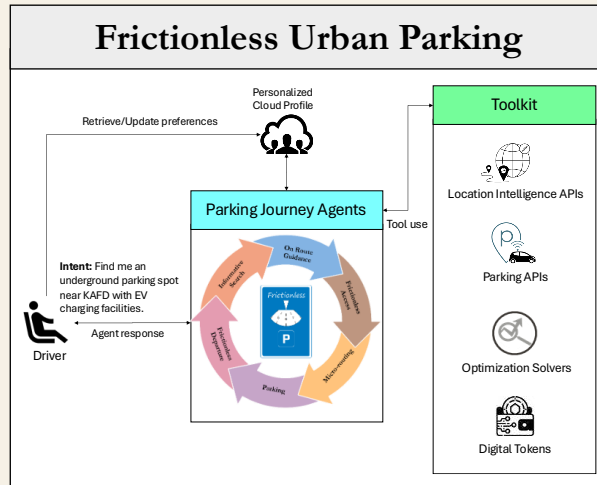
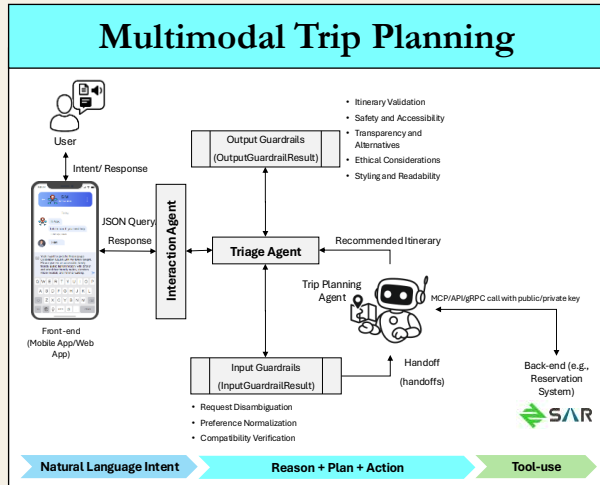
Developing optimization methods for complex design, planning, and control problems across mobility, logistics, and transportation infrastructure.



Agentic AI for Seamless Integrated Mobility

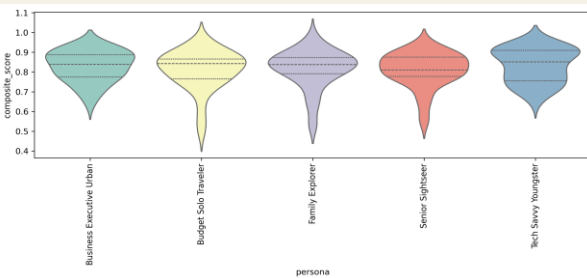


Agentic AI for Seamless Integrated Mobility



EVALUATION METRICS AVERAGED ACROSS MODELS FOR EACH PERSONA.

Persona	Time (s)	Steps	Semantic Similarity	Composite Score
Budget Solo Traveler	69.7	12.9	0.77	0.81
Business Executive Urban	66.1	12.8	0.81	0.83
Family Explorer	75.6	14.0	0.78	0.81
Senior Sightseer	56.7	11.6	0.76	0.81
Tech-Savvy Youngster	68.9	15.7	0.83	0.83



Demonstrates personalized trip planning, where AI agents generate adaptive multimodal journeys based on user preferences, constraints, real-time conditions, and travel context.

Presents an Agentic AI architecture and evaluation framework and demonstrates frictionless urban parking, where AI agents autonomously coordinate parking discovery, reservation, navigation, access, and payment.

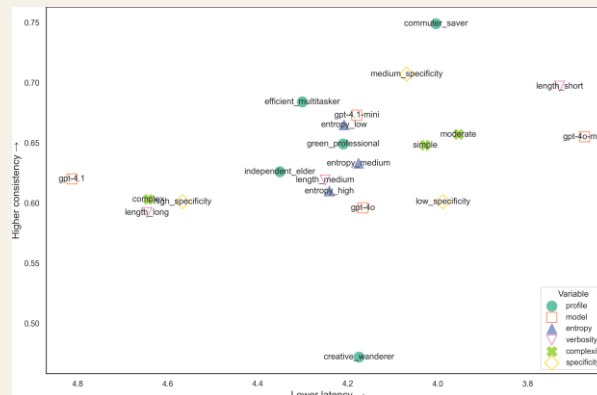


TABLE IV
TOP GENERATED CANDIDATE BUNDLES BY PERSONA.

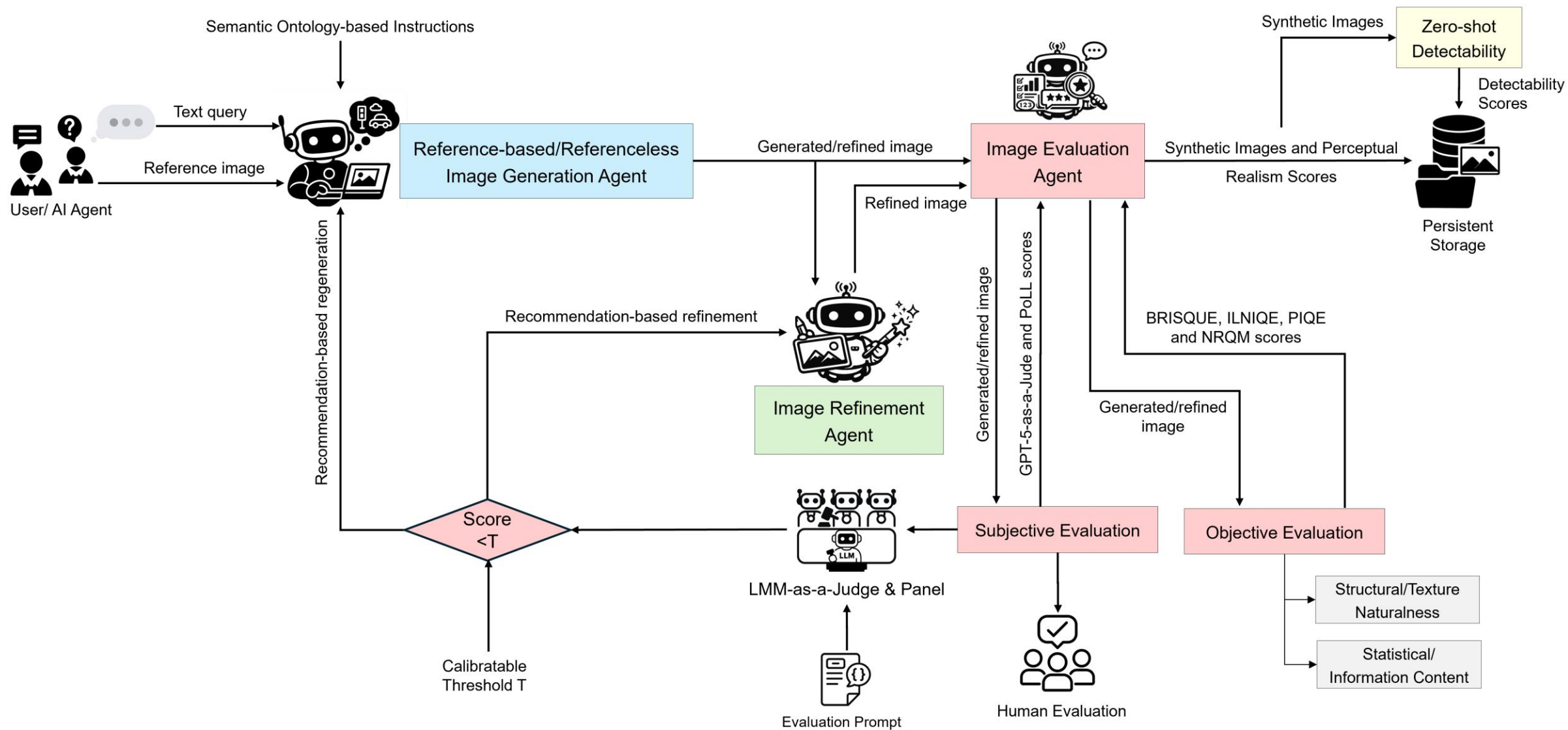
Persona	Services	p	$C(B)$	$A(p)$	Obj.
Student	PT, SC	31.55	24.27	0.5032	7.9973
Student	PT, SC, EB	31.48	24.22	0.5055	7.9957
Student	PT, BK	31.71	24.40	0.4981	7.9941
Family	PT, VP, PK	114.40	88.00	0.4832	12.7931
Family	PT, PK	81.90	63.00	0.7363	11.1224
Family	PT, VP	68.90	53.00	0.8090	10.8533
Professional	RH, CRL, AS	85.23	65.56	0.2711	11.1522
Professional	RH, CRL	71.24	54.80	0.4870	10.5094
Professional	CRL, AS	69.42	53.40	0.5134	10.3006

TABLE V
CONTEXTUAL BANDIT PERFORMANCE BY PERSONA.

Persona	Policy	Avg. reward	Acceptance rate	Avg. price	Avg. S_b	Avg. expected reward
Professional	LinUCB	2.6374	0.374	67.08	4.999	2.6764
Professional	Random	2.3911	0.337	68.05	4.999	2.4133
Student	LinUCB	1.6004	0.405	31.85	3.0152	1.6843
Student	Random	1.4519	0.443	31.63	2.2868	1.4331
Family	LinUCB	0.8675	0.244	31.72	2.5939	0.8956
Family	Random	0.7456	0.229	31.62	2.2565	0.8135

Provides personalized mobility service bundles tailored to diverse user personas, mobility needs, and special events, seamlessly integrating multiple transport modes and services.

Rare-Event Data Augmentation



Rare-Event Data Augmentation

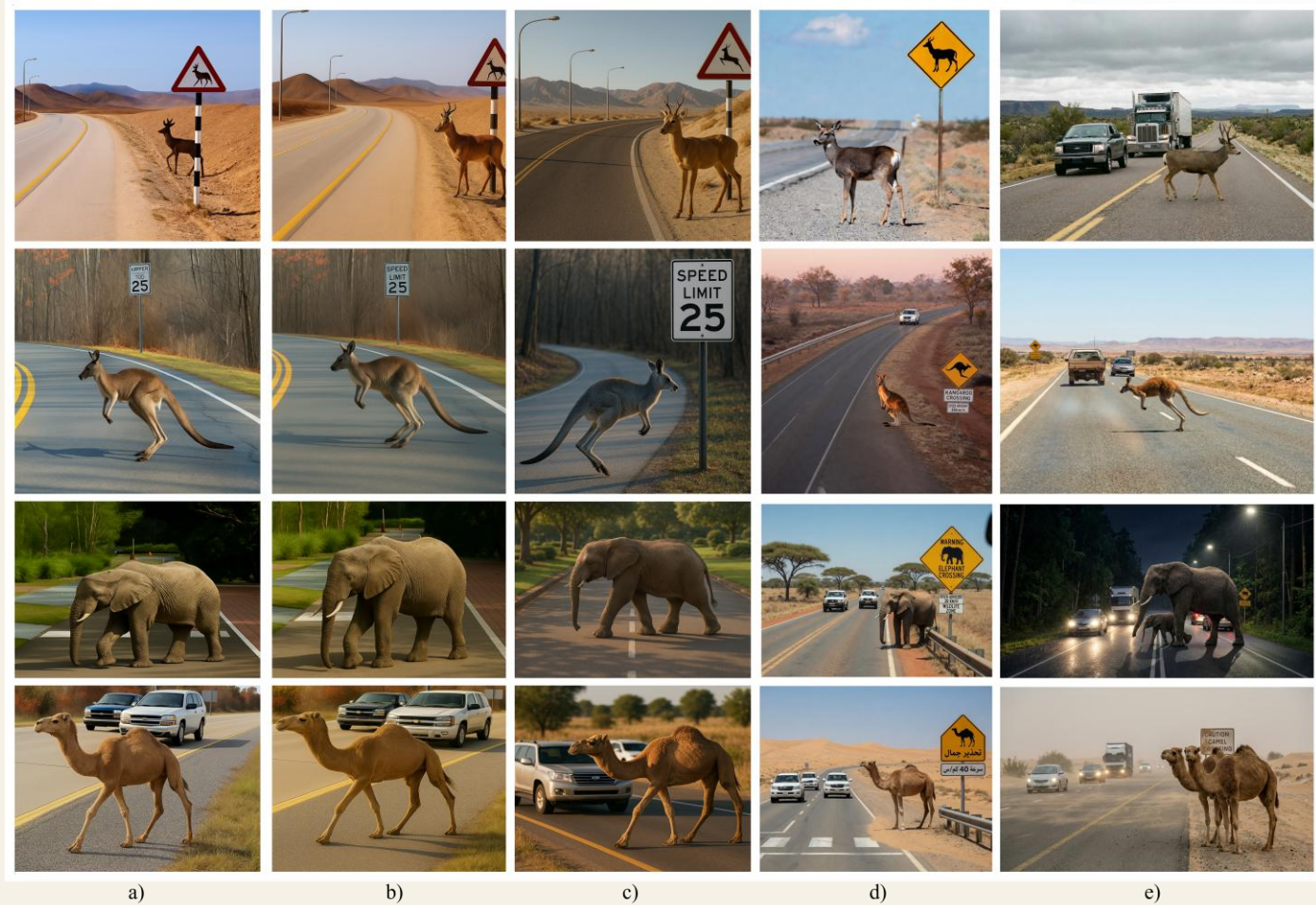
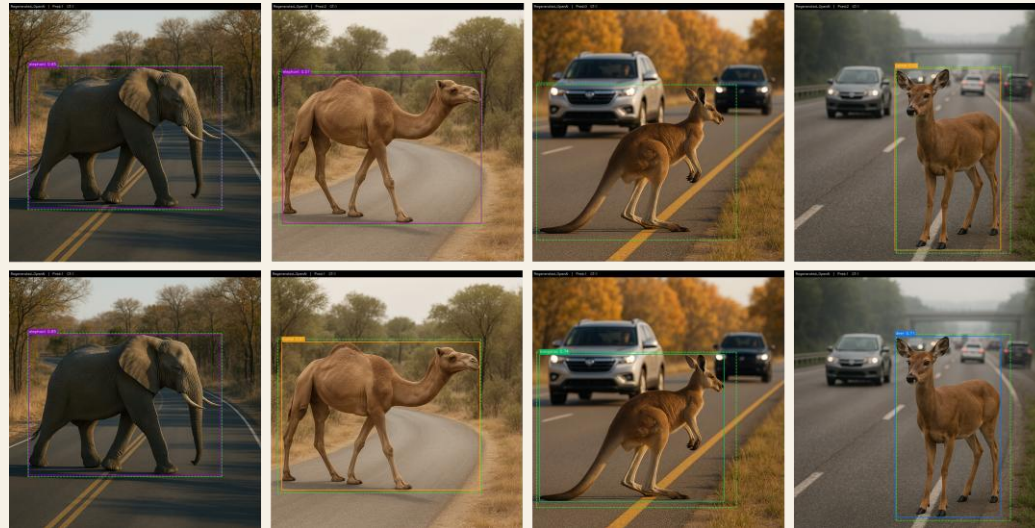


Figure: Examples of generated images using a referenceless ontology-guided approach. (1) GPT-5 synthesis, (b): gpt-image-1 refinement, (c) Recommendation-driven regeneration, (d) Gemini 3 Pro synthesis, (e) Grok 4.20

Method	BRISQUE ↓	ILNIQE ↓	PIQE ↓	NRQM ↑
Referenceless OpenAI (Regenerated)	19.47	48.76	34.10	7.39
Referenceless Gemini 3 Pro	9.67	41.31	32.12	8.36
Referenceless Grok 4.20	11.55	42.99	34.48	8.13

Method	LMM-as-a-Judge ↑	Panel of LMMs ↑	Human Evaluation ↑
OpenAI (Regenerated)	4.38	3.90	2.45
Gemini 3 Pro (Referenceless)	4.74	5.00	3.54
Grok 4.20	3.60	4.00	2.31



Khamis, A. (2026). Agentic Ontology-guided Image Generation and Evaluation for Rare-Event Data Augmentation in Safety-Critical Perception. Array. <https://doi.org/10.1016/j.array.2026.100932>. [GitHub](#)

SDV Code Generation



Orchestrator Agent LLM planner · shared memory & context · tool routing · traceability log · human-in-the-loop approval gates at design, security & release



Requirements

- Stakeholder & feature requirements
- Safety (ISO 26262) & cyber (ISO/SAE 21434) goals
- Regulatory: UN R155 / R156



Requirements Agent

- Refine into user stories
- Acceptance criteria
- Map to COVESA VSS signals & APIs



Architect & Threat-Model Agent

- Service-oriented design
- TARA → security requirements
- Traceability matrix



Code Generation Agent

- Vehicle apps (Python / C++) on Velocitas & KUKSA SDK
- RAG over VSS & SDK docs
- Generates unit tests



Security & Review Agent

- SAST, MISRA / lint
- SCA, SBOM, secrets scan
- LLM peer code review



Build & Test Agent

- CI pipeline & containerize
- Unit / integration tests, coverage
- Sign artifacts()



Release & Operate

RELEASE · DEPLOY · OPERATE

- Signed OTA roll-out (UN R156)
- Runtime monitoring & IDS (UN R155)
- Fleet telemetry → continuous improvement

VERIFICATION & VALIDATION



dreamKIT

TEST · HARDWARE-IN-THE-LOOP

- Deploy vehicle app to dreamKIT edge hardware
- Real-time run with KUKSA databroker & CAN / ECU I/O
- Performance, safety & penetration tests (HIL)



digital.auto Playground

TEST · VIRTUAL (SIL)

- Deploy prototype to the web-based playground
- Simulate VSS signals & vehicle APIs (SIL)
- Validate behaviour & UX vs. acceptance criteria

Field feedback → new requirements

Defects & failed tests → auto-fix loop

Signed build

approve

pass



Continuous security (shift-left)

TARA · ISO/SAE 21434

SAST / DAST

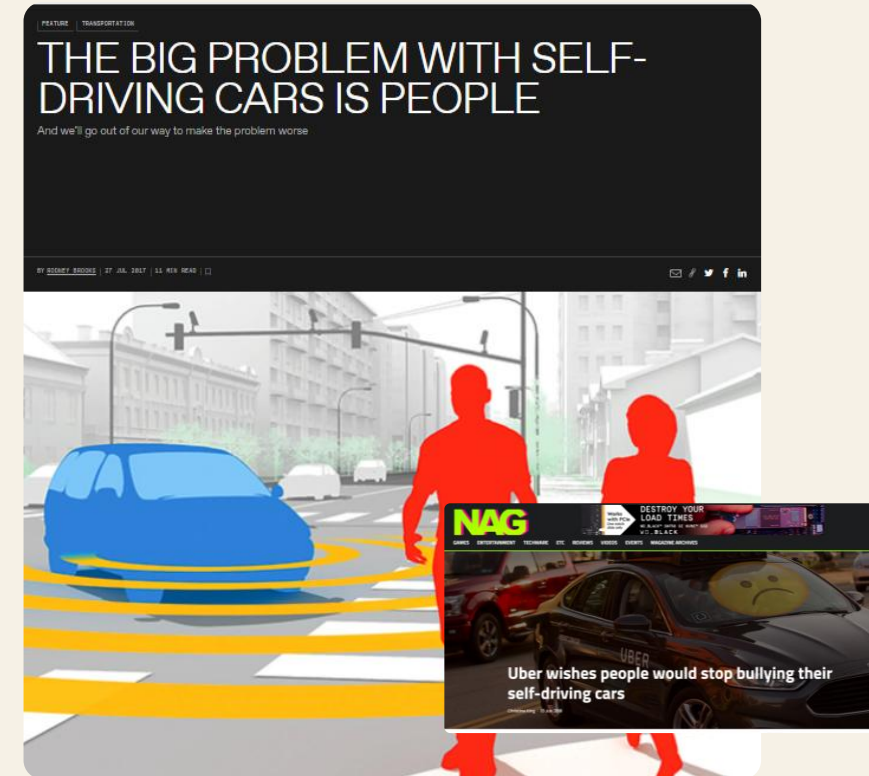
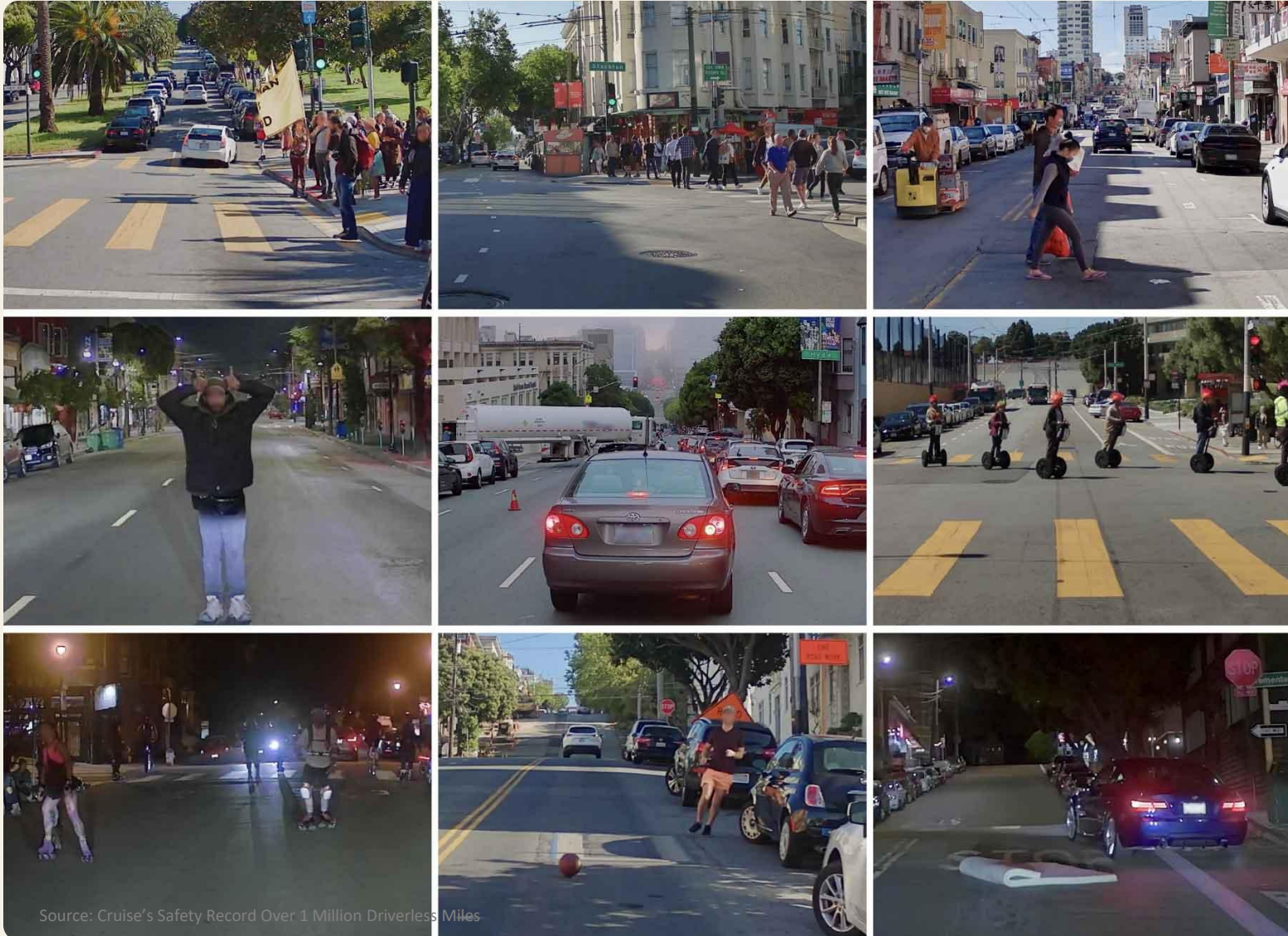
SCA & SBOM

Secrets & policy-as-code

Code & image signing

Audit trail · UN R155/R156

VRU Adversarial Behavior Recognition



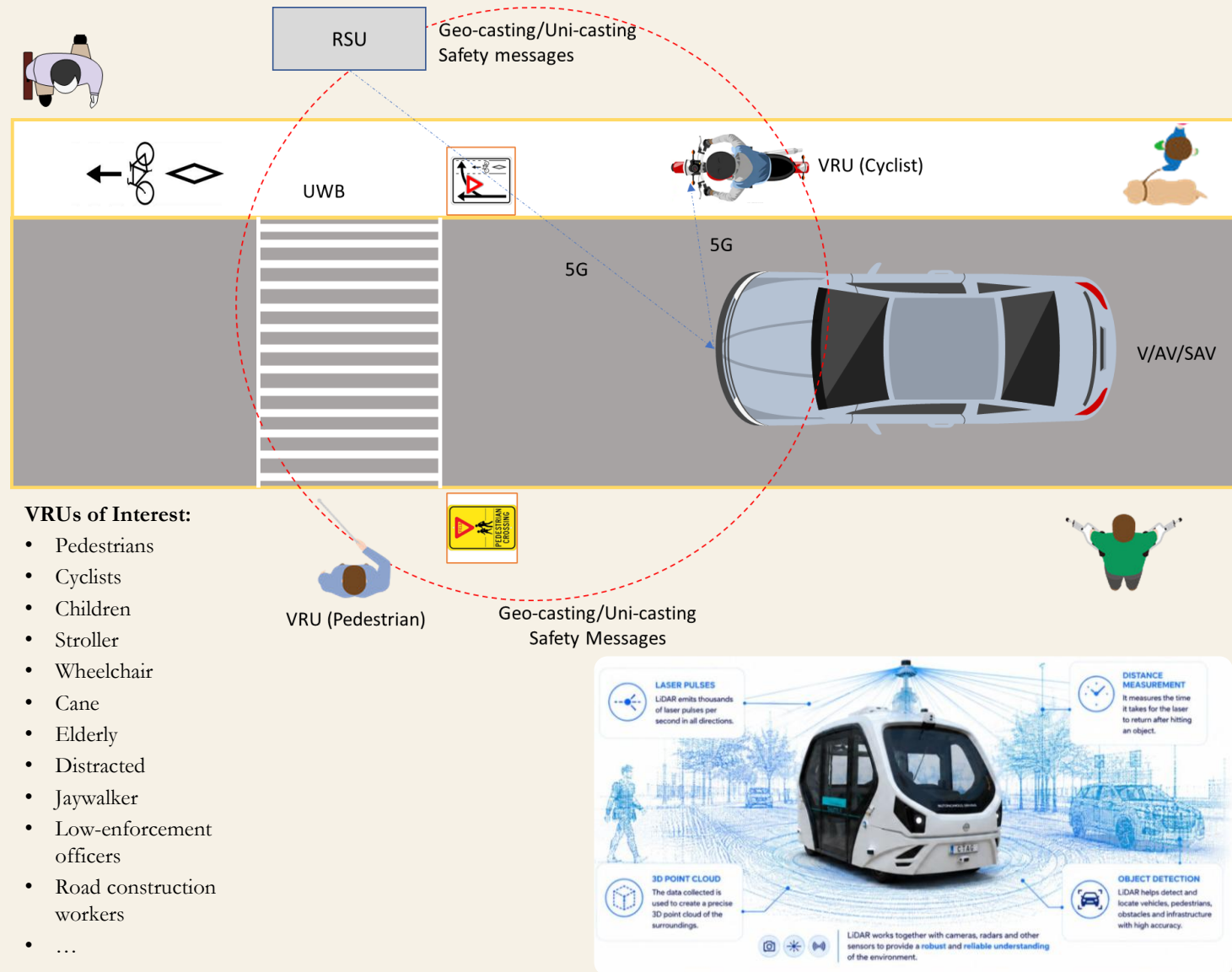
1. Alaa Khamis, et al. VULNERABLE ROAD USER'S ADVERSARIAL BEHAVIOR RECOGNITION, P104517-PRI-NP-US01.
2. Alaa Khamis, et al. VULNERABLE ROAD USER'S ADVERSARIAL BEHAVIOR OPPORTUNITY AND CAPABILITY, P104518-PRI-NP-US01.
3. Alaa Khamis, et al. RESPONSES TO VULNERABLE ROAD USER'S ADVERSARIAL BEHAVIOR, P104516-PRI-NP-US01.

VRU Adversarial Behavior Recognition



Decision Making/Support

Decision Agent: brake, slow down or warn driver / VRU



Situation Awareness



L3: Projection

Status of entities in the near future and possible consequences



L2: Comprehension

Relate situation entities



L1: Perception

Identify situation entities



L0: Sensing

Collect relevant data

VLA Model

L3 · Predicts intent & path
“Will cross in ~2 s → high risk”

L2 · Reasons about behavior
“Distracted, looking at phone”

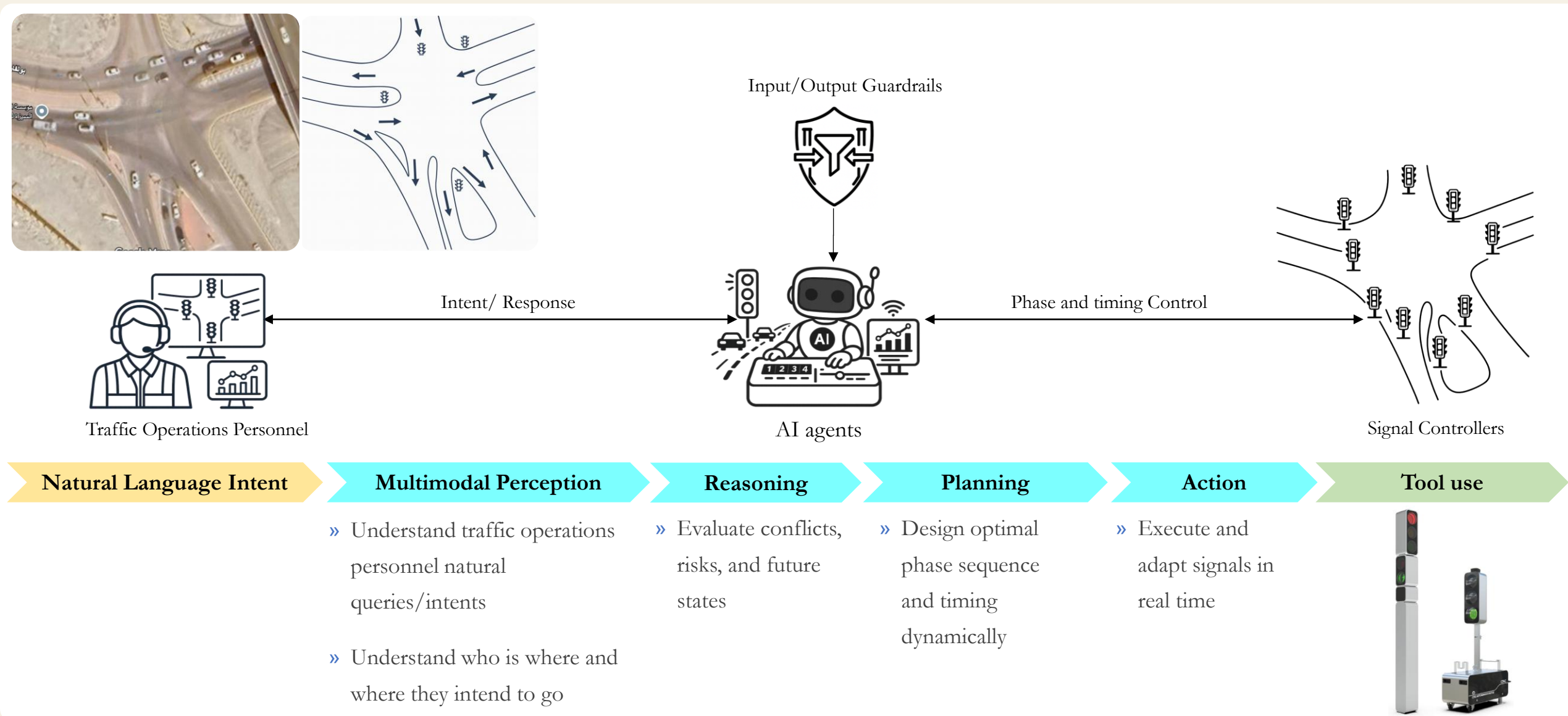
Perception Agent

Detects & tracks pedestrians and cyclists

Sensor Agent

Ego vehicle and intersection cameras, LiDAR, radar & V2X

Agentic AI for Adaptive Traffic Signal Control



Thank you!

Human-Gated Agentic AI for HAZOP

Sunhwa Park, Ph.D.

Chemical Engineering, KFUPM



Why We Exist

We protect **people and property** from process-safety incidents.

HAZOP is the last safety net that finds hazards at the design stage. Making that HAZOP deeper and more consistent is what SERiD AI does.



People

Protecting lives

- Major process incidents cause irreversible loss of life
- Texas City refinery explosion (2005): 15 fatalities, 180 injured — gaps in design and hazard review (US CSB)
- Finding more hazard scenarios, earlier, is the only prevention



Property

Protecting plant and production

- One release, fire or explosion = equipment loss + months or years of lost production + recovery cost
- Repeated groundwork cost — and missed hazards — at every revalidation and change
- A consistent HAZOP is the cheapest asset insurance



Trust

Environment and license to operate

- A plant without releases is a plant without flaring, emissions and clean-up
- Audit-ready records with evidence and approval history (PSM · IEC 61511 · EU AI Act)
- Process-safety knowledge that survives workforce turnover

SERiD AI — the first human-gated agentic AI platform in process safety. AI proposes, experts approve, every decision is traceable.

Who I Am

SERiD AI
Safety and Energy Risk Intelligence via Data



Sunhwa Park, Ph.D.

Key experience

- Assistant Professor, KFUPM — founded and coordinating the Process Safety M.Sc. program
- M.Sc. Safety Engineering and Ph.D. Chemical Engineering, Texas A&M (Mary Kay O'Connor Process Safety Center)
- Risk consultant (Samsung Fire & Marine), senior researcher (LG Chem), design engineer (DL E&C), field engineer (BASF)

What We Solve

50–60 % of HAZOP preparation and meeting time goes into interpreting P&IDs and piecing process data together. Time buried in groundwork is time taken from finding hazards.

HAZOP time allocation

Typical HAZOP

Reading P&IDs and piecing process data (50–60 %)

Safety discussion

With SERiD AI — target: groundwork cut by more than 50 %, safety discussion up by 50 %+

Groundwork

Safety discussion — hazard scenarios · safeguard independence · residual risk



Time and cost

Days to weeks per unit; a single day of remote HAZOP costs about £2,500–4,000¹ — preparation is the largest share



Expert dependence and variance

Results and recommendations vary with team composition and facilitator skill — poor reproducibility



Disconnected data

Process data, regulations and past cases are not linked, so reuse and digital integration are hard



AI hallucination concerns

Consistency and reliability of generative-AI output — plants will not hand safety judgment to an AI

SERiD's answer — Human-Gated: AI does the groundwork and the drafts; engineers keep the authority over process understanding and safety judgment; every decision is traceable to its evidence and approvals.

¹ OTECSA Consulting, How much does a HAZOP cost?

What We Do

SERid HAZOP — Steps 1 to 5 with Human Gates A · B · C

1 Study setup

Setting up the HAZOP study foundations

2 Process analysis

Process information linked and knowledge graph

A
Human gate

3 Node setup

Design-intent-based node analysis

B
Human gate

4 Scenario analysis

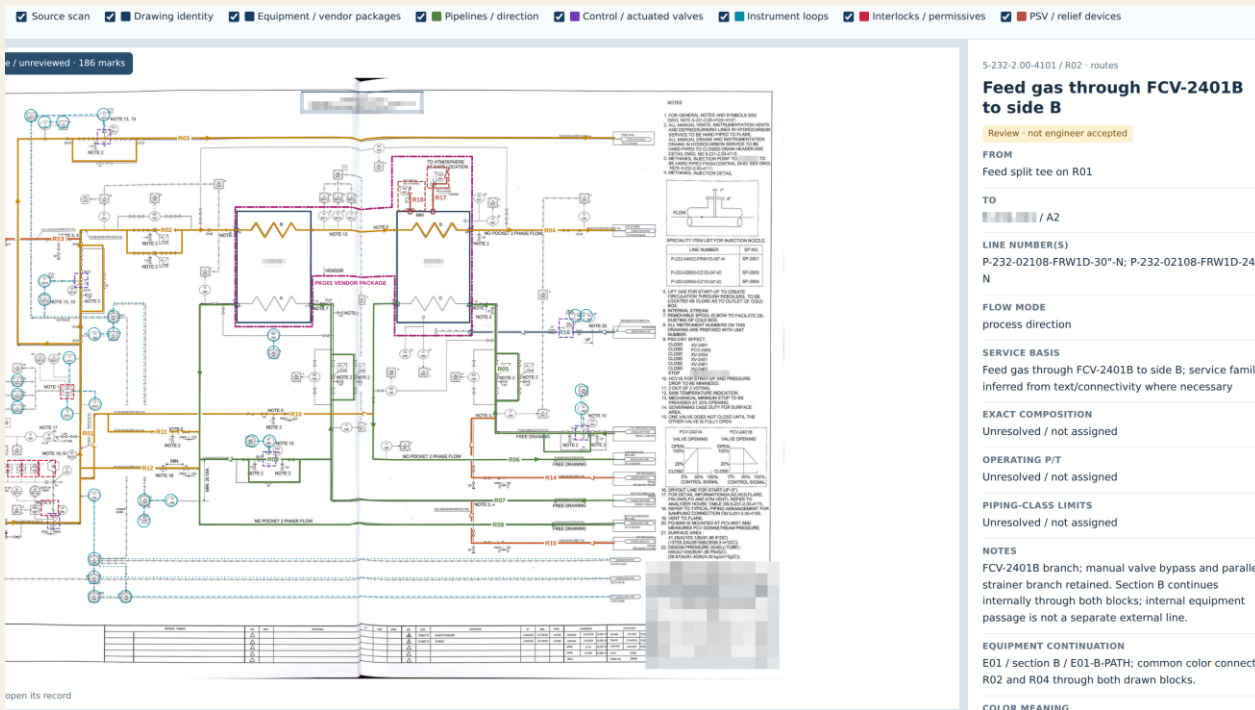
HAZOP performance

C
Human gate

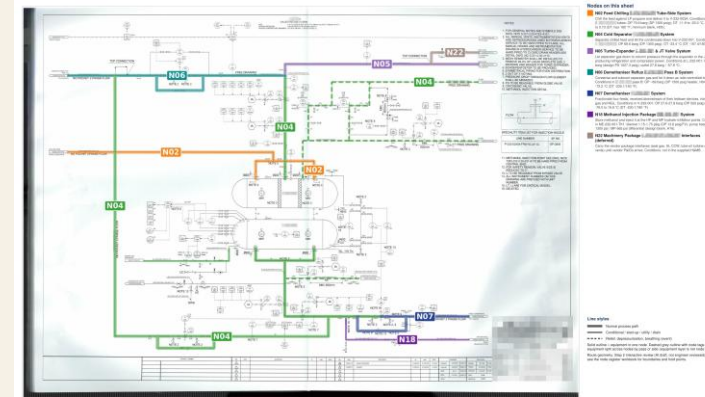
5 Reporting and follow-up

Study report and action register

Process understanding, node and design intent, and HAZOP risk do not move to the next step without an engineer's approval. * With AI assist · with human validation



Step 2 · Process interpretation — every record on the drawing carries position, source and confidence (client identifiers redacted)



Step 3 · Node-marked P&ID

SCENARIO	PARAM	MODE	DEVIATION	CAUSE	CONSEQUENCE (UNMITIGATED)	MITIGATIONS - AS DESIGNED, NOT DESIGNED	P	E	A	R	R
MAX 001.0	Flow	No	No flow (gas outlet)	HPPS spurious trip (C003) closes both gas-outlet valves	Separator blocked in with feed running; vessel out-rates the source — unit trip	Inherent: vessel rating > max feed pressure; PSDs - feed trip - alarms	1	1	2	C	Medium
MAX 001.0	Flow	No	No flow (gas outlet)	ESD fault or sensor/valve failure closes the gas-outlet valves	Same blocked-in trip outcome	As above	1	1	2	D	High
MAX 001.0	Flow	No	No flow (condensate)	Level control valve fails closed / level loop fault	Level reaches the gas outlet; liquid slug to the turbo-expander — blade / bearing wreck	PSD (C003 HRT) trip - alarm; vendor vibration trips	2	1	4	D	Extreme
MAX 001.0	Flow	No	No flow (condensate)	Shutdown isolation valves spurious closure	Same level-rise chain toward the expander	As above - valve position indication	2	1	4	D	Extreme

Step 4 · Worksheet — one row per cause

90 %+

Component recognition accuracy

95 %+

Design intent matching

100 %

Results and evidence

ZDR

DB · zero data retention

PoC targets

What We Do

Process knowledge graph · Neo4j

Every element of a P&ID is stored as an ontology instance with its relations.

Class 32

Family 8

Instance 24

Equipment 2

FinalElement 5

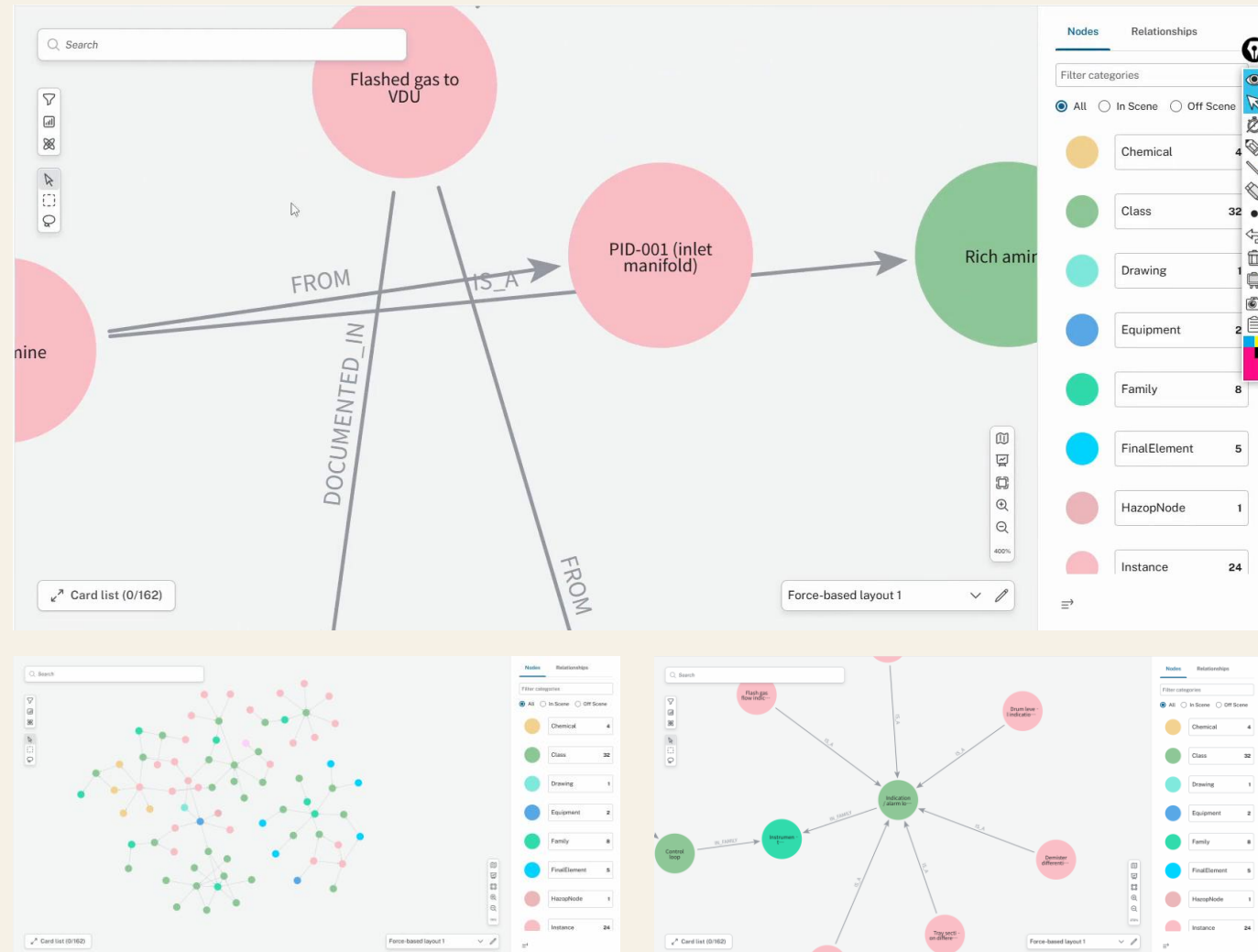
HazopNode 1

Chemical 4

Drawing 1

- Drawing → ontology → node → scenario in one graph — every record with source, confidence, approval and revision
- Management of change
- LOPA and SIL verification inputs
- The safety layer of the enterprise digital twin

Dedicated network and DB · Zero Data Retention (ZDR)



What You Get

HAZOP that protects people and property — lower HAZOP cost, more safety discussion, **focus on the measures that matter.**

1 Protecting people and property

- More hazard scenarios, found earlier — time moves from groundwork to hazards, safeguards and residual risk
- Consistent hazard coverage: fewer hazards missed because of team or facilitator

2 Total cost ↓ Safety discussion ↑

- P&ID and process-data groundwork cut by 50 %+ (target)
- Owners and operators: verified process knowledge reused at every revalidation
- EPC: HAZOP off the critical path, fewer late safeguard changes and rework

3 Consistent data, excellent reports

- Standardized, structured, traceable process-safety data → digital-twin safety layer
- Reused for MOC, LOPA/SIL, alarm rationalization and safe operating limits

≈ 110 engineer-days

8 specialists × 10 session days, plus preparation

»

≈ 60 engineer-days

50–60 % groundwork share

»

≈ 30 engineer-days returned

groundwork cut by more than half

»

≈ USD 35–45 k / unit

at USD 1,200–1,500 per engineer-day

»

≈ USD 0.7–0.9 M

20-unit revalidation program per cycle

Human-gated, not autonomous. SERiD AI does not shorten HAZOP — it gives HAZOP its time back, to protect people and property.



SERiDAI

Safety and Energy Risk Intelligence via Data

HUMAN-GATED AGENTIC AI FOR HAZOP

Redefining HAZOP: agents prepare, experts validate, every decision remains traceable

Closing



SERiD AI contributes to the digital transformation of process safety — **protecting people and property** with Human-Gated Agentic AI for HAZOP.

Thank you!

Agentic AI for Construction Scheduling and Cost Estimating

Dr. Ibrahim Yahaya Wuni

AECEM Department and Research Scholar, IRC-CBM



Case Study 1: Agentic AI for Construction Scheduling

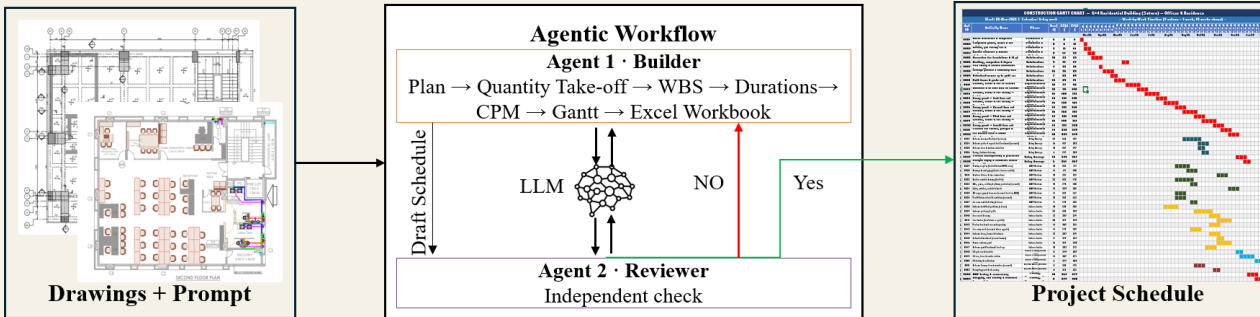


Figure 1. Overview of the Agentic Workflow for Construction Schedule Generation

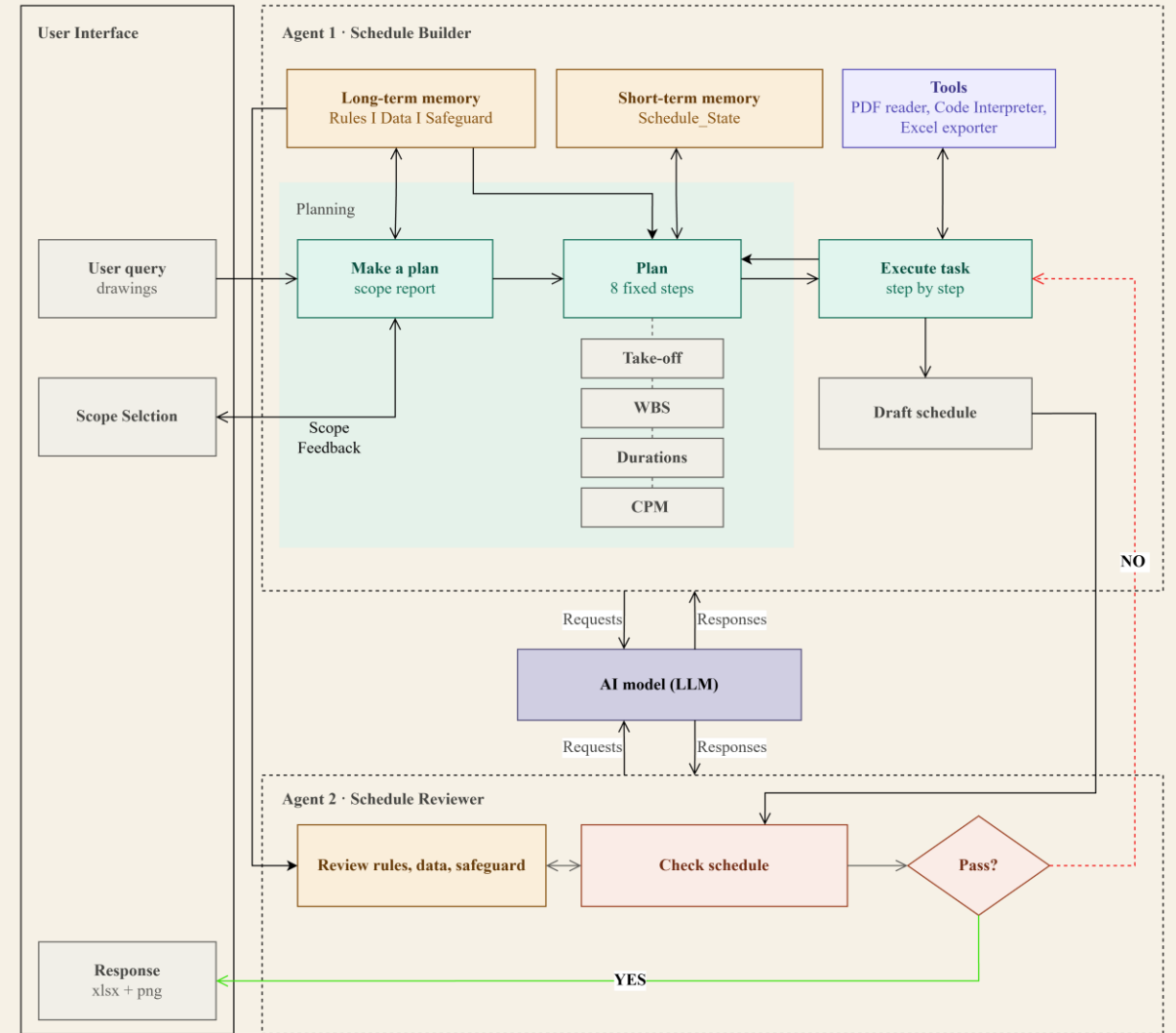


Figure 2. Architecture the Intelligent Construction Scheduling Agent

Case Study 2: Agentic AI for Construction Cost Estimating

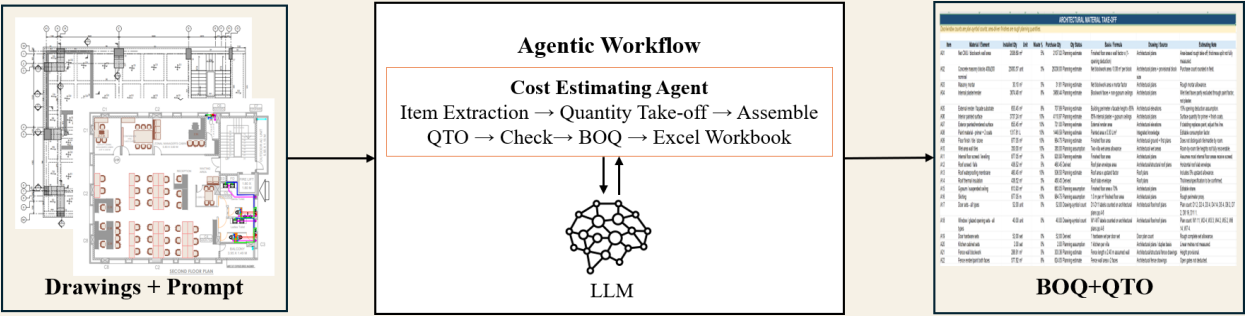


Figure 3. Overview of the Agentic Workflow from Construction Drawings to a Priced Bill of Quantities

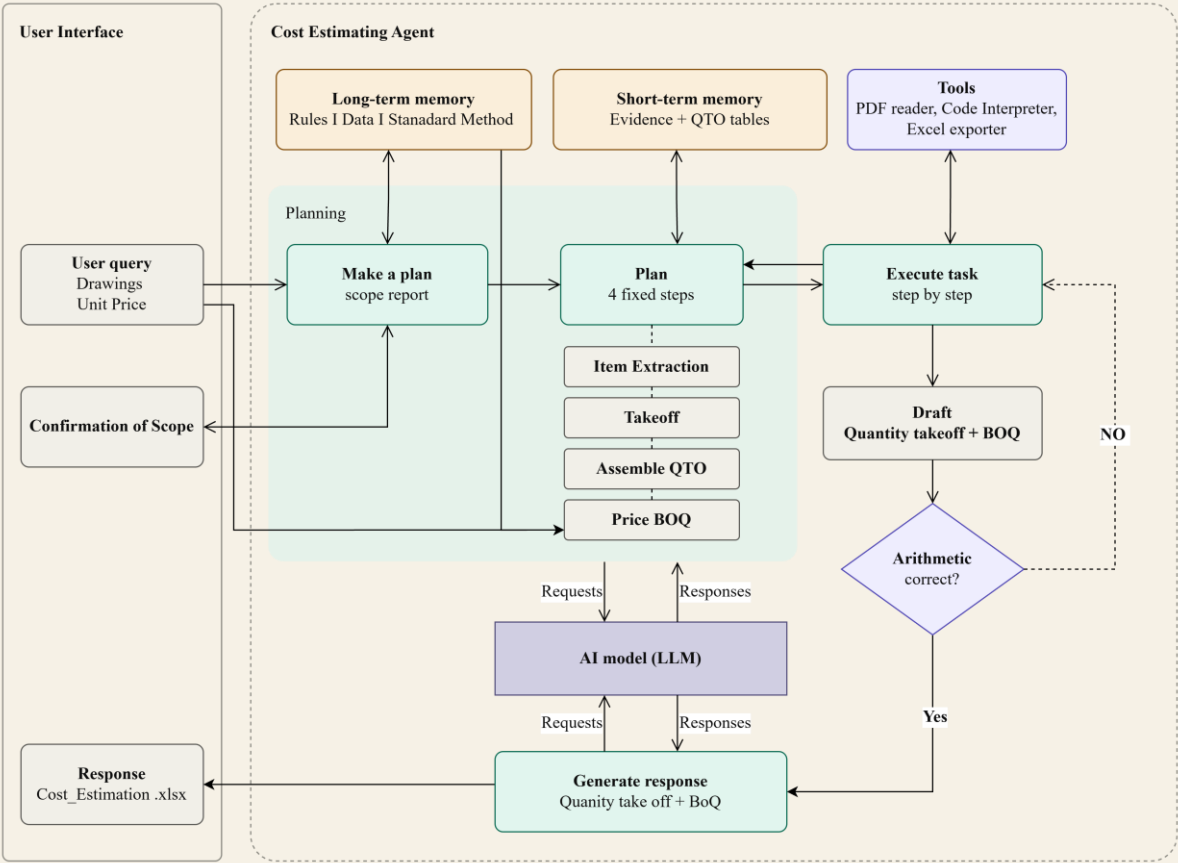


Figure 4. Architecture of the Cost Estimating Agent

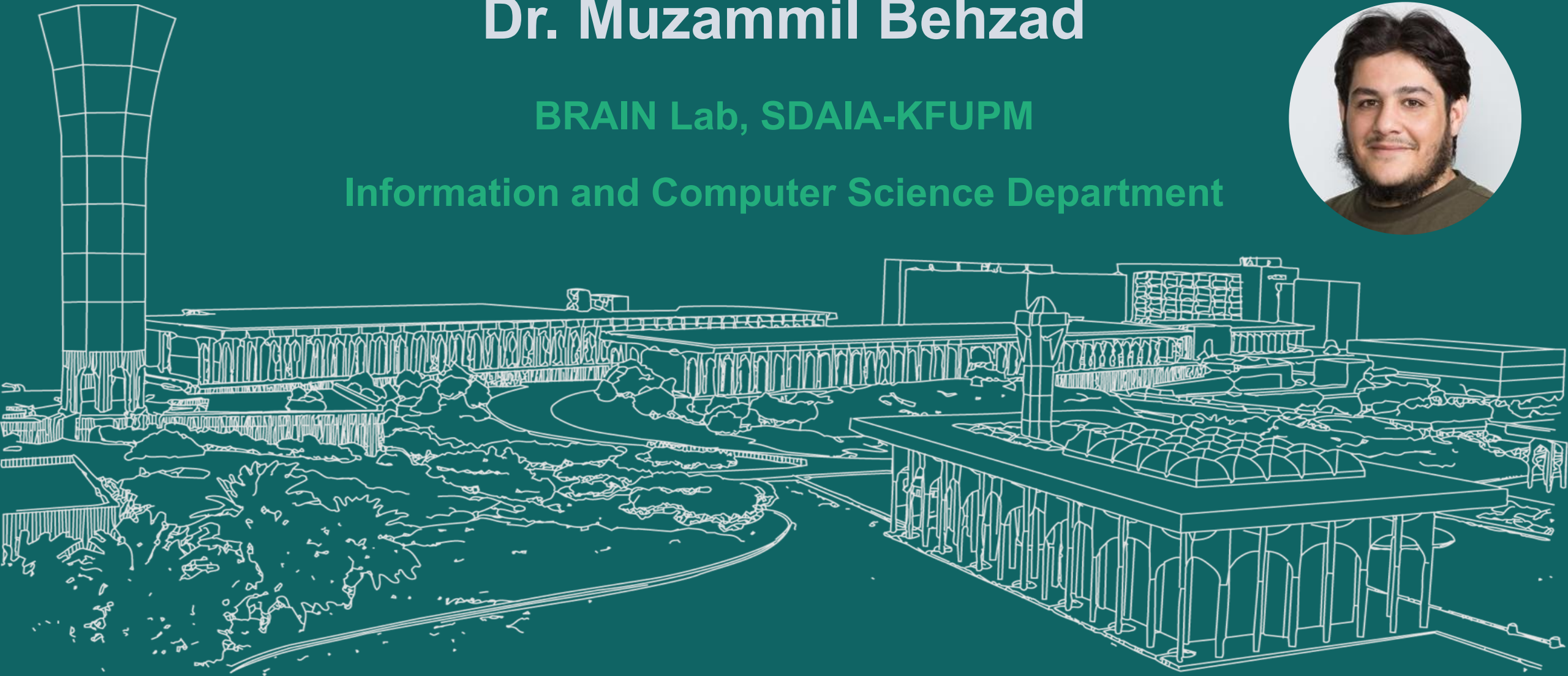
Thank you!

Multimodal Agentic AI for Healthcare

Dr. Muzammil Behzad

BRAIN Lab, SDAIA-KFUPM

Information and Computer Science Department





Multimodal Agentic AI for Healthcare

Dr. Muzammil Behzad

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Officialpage



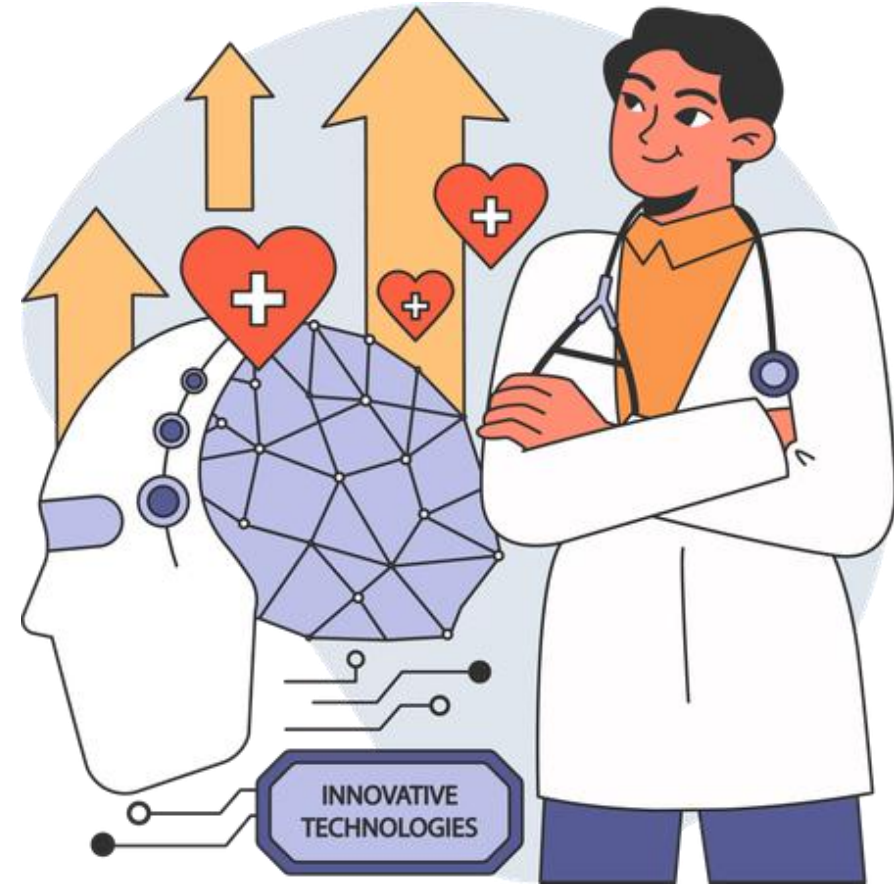
Homepage



BRAIN Lab



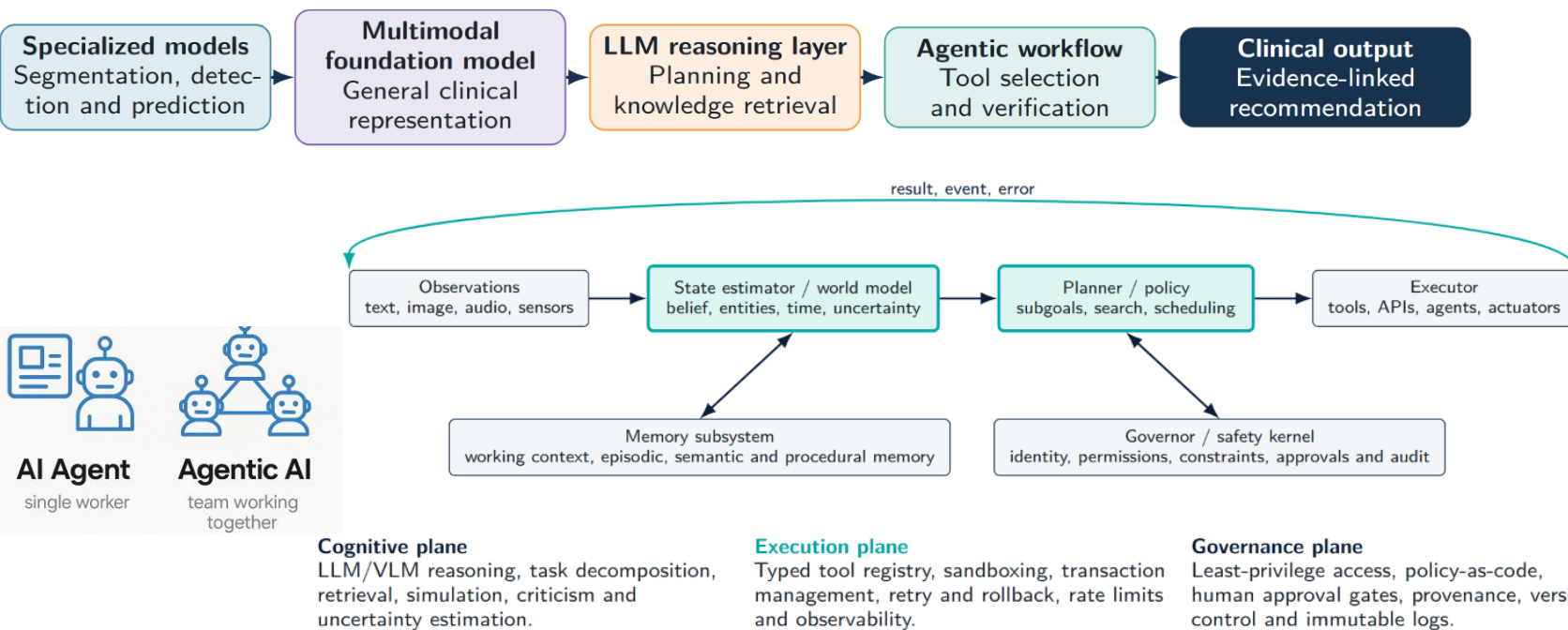
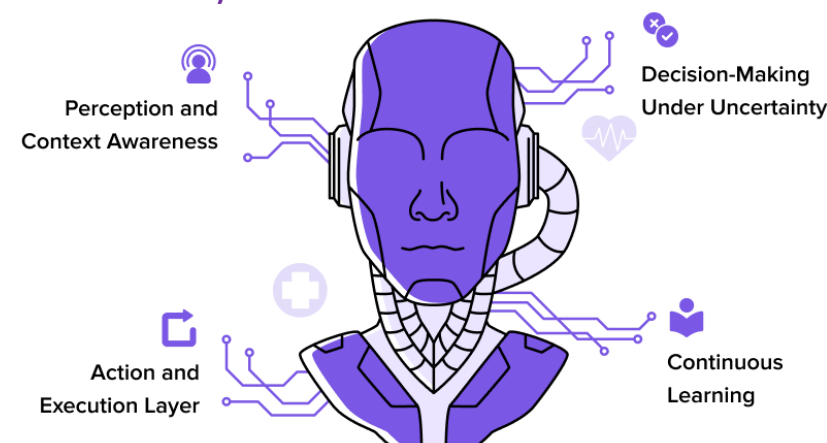
LinkedIn



Multimodal Agentic AI in Healthcare

AI Model	Foundation Model	Large Language Model	Agentic AI
A trained model that maps a specific input to an output. Clinical role: Disease classification, organ segmentation, risk prediction, or signal analysis.	A broadly pretrained model adaptable to multiple tasks and modalities. Clinical role: Interprets medical images, text, signals, and structured patient data.	A language-focused foundation model for understanding and generating text. Clinical role: Summarizes records, retrieves knowledge, interprets notes, and explains findings.	A goal-driven system that plans, uses tools, maintains state, and verifies results. Clinical role: Coordinates evidence, specialized models, guidelines, and human review.

A closed-loop clinical reasoning system
An agent does more than generate a response. It maintains a clinical goal, selects evidence, calls validated tools, checks intermediate results, and escalates uncertainty.



Core design principle

The foundation model orchestrates the workflow. Specialized models perform measurements such as segmentation, detection, risk scoring, and physiological signal analysis.

Case Study: Evidence-based Multimodal Diagnosis

Case study: MedAgent-Pro

- Disease-level plan:** retrieve diagnostic criteria, define required indicators, and encode a structured decision path.
- Patient-level execution:** route fundus images or 3D echocardiography plus demographics and measurements to task-specific tool agents.
- Evidence verification:** preserve each quantitative or qualitative finding before the final diagnostic synthesis.

Representative clinical workflows

Setting	Agent actions
Glaucoma	Inspect fundus image, quantify optic-disc features, combine intraocular pressure and visual-field evidence
Cardiology	Analyze 3D echo, extract structural and functional indicators, integrate patient risk factors
Radiology	Select detection or segmentation tools, compare prior studies, draft evidence-linked findings
Pathology	Localize regions of interest, invoke cell or tissue classifiers, retrieve disease-specific criteria

Failure mode: an incorrect plan can propagate across every downstream tool. Require criterion-level traceability and clinician override.

Why the agentic layer matters

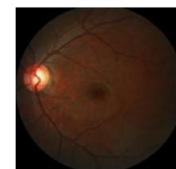
A general MLLM may describe an image but cannot reliably compute clinical measurements. Tool calls convert pixels into auditable indicators, while guideline retrieval constrains the reasoning path.

MedAgent-Pro: Towards Evidence-based Multi-modal Medical Diagnosis via Reasoning Agentic Workflow

Ziyue Wang¹, Junde Wu², Linghan Cai³, Chang Han Low¹,
Xihong Yang^{1,4}, Qiaxuan Li², Yueming Jin¹

¹National University of Singapore, ²University of Oxford,
³Harbin Institute of Technology (Shenzhen) ⁴National University of Defense Technology
⁵The Second Affiliated Hospital Zhejiang University School of Medicine

Example 1: Glaucoma



Fundus Image

Answers from VLMs/Medical Agents

GPT-4o: To confirm whether a patient has glaucoma, it is usually necessary to comprehensively analyze **multiple factors** such as the morphology of the optic disc, the thickness of the optic nerve fiber layer, intraocular pressure, and visual field test. It is **difficult to diagnose glaucoma** based on this fundus photo alone.

MMedAgent: This is a **VQA** task and should be handled by LLaVa-Med. According to the fundus image, the patient **does not have glaucoma**. The image shows a **normal optic nerve**.

Answers from MedAgent-Pro

According to **relevant documents**, the key indicators for diagnosing glaucoma from fundus images are as follows:

According to the **segmentation result** of optic cup and disc, the **vCDR is 0.681**, higher than the risk threshold 0.6.

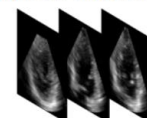
The thickness of the patient's neuroretinal rim **deviates from the normal ISNT rule**, with a much thinner temporal margin.

The patient have **optic disc hemorrhage** with associated cup-to-disc and retinal pigmentation.

The image shows an **unclear optic disc boundary**, which may related to peripapillary atrophy.

Based on above findings, the patient are very likely to **have glaucoma**

Example 2: Heart Disease



3D Echocardiography

Age	height	weight	Sex
47	170cm	96kg	M

Patient Information

Answers from VLMs/Medical Agents

GPT-4o: I can help you analyze the features of this 3D echocardiography image, but I **cannot directly diagnose heart disease**. To determine whether a patient has heart disease, professional medical image analysis and a doctor's evaluation are required.

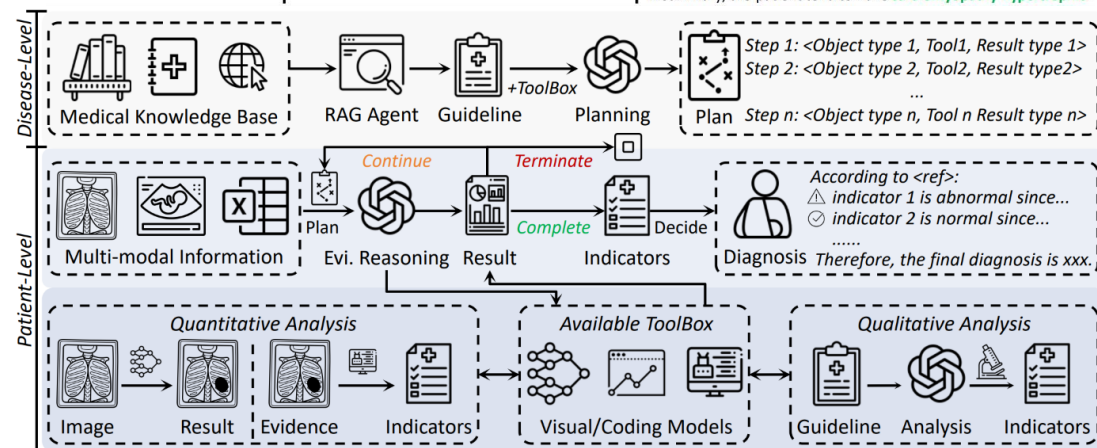
MMedAgent: This is a **VQA** task and should be handled by LLaVa-Med. The echocardiography image shows **normal cardiac function**, which means that the patient does **not have heart disease** according to the information provided.

Answers from MedAgent-Pro

According to **relevant documents**, the key indicators for diagnosing heart disease from 3D echocardiography images are as follows:

According to the **segmentation result** of leftventricle and myocardium, the **Left Ventricular Ejection Fraction (LVEF)** of the patient is 26%, significant below the normal value. The **Left Ventricular End-Diastolic Diameter (LVEDD)** is 62mm, which is larger than the normal value of male (42-58mm).

Considering the given patient information, the patient's heart burden may increase (**BMI=33.2**). The **Left Ventricular Mass Index (LVMI)** is 130g/m², which is larger than the normal value of male (50-118mm). In summary, the patient tend to have **cardiomyopathy hypertrophic**.



Case Study: Intervention Planning and Longitudinal Care

Case study: FUAS-Agents

For focused ultrasound ablation, the system combines patient profiles and MRI with specialist tools for segmentation, dose prediction, and guideline retrieval. The output is a personalized plan containing image-based targets, dose parameters, and treatment strategy.

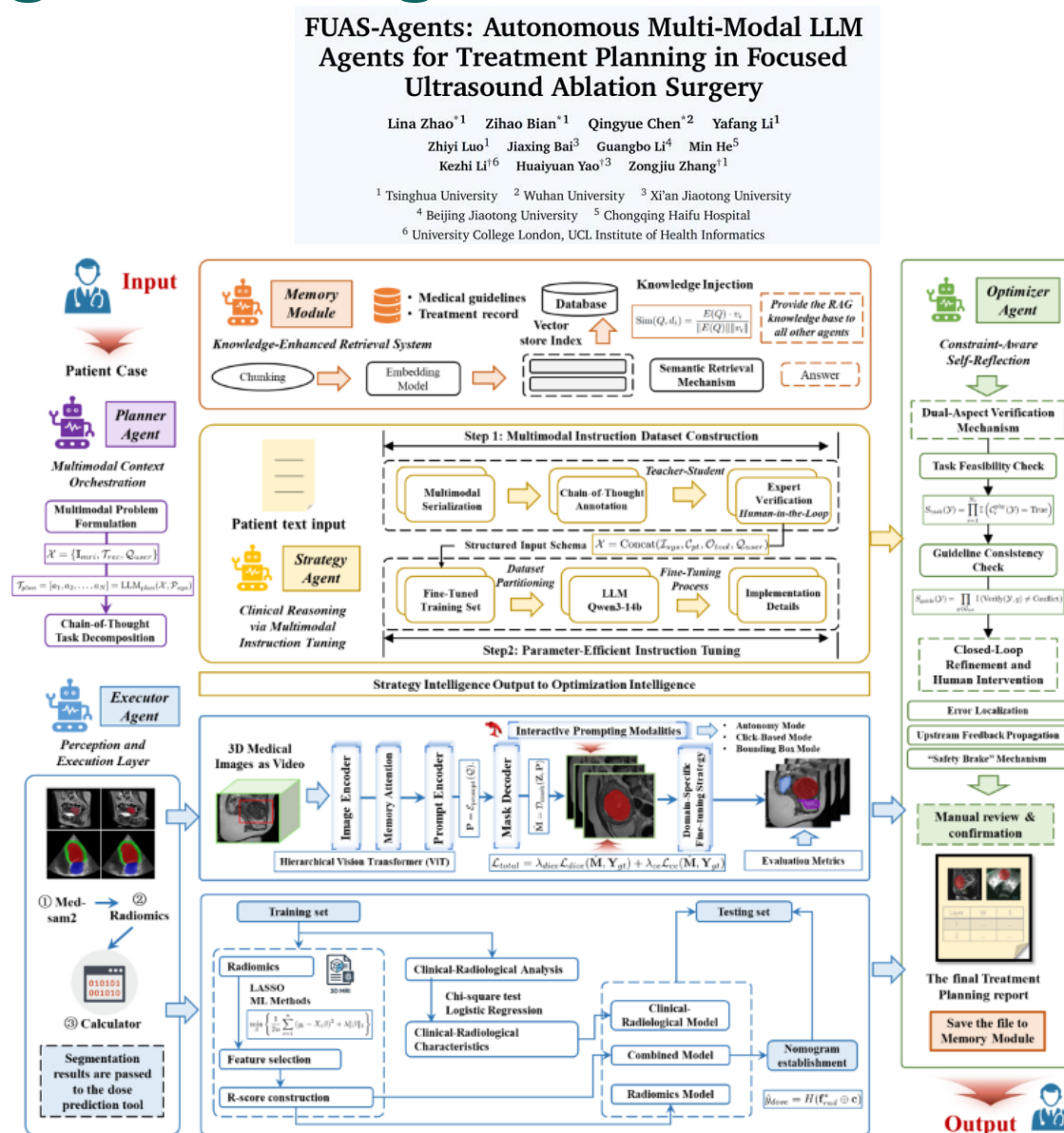
Critical constraint: agents that can recommend or act need bounded permissions, reversible actions, provenance logs, and explicit human authorization at high-risk steps.

Agentic treatment-planning loop

Eligibility check → target segmentation → safety-structure analysis → dose proposal → guideline verification → clinician approval.

Zhang et al., FUAS-Agents: Autonomous Multi-Modal LLM Agents for Focused Ultrasound Surgery, arXiv:2505.21418 (2025); Yue et al., ClinicalAgent, arXiv:2404.14777 (2024).

Extension to longitudinal care	
Use case	Multimodal agent behavior
Ward monitoring	Fuse trends in labs, notes, vitals, waveforms, and medication changes; trigger a clinician-defined escalation pathway
Chronic disease	Combine glucose or cardiac wearables with adherence, symptoms, and EHR context; adapt follow-up intensity
Care coordination	Reconcile discharge plans, appointments, medications, and home-monitoring signals; route unresolved tasks to the responsible team
Clinical trials	Match phenotype, imaging, biomarkers, and eligibility criteria; retrieve supporting evidence for coordinator review



Agentic Healthcare: Build for Evidence, Not Autonomy

Minimum Safe Deployment Stack

1. **Constrain multimodal inputs:** Define supported modalities, input quality requirements, permitted tools, action boundaries, and uncertainty thresholds.
2. **Ground cross-modal conclusions:** Link every output to the relevant image region, video frame, signal interval, text passage, or structured feature.
3. **Validate multimodal reasoning:** Test missing or corrupted modalities, cross-modal conflicts, distribution shifts, tool failures, and modality dominance.
4. **Evaluate the complete pipeline:** Measure perception accuracy, cross-modal alignment, grounding quality, calibration, tool-selection accuracy, latency, and end-task performance.
5. **Monitor after deployment:** Log multimodal inputs, reasoning traces, tool calls, model versions, confidence changes, subgroup performance, and data drift.

Selected References

- Li, B. et al. *MMedAgent: Learning to Use Medical Tools with Multimodal Agent*. arXiv:2407.02483, 2024.
- Schmidgall, S. et al. *AgentClinic: A Multimodal Agent Benchmark to Evaluate AI in Simulated Clinical Environments*. arXiv:2405.07960, 2024.
- Wang, Z. et al. *MedAgent-Pro: Towards Evidence-based Multimodal Medical Diagnosis via Reasoning Agentic Workflow*. arXiv:2503.18968, 2025.
- Xia, P. et al. *MMedAgent-RL: Optimizing Multi-Agent Collaboration for Multimodal Medical Reasoning*. arXiv:2506.00555, 2025.
- Zhang, Y. et al. *FUAS-Agents: Autonomous Multi-Modal LLM Agents for Focused Ultrasound Surgery*. arXiv:2505.21418, 2025.
- *MedAgentBoard: Benchmarking Multi-Agent Collaboration with Conventional Methods for Diverse Medical Tasks*. arXiv:2505.12371, 2025.

Near-term role

The strongest deployment target is a clinician-supervised copilot that assembles evidence and coordinates bounded tasks. Full autonomous diagnosis or treatment requires substantially stronger prospective evidence and governance.

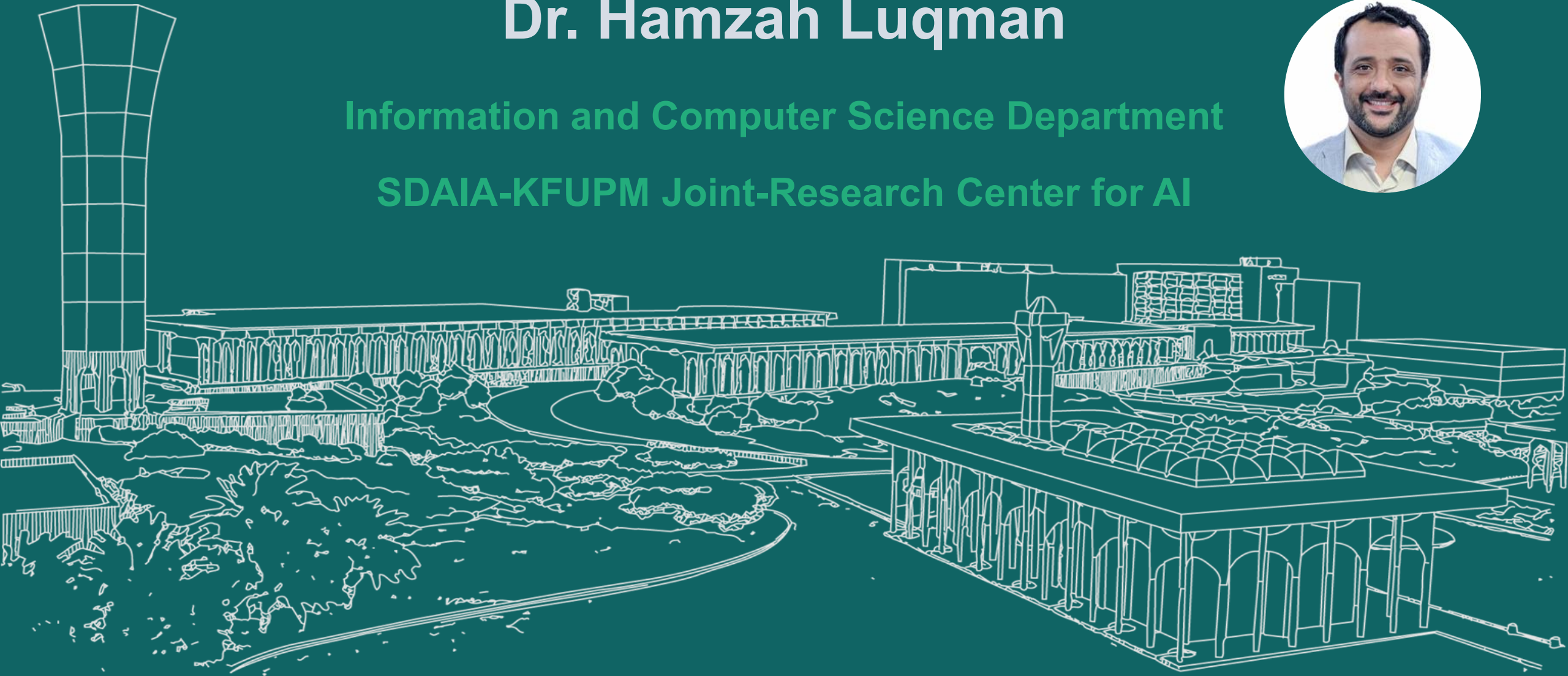
Thank you!

Agentic AI for Natural Language Processing

Dr. Hamzah Luqman

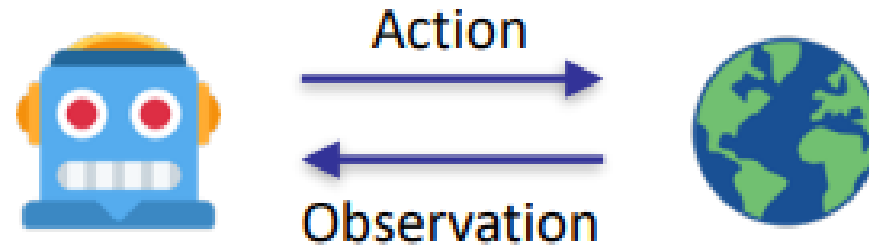
Information and Computer Science Department

SDAIA-KFUPM Joint-Research Center for AI

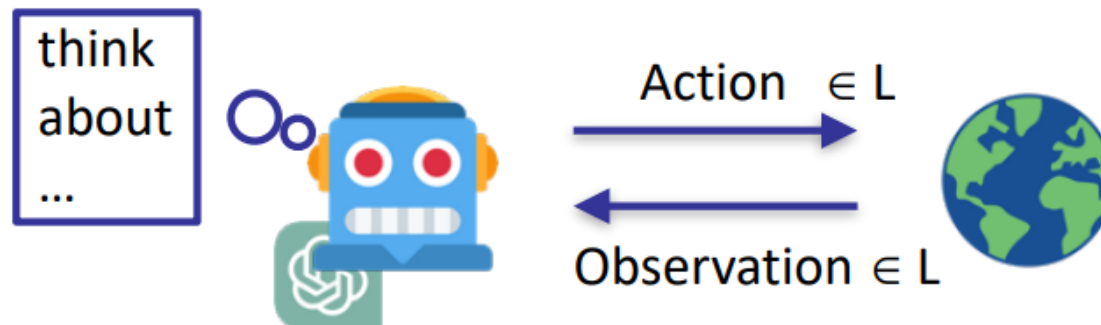


AI agent and LLM agent

- An agent is an “intelligent” system that interacts with some “environment”



- An LLM agent is an agent that uses LLM to act

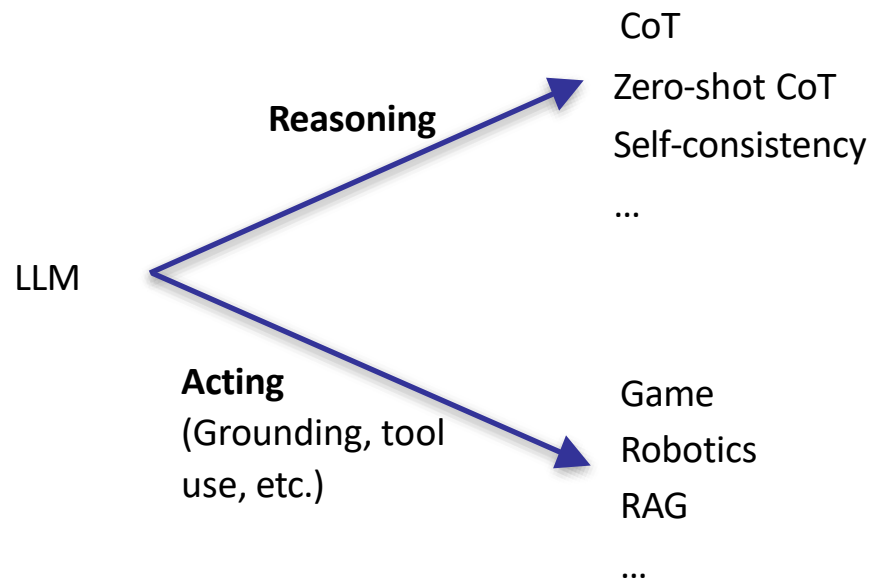
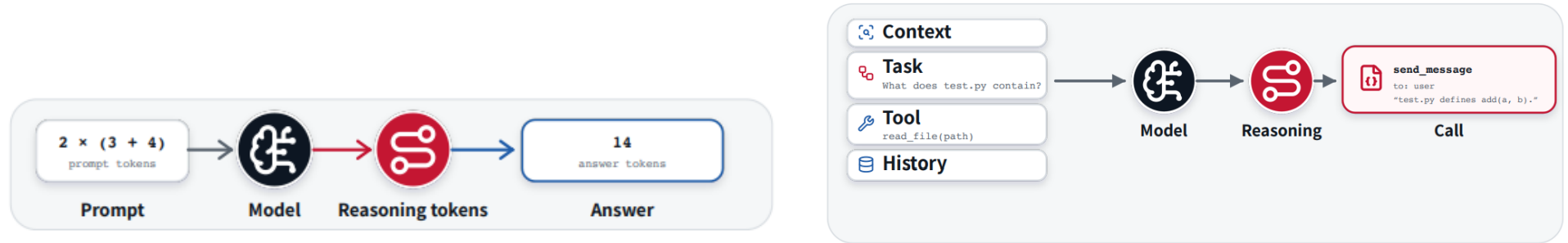


From prompting to trained agents

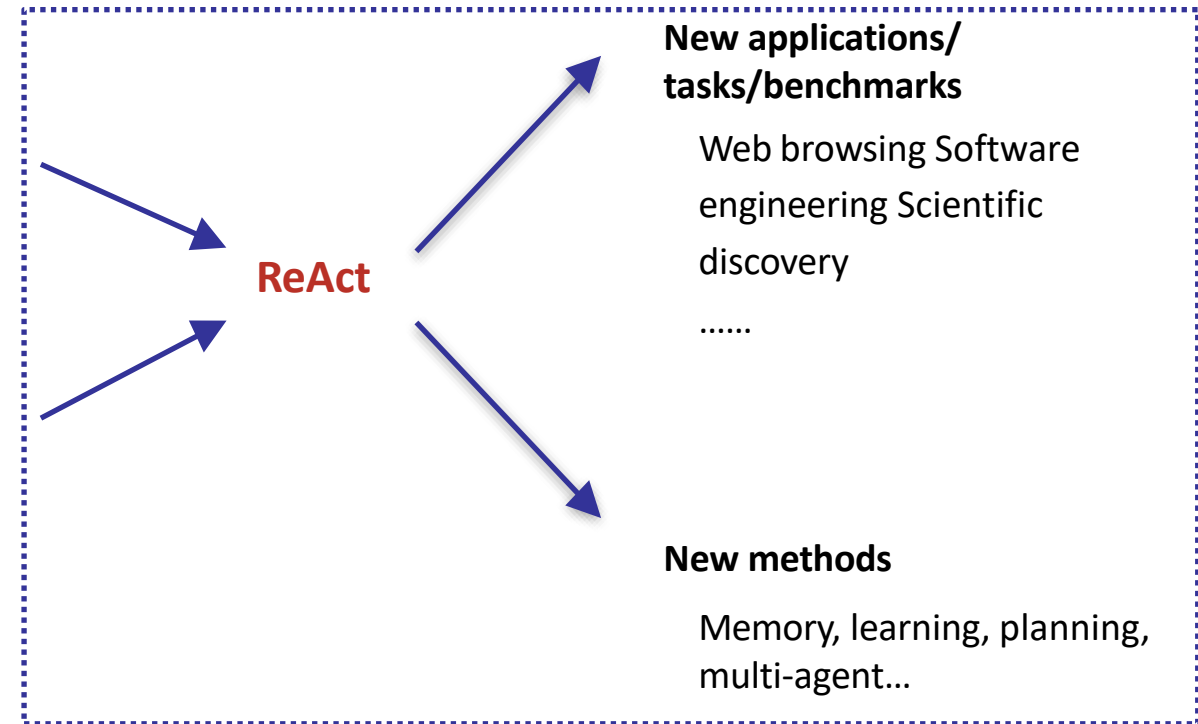


The shift: the agent loop moved from hand-written prompts toward behavior optimized with feedback.

ReAct: Agents that reason and act



LLM agent (but not reasoning agent)



Reasoning agent

ReAct is general and effective

(NLP tasks)

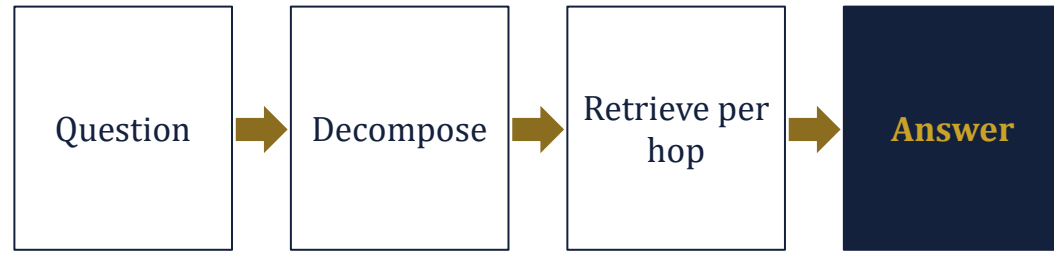
(RL tasks)

PaLM-540B	HotpotQA (QA)	FEVER (fact check)	ALFWorld (Text game)
Reason	29.4	56.3	N/A
Act	25.7	58.9	45
ReAct	35.1	64.6	71

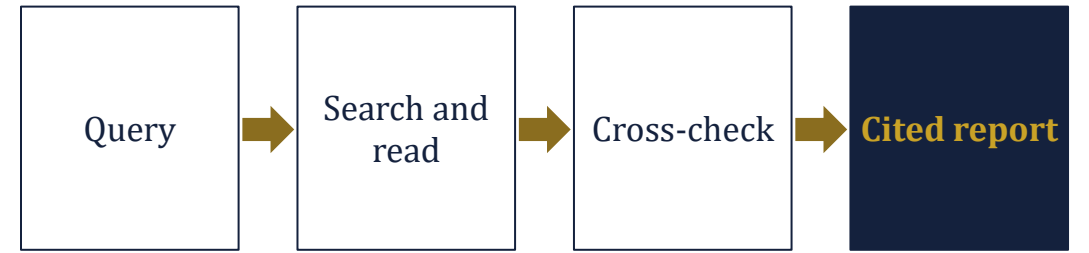
Activate Window

NLP use cases: information access

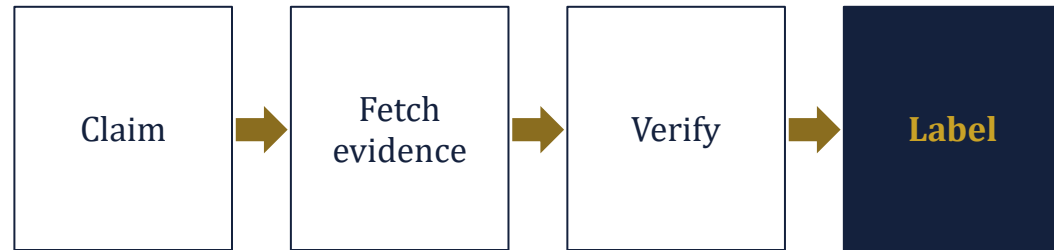
Multi-hop question answering



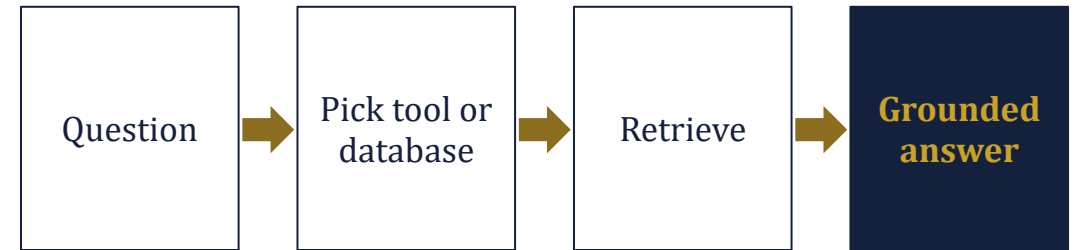
Deep research



Fact-checking, hallucination detection

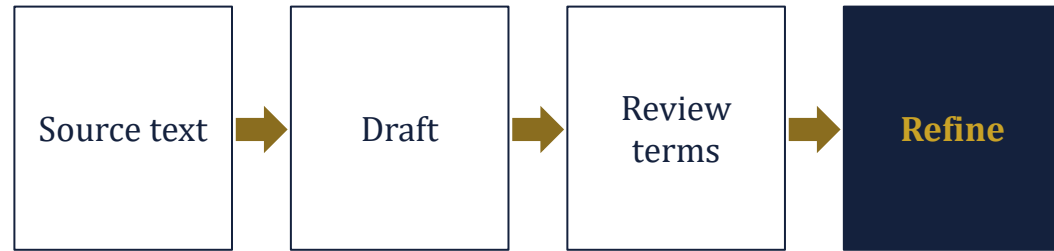


Enterprise and domain search

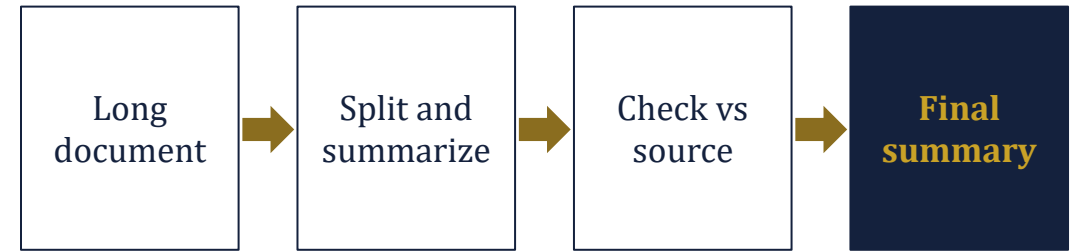


NLP use cases: generation and data

Machine translation



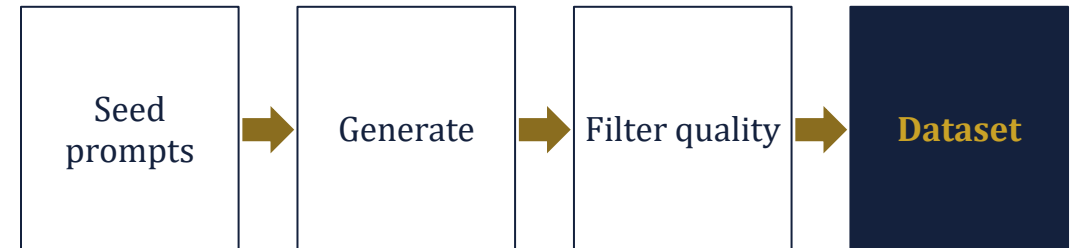
Summarization



Data annotation

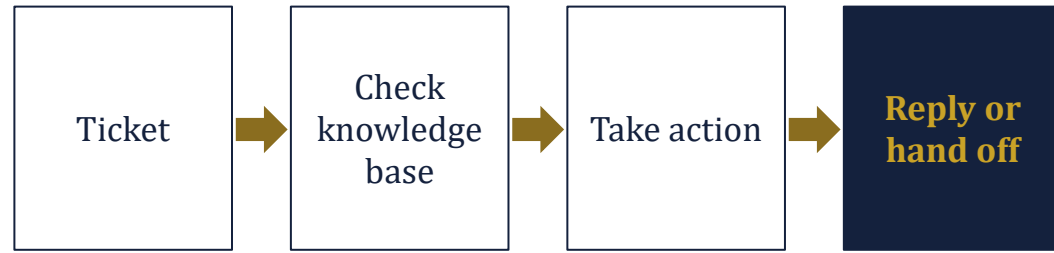


Synthetic data creation

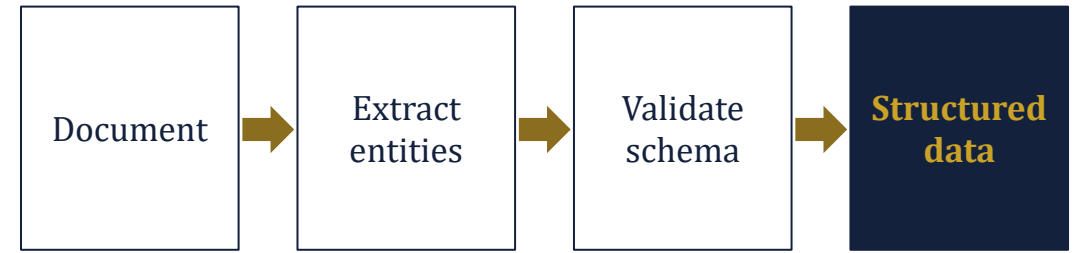


NLP use cases: interaction and analysis

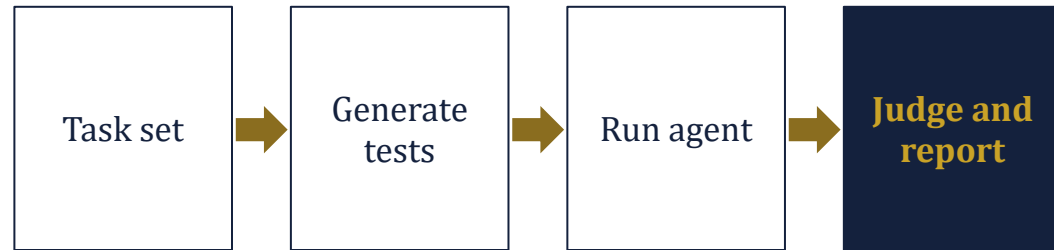
Task-oriented support dialogue



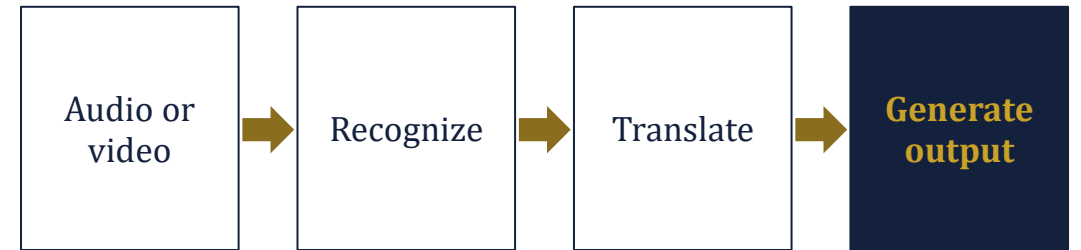
Information extraction



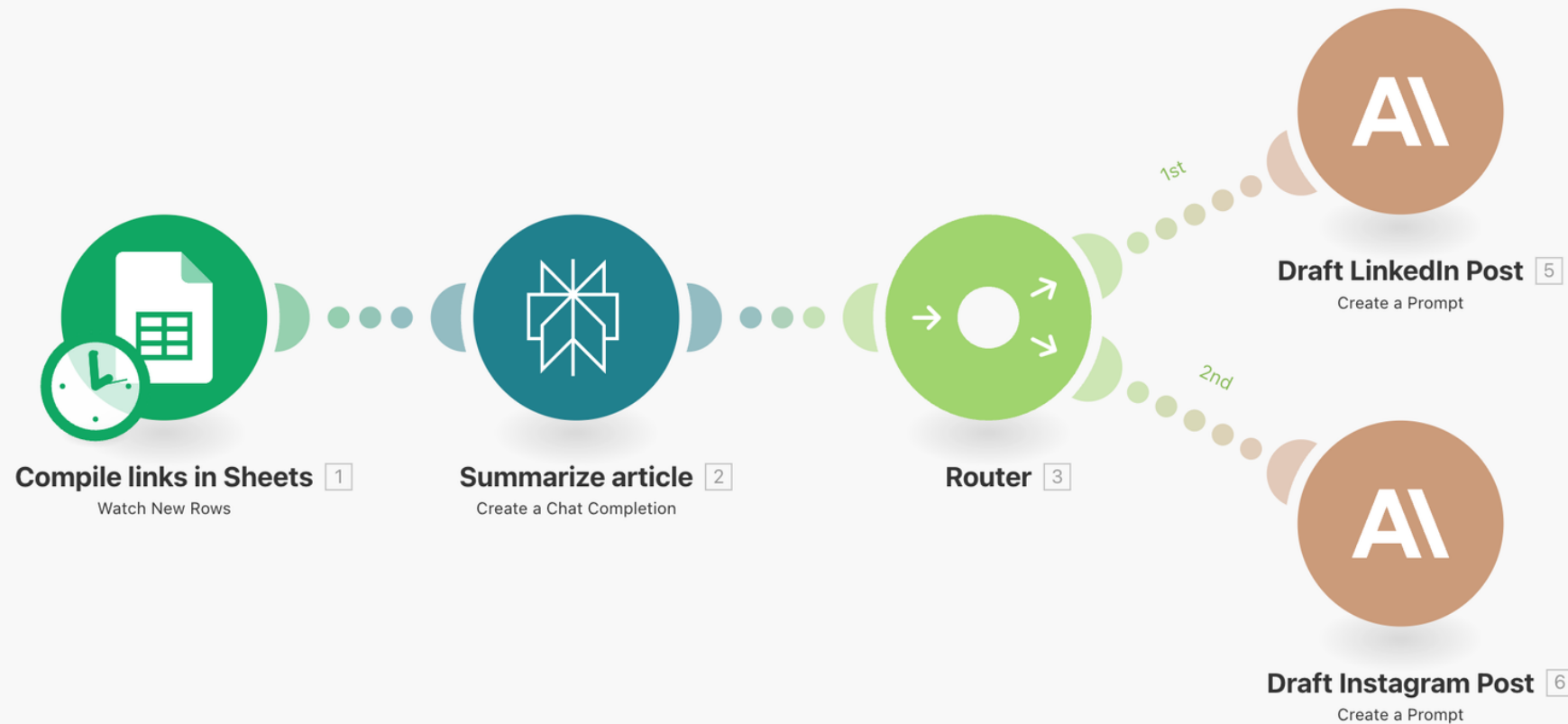
LLM-as-judge evaluation



Speech and sign pipelines



NLP use cases: Generating Social Media Posts





Create a Prompt

NLP use cases: Generating Social Media Posts



Compile links in Sheets 1
Watch New Rows



Summarize article
Create a Chat Completion

1 Collection
Type: text
Text: (Long String)
LinkedIn Post

"The Evolution of Short-Form News: Reflecting on BBC News Summary's Legacy"

For over three decades, the BBC News Summary served as a cornerstone of quick, reliable information for millions. This 90-second bulletin, delivered by a single newsreader, managed to compress national headlines, international developments, regional updates, and weather forecasts into a concise package that kept us informed without overwhelming us.

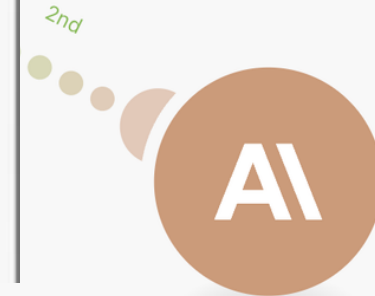
As our media landscape continues to transform, I find myself reflecting on how these efficient information formats shaped our consumption habits. From 1986 to 2018, this brief but mighty program demonstrated that quality news delivery doesn't always require lengthy segments.

What short-form news sources do you rely on today? Has the way you consume news changed significantly in recent years?

Let's discuss how we balance brevity with depth in our increasingly fast-paced information environment.

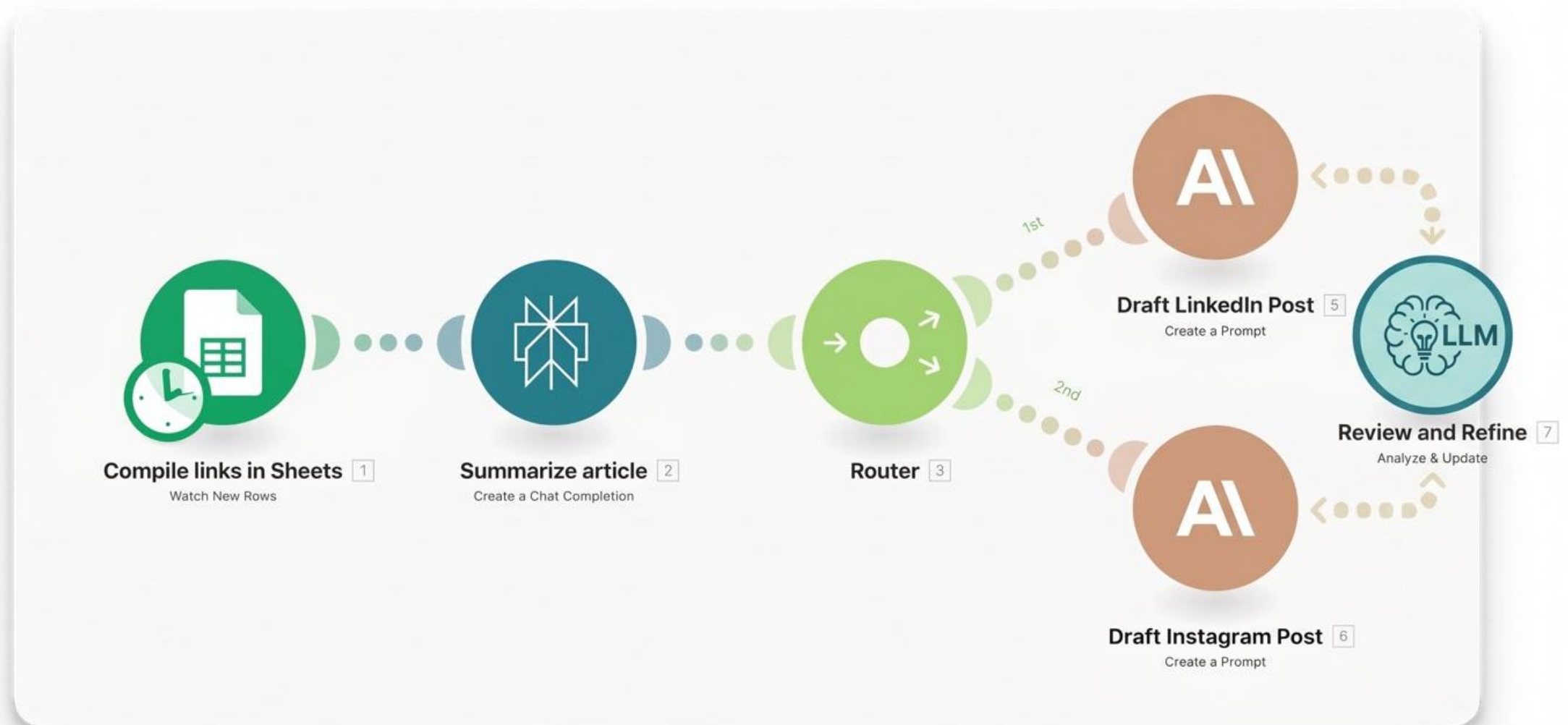


Draft LinkedIn Post 5
Create a Prompt



Draft Instagram Post 6
Create a Prompt

NLP use cases: Generating LinkedIn Post



Thank you!

Agentic AI for Scientific Discovery and Lab Automation

Dr. Burhan SaifAddin

Physics Department, College of Engineering and Physics



Agentic AI is doing to science what photography did to painting

Photography, 1839

THE MACHINE TOOK OVER

Likeness



PEOPLE MOVED TO

- Interpretation, feelings, etc.
- Skill (record keeping, etc.) became accessible for everyone



≈

Agentic workflows, now ?

THE MACHINE TOOK OVER

Routine execution of literature survey, collecting data, code writing, carrying out experiments

PEOPLE MOVED TO

- Human understanding, Judgment, Taste,
- Acceleration
- Skill became accessible for everyone

Photography became accessible to everyone. Most agents skills should be accessible to everyone

AI mathematics, 2022–2026: exponential improvement

2026

Open research problems^[1–4]

2025

Olympiad gold: 35 of 42^[5,6]

2024

Olympiad silver: 28 of 42^[5,7]

2022

Struggle at high-school competition problems^[8]

What it is the cost?

9–900× cheaper per year^[9]

\$2,000 for ten Astra results^[3]

\$1–2k for the nine-loop result^[10]

Assuming the exponential continues, in four years the abilities are unimaginable.

Sources: [1] L. Alpöge, announcement on X (20 Jul 2026) · [2] Anthropic, Riemann zeta (Aug 2026) · [3] OpenAI, Ten advances (Aug 2026) · [4] OpenAI on X (8 Sep 2026) · [5] The Decoder, IMO 2025 · [6] Analytics India Magazine (2025) · [7] Google DeepMind, AlphaProof (2024) · [8] Google Research, Minerva (2022) · [9] Epoch AI, LLM price trends · [10] Anthropic, Nine Loops (25 Sep 2026). Figures are company-reported unless noted. Inference at API rates; training cost not included.

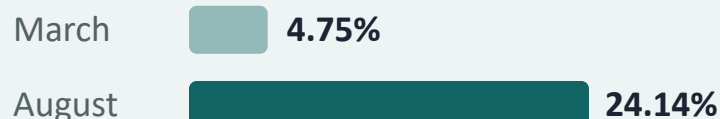
Mathematics and AI: adoption, concern, alignment

Two meanings of “alignment”: mathematicians and AI companies, and mathematics for AI itself

Already here

1 in 4

new arXiv math papers disclosed AI use
in August 2026 (through the 20th)^[1]



A warning and feedback (Fields medallists):

“A Severe Misalignment of AI in Mathematics”^[2]

- Solving famous problems is a proxy for human understanding, not the goal.
- Ideas need people to carry them into the human mathematical canon.
- Rushed announcements leave no time for write-ups or credit to earlier work.

Math for AI alignment

- AI alignment mainly a mathematical problem^[3].
- *We need to figure out a way to align the objective function of AI systems with human values.*
- *Many pure mathematicians are switching to AI alignment and IA safety research.*



math for alignment

Takeaway: the job shifts from solving to understanding, checking, and choosing what to ask.

Biology: agent-led discovery

950
agents

21 h
of searching

210M
tokens

A new CRISPR-like enzyme system in phage DNA: ART^[1]

200,000+
reverse transcriptases gathered

Agents survey a protein family

3,500
new candidate systems

A second pass drops most candidates

20
reports written

One agent spots a CRISPR-like repeat array

1
ART

function still unknown

preprint

all lab work done by people

Speed

≈4×

faster, across 30+ models^[2]

Binder design

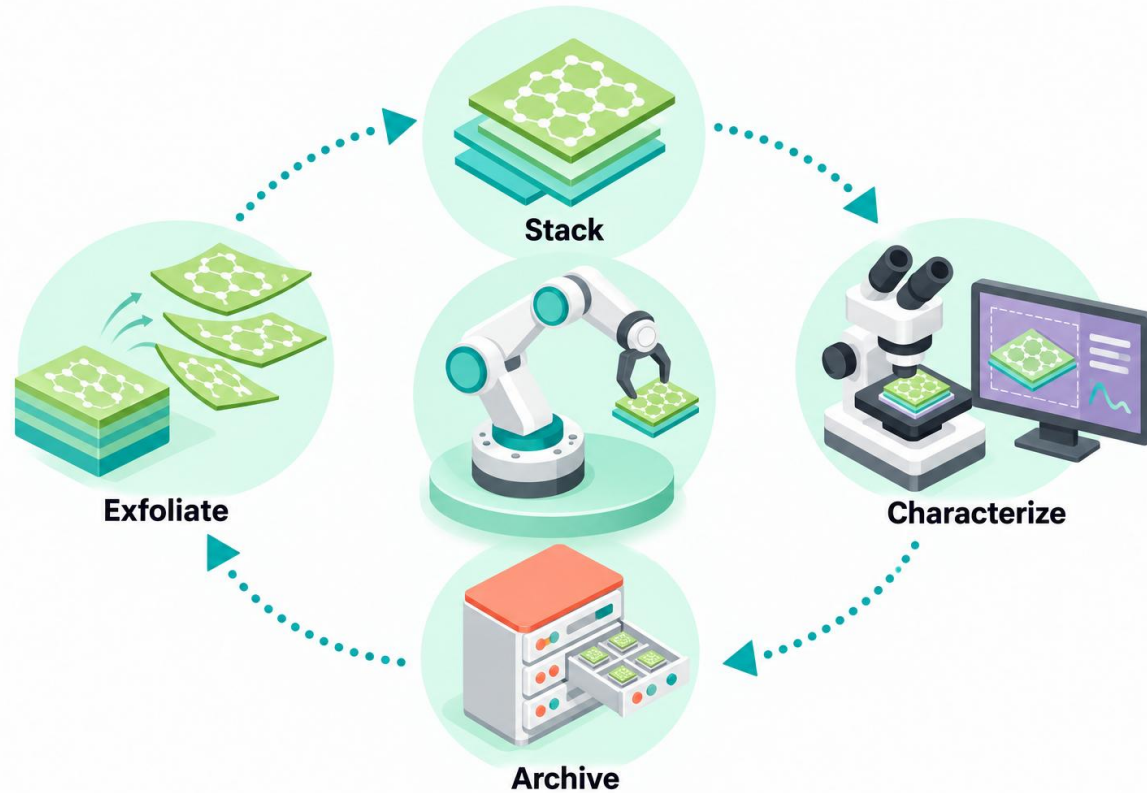
≈100×

fewer GPU hours for the same scores^[2]

Agents search at scale; people run and check.

Sources: [1] Anthropic, ART (23 Sep 2026) · [2] Anthropic, biomolecular modeling (Sep 2026). Company reports; ART is a preprint. ART = array-associated reverse transcriptases.

Robotic labs: accelerated discovery & development of materials



HOW THE AGENT RUNS THE LAB

AI agent



MCP · calls tools^[1]



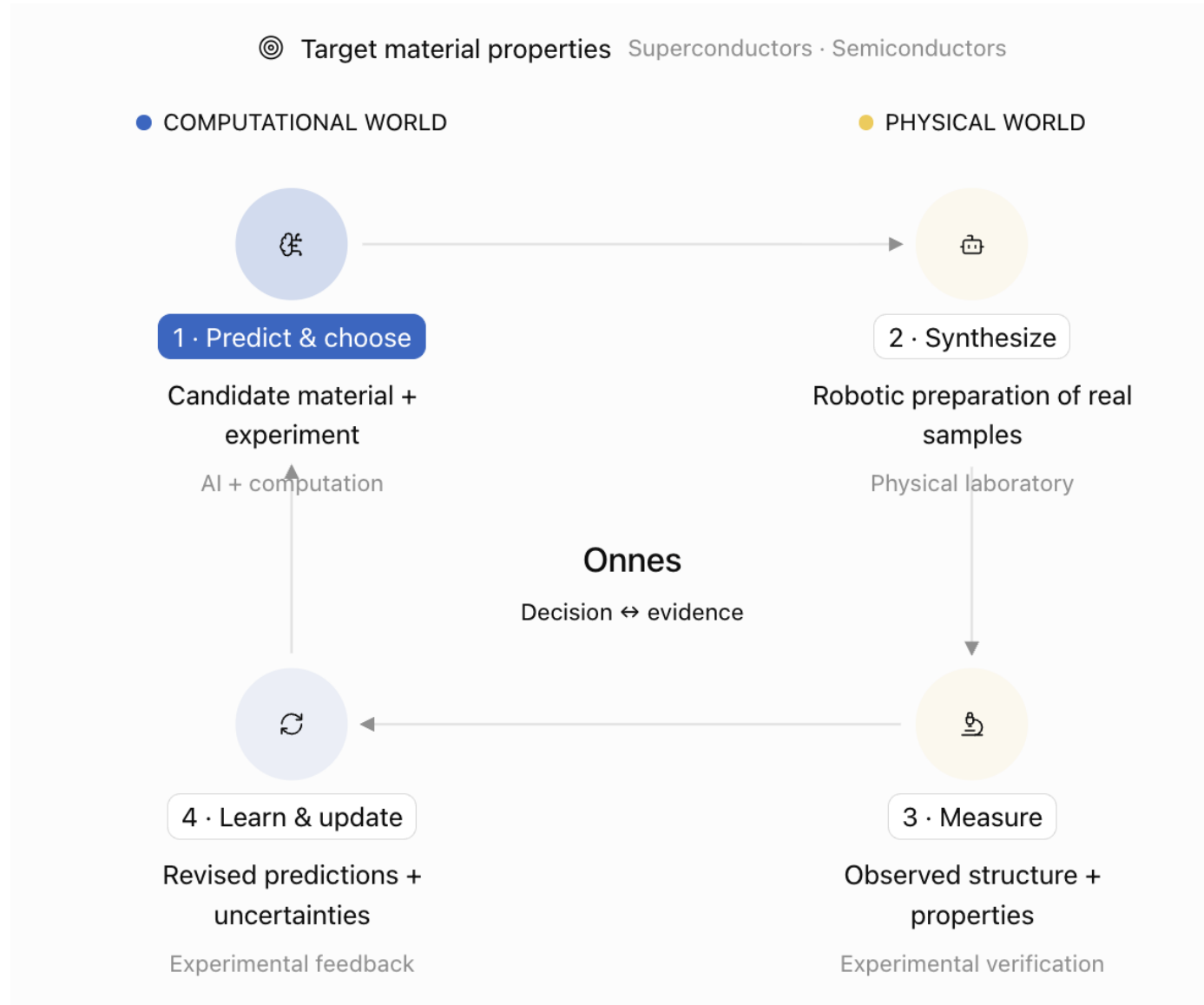
MHS · drives instruments^[2]



Robotic lab · step 3

MCP and MHS make instruments callable, so checking becomes a loop.

Periodic Labs: a lab that trains the model



PERIODIC NEON on FrontierXRD

55.3%

on X-ray analysis, up from 2.7%^[2]

1T parameters^[1]

Released Sept 2026

Results feed the next round of predictions

The lab is the training environment, so the model learns from real atoms.

[1] Periodic Labs, [Introducing Periodic Neon](#) (15 Sep 2026) · [2] [Periodic Labs via L. Fedus on X, FrontierXRD, 134 samples](#) (15 Sep 2026). Company-reported, internal benchmark.

NSF cloud laboratories: awardees

\$400M
NSF funding

20
nodes (teams)

4 years
per award

Biology & biotech

3

- Northwestern · DREAM
- Boston University · CLAIRE
- Maryland · CRAB Lab

2D, thin-film & quantum

4

- Penn State · LATTICE
- Utah · AURORA
- Rice · READINESS
- Purdue · ICoN-PCL

Electronics & manufacturing

2

- Virginia Tech
- Georgia Tech · AIMMPF

Chemistry

3

- Scripps Research
- NC State · SPEED
- Caltech · ELECTRA

Polymers & soft materials

2

- UC Santa Barbara · COAST
- UChicago · PoLARIS

Multi-domain & standards

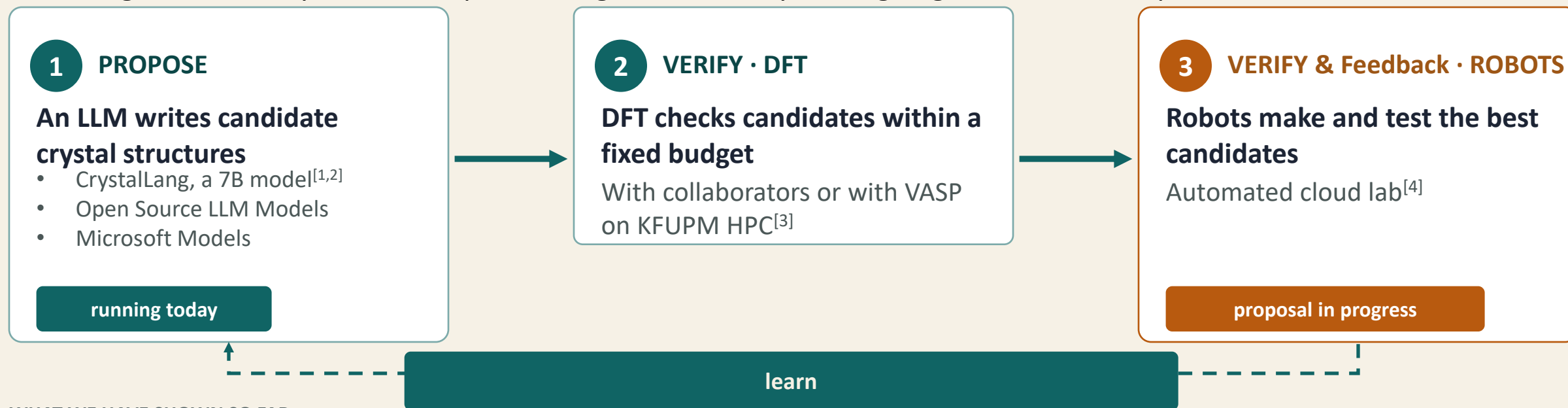
2

- Carnegie Mellon
- Stanford · GEMSTONE

+4 more nodes not yet identified

Materials discovery: propose, feedback, learn

Generating candidate crystals is cheap. Checking them is costly but is going to become cheaper.



WHAT WE HAVE SHOWN SO FAR

NeurIPS 2026 workshop · AI for Science Workshop at NeurIPS 2026 accepted^[5]

What Does a Verification Budget Buy? Sampling and Training Levers in LLM-Driven Crystal Structure Generation

Crystals 2026 · published^[6]

Crystal-X: edge-aware graph network for band gaps

ACM ICPS + MRS Fall 2026 · accepted^[7]

Is Fine-Tuning Worth It?

Other emerging areas in physics and engineering : from high energy physics to chip design.

Thank you!

Building Software Systems using Agentic AI

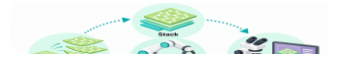
Dr. Omar Jamal Hammad, ICS, SDAIA



إتلاق
etlaq



HOW MUCH DOES IT TAKE



Scriptorium / Python / CS-101: Python Memory Model, References & Mutability

INSTRUCTOR GUIDE + NEW FOLIO PODIUM MODE

CS-101 · Module 02 · Memory & Data Model · 75 Min Run

Python Memory Model, References & Mutability

Variables as Sticky Labels, Heap Allocation, and the Mutable Default Argument Trap – Prof. Alex Mercer (Department of Computer Science)

SPLIT STAGING ENGAGEMENT KIT SPINE TEXT + ADD CHECKPOINT

CS-101 CHECKPOINTS (4 STAGED)

ALL (4) Prompts (2) Polls (1) Quizzes (1)

STAGE 01 · CONCEPTUAL FOUNDATIONS MIN 15:00

"Why does Python use pass-by-object-reference instead of classic pass-by-value or pass-by-reference?"

SURFACE

GIVEN `x = [1, 2]; def modify(lst): lst.append(3); lst = [9]; modify(x)`, what is the final value of `x` and why?

CONCEPTUAL

How does rebinding a local identifier differ from invoking in-place mutator methods on an existing object reference?

COUNTER

Functional languages mandate immutable data structures by default. What performance and safety sacrifices did Python accept by making list, dict, and set mutable?

COLD-CALL KEY:

Call on students with prior C/C++ experience to explain how Python references avoid raw pointers while still exhibiting pointer aliasing.

3 Tiers Calibrated Podium Presentation

STAGE 02 · LIVE CODE PREDICTION POLL MIN 32:00

Given `def add_item(i, box=[]): box.append(i); return box`. What is returned by calling `add_item(1)` followed by `add_item(2)`?

Turnout: 42 / 42 (100%) Quorum Met

[2] – Each call receives an isolated, freshly constructed empty list 7 (17%)

[1, 2] – Default list persists across calls in `fn__defaults__` 32 (76%)

`TypeError` – Python disallows mutable default parameter values at compile time 3 (7%)

Simulate Reset Present Poll

STAGE 03 · CPYTHON INTERNALS DIAGNOSTIC MIN 50:00

Diagnostic: Identity vs Equality in CPython Integer Optimization

In CPython interactive session:

```
>>> a = 256; b = 256; a is b
True
>>> x = 257; y = 257; x is y
False
```

Window: 3:00 START

CPython pre-allocates a singleton array for small integers in the range [-5, 256] during interpreter startup.

STAGE 04 · DEFENSIVE COPYING CHALLENGE MIN 65:00

"When should production Python APIs use immutable tuples/frozensets instead of defensive `copy.deepcopy()`?"

SURFACE

Explain what happens if you attempt to use a list as a dictionary key or set element.

CONCEPTUAL

Why does `deepcopy()` incur severe CPU and memory overhead on recursive or graph-structured objects?

COUNTER

Can dataclasses with `frozen=True` or `NamedTuple` eliminate defensive copying altogether in domain-driven systems?

<https://lecture-engagement.etlaq.app>

A lot is still manual

Course files

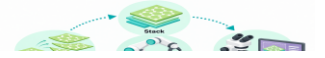
Grading

Lecture preparation

Committee work

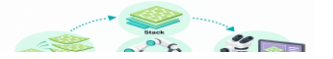
Advising

Flipped classroom activities



We already have ideas

The technical barrier



What comes after the idea?

How do I build it?

Is the design good?

Where do I publish it?

How are people using it?

How do I test it?

How can I improve it?

TODAY

^{AY}**AI makes this possible**

Different ways to build with AI

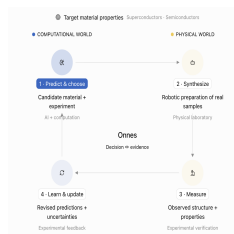
Chat + code

More manual

App builders

More automated

Ask for code. Copy it. Run it.



ChatGPT



Claude



Gemini

Site-building skills

Backend and analytics

More setup to manage

The full development cycle

Requirements

Design

Implementation

Testing

Publishing

Analytics

Integrations

Continuous improvement

Tools you may already know



Etlaq



Replit



Lovable



جامعة الملك فهد للبترول والمعادن
King Fahd University of Petroleum & Minerals

v0

EtlAQ is a KFUPM product

By colleagues and students you have taught

04/06/2010

 CAMBRIDGE <http://www.cambridge.org/9780521876223>

Learning Goal: I will be able to determine the magnitude of the force exerted by a spring.

• LIVE LECTURE: POST-GRADUATE STUDY / ACTING LEARNING DAY

► Assembly Line: Bent

• Flag-Cell: same action

1928-194. IV. INTRODUCTION TO ALGEBRAIC GAME THEORY

FINLLING GROUP: 10 010

FULL TV BOND ISSUE NEW BOND - 10 YEAR BOND ISSUE REDEMPTION TV BOND & NEW

• SUMMARY

© 2004 The Authors

+ google/gemini-3.8-flash

00:13.49

00:13.49

00:13.49

Local support. Local fit.

Local data

1-to-1 technical support

Local integrations

Lower-cost enterprise deals

Thank you!

Etlaq.sa



Breakout Sessions